HEUTHESIS：哈尔滨工程大学学位论文模板

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摘要

该宏包为哈尔滨工程大学本、硕、博中英文毕业论文、开题和中期报告以及博后出站报告模板。

免责声明

1. 本模板的发布遵守 LaTeX Project Public License，使用前请认真阅读协议内容。
2. 本模板为作者根据 哈尔滨工程大学本科生院颁发的《哈尔滨工程大学本科生毕业论文撰写规范》，哈尔滨工程大学研究生院颁发的 《哈尔滨工程大学研究生学位论文撰写规范》编写而成，为 哈尔滨工程大学学生撰写毕业论文使用。
3. 哈尔滨工程大学教务处和研究生院只提供毕业论文写作指南，不提供官方模板，也未授权第三方模板为官方模板，所以此模板仅为写作指南的参考实现，不保证格式审查老师不提意见。任何由于使用本模板而引起的论文格式审查问题均与本模板作者无关。
4. 任何个人或组织以本模板为基础进行修改、扩展而生成的新的专用模板，请严格遵守 LaTeX Project Public License 协议。由于违犯协议而引起的任何纠纷争端均与本模板作者无关。

目录

1 模板介绍 3

2 安装 3

2.1 模板的组成 3
2.2 生成模板 4
2.3 生成论文 4
2.3.1 XƎLaTeX 4
2.3.2 latexmk 5
2.3.3 make 5
2.3.4 buidl.bat 5
2.4 生成封面 5

3 使用说明 6

3.1 示例文件 6
3.2 论文选项 18
3.3 中文字体 18
3.4 前文 18
3.4.1 密级 18
3.4.2 论文标题 19
3.4.3 作者姓名 19
3.4.4 申请学位名称 19
3.4.5 院系名称 19
3.4.6 专业名称 19
3.4.7 导师 20
3.4.8 日期 20
3.4.9 学生类型 20
3.4.10 学号 21
3.4.11 原创性与授权声明 . . 21
3.4.12 摘要 21
3.4.13 符号对照表 21
3.4.14 目录 22
3.5 正文 22
3.5.1 图和表 22
3.5.2 公式 25
3.5.3 数学环境 25
3.5.4 算法 25
3.5.5 引用参考文献 25
3.5.6 列表环境 26
3.6 后文 26
3.6.1 结论 26
3.6.2 参考文献 26
3.6.3 附录 27
3.6.4 所发表文章 27
3.6.5 索引 27
3.6.6 致谢声明 28
3.6.7 简历 28
3.7 其它 28

4 实现细节 29

4.1 基本信息 29
4.2 定义选项 77
4.3 装载宏包 81
4.4 页面设置 83
4.5 主文档格式 85
4.5.1 页面选项 85
4.5.2 字体 86
4.5.3 页眉页脚 88
4.5.4 段落 90
4.5.5 脚注 91
4.5.6 数学相关 91
4.5.7 浮动对象以及表格 . . 96
4.5.8 章节标题 99
4.5.9 定义封面 107
4.5.10 目录 125
4.6 其它 131

5 修改记录 131

6 索引 132

1 模板介绍

HeuThesis (**H**arbin **E**ngineering **U**niversity LaTeX **Thesis** Template) 是为了帮助 哈尔滨工程大学毕业生撰写毕业论文而编写的 LaTeX 论文模板。

本模板是基于哈尔滨工业大学学位论文模板 hiThesis 修改而来。

本文档将尽量完整的介绍模板的使用方法，如有不清楚之处可以参考示例文档，有兴趣者都可以参与完善此手册，也非常欢迎对代码的贡献。

文档记号的说明

本文采用不同字体表示不同内容。无衬线字体表示宏包名称，如 `xeCJK` 宏包、heuhesis-book 文档类等；等宽字体表示代码或文件名，如 `\heusetup` 命令、`thesis.tex` 等；带有尖括号的楷体（或西文斜体）表示命令参数，如 **模板选项**、English title 等。在使用时，参数两侧的尖括号不必输入。示例代码进行了语法高亮处理，以方便阅读。在用户手册中，带有紫色侧边线的为 LaTeX 代码，而带有红色侧边线的则为命令行代码，请注意区分。

注意：模板的作用在于减少论文写作过程中格式调整的时间。前提是遵守模板的用法，否则即使用了 *HeuThesis* 也难以保证输出的论文符合学校规范。

2 安装

HeuThesis 一般不需要进行安装，直接将压缩包解压后就可以使用。

2.1 模板的组成

下表列出了 HeuThesis 的主要文件及其功能介绍：

文件（夹）	功能描述
heuthesis.ins	DocStrip 驱动文件（开发用）
heuthesis.dtx	DocStrip 源文件（开发用）
heuthesisbook.cls	模板类文件
heuthesisbook.cfg	模板配置文件
heuthesis.bst	参考文献样式文件
heuthesis.ist	索引样式文件
reference.bib	文档参考文献
main.tex	示例文档主文件
front/	正文之前内容
body/	正文内容
back/	正文之后内容
data/	TiZk 绘图数据文件
figures/	插图图片路径
heuthesis.sty	为示例文档加载其它宏包
Makefile	Makefile

文件（夹）	功能描述
latexmkrc	latexmk 配置文件
README.md	Readme
heuthesis.pdf	用户手册（本文档）

几点说明：

- `heuthesisbook.cls` 和 `heuthesisbook.cfg` 可由 `heuthesis.ins` 和 `heuthesis.dtx` 生成。
- 使用前阅读文档：`heuthesis.pdf`。
- 默认的生成的论文中含有丰富的格式示例，使用前请仔细阅读 `main.pdf`。

2.2 生成模板

将模板解压缩到文件夹，其中包括：模板源文件（`heuthesis.ins` 和 `heuthesis.dtx`），参考文献样式 `heuthesis.bst` 和示例文档。在使用之前需要先生成模板文件和配置文件（具体命令细节请参考 `README.md` 和 `Makefile`）：

如果是直接使用编译后的模板文件，可以忽略下述操作。

```
$ cd heuthesis
# 生成 heuthesisbook.cls 和 heuthesisbook.cfg
$ xelatex heuthesis.ins

# 下面的命令用来生成用户手册，可以不执行
$ xelatex heuthesis.dtx
$ makeindex -s gind.ist -o heuthesis.ind heuthesis.idx
$ makeindex -s gglo.ist -o heuthesis.gls heuthesis.glo
$ xelatex heuthesis.dtx
$ xelatex heuthesis.dtx  % 生成说明文档 heuthesis.pdf
```

2.3 生成论文

本节介绍几种常见的生成论文的方法。用户可根据自己的情况选择。

2.3.1 XeLaTeX

很多用户对 LaTeX 命令执行的次数不太清楚。一个基本的原则是多次运行 LaTeX 命令直至不再出现警告。下面给出生成示例文档的详细过程（# 开头的行为注释），首先来看推荐的 `xelatex` 方式：

```
# 1. 发现里面的引用关系，文件后缀 .tex 可以省略
$ xelatex main

# 2. 编译参考文件源文件，生成 bbl 文件
$ bibtex main

# 3. 下面解决引用
$ xelatex main
```

```
$ xelatex main    # 如果不需要生成索引此时生成完整的 pdf 文件
$ splitindex main -- -s heuthesis.ist   # 自动生成索引
$ xelatex main.tex
```

2.3.2 latexmk

latexmk 命令支持全自动生成 LaTeX 编写的文档，并且支持使用不同的工具链来进行生成，它会自动运行多次工具直到交叉引用都被解决。下面给出了一个用 latexmk 调用 xelatex 生成最终文档的示例：

```
# 一句话就够了！
$ latexmk -xelatex main
```

2.3.3 make

上面的方法虽然不复杂，但是每次都输入还是非常罗嗦，所以 HeuThesis 提供了一个 Makefile：

```
$ make clean
$ make cls        # 生成 heuthesisbook.cls 和 heuthesisbook.cfg
$ make doc        # 生成说明文档 heuthesis.pdf
$ make main       # 生成示例文档 main.pdf
```

HeuThesis 的 Makefile 默认用 latexmk 调用 xelatex 编译，此外还支持直接用 xelatex 编译。如有需要可修改 Makefile 开头的参数或通过命令行传递参数（请参看 README.md），进一步还可以修改 latexmkrc 进行定制。

生成新的类文件和配置文件即可。也可以直接拷贝 heuthesisbook.cls,heuthesisbook.cfg 和 heuthesis.ist，免去上面命令的执行。

2.3.4 buidl.bat

示例文件夹中也包含了 build.bat 批处理文件，在 Windows 环境下可以直接使用改批处理文件生成最终 PDF 格式的论文 main.pdf。

2.4 生成封面

模板还提供了用于生成论文 A3 页面的封面定义文档。用户可根据自己的情况选择。

A3 封面的生成是利用上述论文的首页封面和书脊拼合而成，因此在生成 A3 封面前需要先生成论文 main.pdf 并在 spine.tex 定义论文标题和作者姓名信息，然后使用 xelatex 依次编译 spine.tex 和 a3cover.tex：

```
# 1. 使用 xelatex
$ xelatex spine
$ xelatex a3cover

# 2. 或者使用 latexmk
$ latexmk spine
$ latexmk a3cover
```

在 Windows 系统下也可以直接使用 build_all.bat 批处理生成 A3 封面。

3 使用说明

本手册假定用户已经能处理一般的 LaTeX 文档，并对 BibTeX 有一定了解。如果从来没有接触过 TeX 和 LaTeX，建议先学习相关的基础知识。

3.1 示例文件

模板核心文件有三个:`heuthesisbook.cls`,`heuthesisbook.cfg` 和 `heuthesis.bst`，但是如果没有示例文档用户会发现很难下手。所以推荐新用户从模板自带的示例文档入手，里面包括了论文写作用到的所有命令及其使用方法，只需要用自己的内容进行相应替换就可以。对于不清楚的命令可以查阅本手册。下面的例子描述了模板中章节的组织形式，来自于示例文档，具体内容可以参考模板附带的 `main.tex`。

```
% !Mode:: "TeX:UTF-8"
%
%%%
```

```
%   set your OS here, or, use open source font set
% openright=true|false
%   Force the Chapter page number to be even or not.
% pageempty=true|false
%   Epmty page without pagehead and pagefoot
% library=true|false
%   Set this option will remove all blank page in your thesis, Library
preferred.
% newtxmath=true|false
%   Math font used in the thesis
%
%%%
```

```latex
\documentclass[type=bachelor,fontset=windows]{heuthesisbook}
% 此处选项中不要有空格
%
%%%
```

```
\tableofcontents   % 论文目录

%%\cleardoublepage % 控制右开页，与前文连续开页时注释掉
%%\listoffigures     % 插图目录，如不需要可用 % 注释掉下面两行

%%\cleardoublepage % 控制右开页，与前文连续开页时注释掉
%%\listoftables      % 表格目录，如不需要可用 % 注释掉下面两行

%%----------  论文主体，可以根据需要添加章节 --------------
\cleardoublepage % 控制右开页，后续论文正文一般采用此格式
\mainmatter
\include{body/chap01}
\include{body/chap02}
\include{body/chap03}
\include{body/chap04}
\include{body/chap05}
\include{body/chap06}
\include{body/chap07}

%%----------  论文结论、参考文献、附录等部分 ------------
\backmatter
\include{back/conclusion}   % 结论

\bibliographystyle{heuthesis}
\bibliography{reference}

\include{back/publications} %成果
\include{back/acknowledgements} %致谢

\include{back/AppendixA}  % 附录A
\include{back/AppendixB}  % 附录B
\include{back/AppendixC}  % 附录C
\include{back/AppendixD}  % 附录D
\include{back/AppendixE}  % 附录E

\end{document}
```

```
% !Mode:: "TeX:UTF-8"
%
%%%
```

```
%
%
%%%
```

```latex
\begin{document}
\frontmatter
\input{front/cover} % 封面和扉页
\makecover

%%\authorization                % 使用模板生成的原始原创性与授权声明页
\authorization[auscan.pdf] % 使用签字扫描版的原创性与授权声明页

\cleardoublepage   % 控制右开页，与前文连续开页时注释掉
\makeabstract      % 封面和摘要

\cleardoublepage   % 控制右开页，与前文连续开页时注释掉
\tableofcontents   % 论文目录

%%\cleardoublepage % 控制右开页，与前文连续开页时注释掉
\listoffigures     % 插图目录，如不需要可用 % 注释掉下面两行

%%\cleardoublepage % 控制右开页，与前文连续开页时注释掉
\listoftables      % 表格目录，如不需要可用 % 注释掉下面两行

%%----------  论文主体，可以根据需要添加章节 ---------------
\cleardoublepage % 控制右开页，后续论文正文一般采用此格式
\mainmatter
\include{body/chap01}
\include{body/chap02}
\include{body/chap03}
\include{body/chap04}
\include{body/chap05}
\include{body/chap06}
\include{body/chap07}

%%----------  论文结论、参考文献、附录等部分 -------------
\backmatter

\include{back/conclusion}   % 结论

\bibliographystyle{heuthesis}
\bibliography{reference}

\include{back/publications}      % 所发文章
\include{back/acknowledgements} % 致谢
\include{back/resume}            % 个人简介

\include{back/AppendixA}  % 附录A
\include{back/AppendixB}  % 附录B
\include{back/AppendixC}  % 附录C
\include{back/AppendixD}  % 附录D
\include{back/AppendixE}  % 附录E

\end{document}
```

```
% !Mode:: "TeX:UTF-8"
%
%%%
```

```latex
%   翻页，将这个选项设置为false，然后在目标位置添加`\cleardoublepage`命令即
%   可。
% pageempty=true|false
%   含义：空白页是否无页眉页脚（默认否）。
% engtoc=true|false
%   含义：是否需要添加英文目录
%
%%%
```

```
\include{back/publications}     % 所发文章
\include{back/acknowledgements} %致谢
\include{back/resume}           % 个人简介

\include{back/AppendixA}  % 附录A
\include{back/AppendixB}  % 附录B
\include{back/AppendixC}  % 附录C
\include{back/AppendixD}  % 附录D
\include{back/AppendixE}  % 附录E

\end{document}
```

```
% !Mode:: "TeX:UTF-8"
%
%%%
```

```
%   题内容，封面中显示。
% openright=true|false
%   含义：博士论文是否要求章节首页必须在奇数页，此选项不在规范要求中，按个
%   人喜好自行决定。 默认否。注意，我校的默认情况是打印版博士论文要求右翻页
%   ，电子版要求非右翻页且无空白页。如果想DIY（或身不由己DIY）在什么地方右
%   翻页，将这个选项设置为false，然后在目标位置添加`\cleardoublepage`命令即
%   可。
% pageempty=true|false
%   含义：空白页是否无页眉页脚（默认否）。
%
%%%
```

```
\bibliographystyle{heuthesis}
\bibliography{reference}

\include{back/publications}     % 所发文章
\include{back/acknowledgements} %致谢
\include{back/resume}           % 个人简介

\include{back/AppendixA}  % 附录A
\include{back/AppendixB}  % 附录B
\include{back/AppendixC}  % 附录C
\include{back/AppendixD}  % 附录D
\include{back/AppendixE}  % 附录E

\end{document}
```

```
% !Mode:: "TeX:UTF-8"
%
%%%
```

```
%   下载安装后设置使用。
% subtitle=true|false
%   含义：论文题目是否含有副标题，缺省值为false，如果有要在cover中设置副标
%   题内容，封面中显示。
% openright=true|false
%   含义：博士论文是否要求章节首页必须在奇数页，此选项不在规范要求中，按个
%   人喜好自行决定。 默认否。注意，我校的默认情况是打印版博士论文要求右翻页
%   ，电子版要求非右翻页且无空白页。如果想DIY（或身不由己DIY）在什么地方右
%   翻页，将这个选项设置为false，然后在目标位置添加`\cleardoublepage`命令即
%   可。
% pageempty=true|false
%   含义：空白页是否无页眉页脚（默认否）。
% engtoc=true|false
%   含义：是否需要添加英文目录
%
%%%
```

```
\include{back/acknowledgements} %致谢
\include{back/publications}     % 所发文章
\include{back/resume}           % 个人简介
\include{back/correspondingaddr} %通信地址

\end{document}
```

3.2 论文选项

论文选项，就是在 main.tex文件的开头，非注释的第一行的方括号中填写的选项，示例见上节。各个选项的含义说明已经在上节中说明，所以这里就不重复了。

3.3 中文字体

正确配置中文字体是使用模板的第一步。模板调用 CTEX 宏包，只提供基于 xeCJK 包，使用 XELATEX 编译的方式。关于如何使用字体命令、字号等等，属于模板格式范畴，在实现细节中讨论。关于中文字体安装、配置的所有问题不在本模板讨论范围。

3.4 前文

前文内容是正文之前，含封面、摘要、目录、符号表。封面信息提供两种配置方法：一是通过统一设置命令 \heusetup 通过 *key=value* 形式完成；二是每个信息利用命令独立设置，其中命令的名字跟 *key* 相同。两种方式可以交叉使用，并按顺序执行（即后来的设置会覆盖前面的）。以 c 开头的命令跟中文相关，e 开头则为对应的英文。

\heusetup \heusetup 用法与常见 *key=value* 命令相同，如下：

```
\heusetup{
  key1 = value1,
  key2 = {a value, with comma},
}
% 可以多次调用
\heusetup{
  key3 = value3,
  key1 = value11, % 覆盖 value1
}
```

注意：*\heusetup 使用 kvoptions 机制，所以配置项之间不能有空行，否则会报错。*

大多数命令的使用方法都是：\command{⟨*arg*⟩}，例外者将具体指出。这些命令都在示例文档的 front/cover.tex 中。

3.4.1 密级

statesecrets 定义秘密级别、编号和国内国际索引号。

cnumber
natclassifiedindex
intclassifiedindex

```
\heusetup{
statesecrets={公开},
cnumber={no9527},
natclassifiedindex={TM301.2},
```

```
intclassifiedindex={62-5},
}
```

3.4.2 论文标题

论文标题
ctitle
etitle
ctitleone
ctitletwo
csubtitle
esubtitle

中英文标题。如果有副标题，需要在封面选项中设置 subtitle=true，否则不显示副标题。博后研究报告封面研究内容标题采用两行格式，需要对 ctitleone 和 ctitletwo 分别定义。

```
\heusetup{
  ctitlecover={封面中文题目（可使用 \cs\cs 断行）},
  ctitle={论文中文题目（页眉等不断行处使用）},
  etitle={Thesis English Title},
  csubtitle={论文中文副题目（如果有）},
  esubtitle={Thesis English Sub-Title (if necessary)},
  ctitleone={博后中文题目上部分},
  ctitletwo={博后中文题目下部分},
}
```

3.4.3 作者姓名

作者姓名
cauthor
eauthor

作者姓名。

```
\heusetup{
  cauthor={中文姓名},
  eauthor={Name in Pinyin}
}
```

3.4.4 申请学位名称

学科名称
cxueke
exueke

按照入学的培养计划中学科自行填写，具体学科名称不是本文档范畴。

```
\heusetup{
  cxueke={工学},
  exueke={Engineering},
}
```

3.4.5 院系名称

院系名称
caffil
eaffil

院系名称，同上，按照入学的培养计划中学科自行填写，具体院系名称不是本文档范畴。

```
\heusetup{
 caffil={动力与能源工程学院},
 eaffil={\emultiline[t]{School of Power and Energy Engineering}},
}
```

注意：个别学院英文名过长，使用以上方法自动换行。

3.4.6 专业名称

专业名称
csubject
esubject

专业名称，同上，按照入学的培养计划中学科自行填写，具体名称不是本文档范畴。

```
\heusetup{
 csubject={动力工程},
 esubject={Engineering of Power Energy},
}
```

3.4.7 导师

导师

csupervisor
esupervisor
cprofessional
creviewer

直接导师。

```
\heusetup{
  csupervisor={导师~教授},
  esupervisor={Supervisor},
  cprofessional={职称（本科）},
  creviewer={论文主审人}
}
```

副导师

cassosupervisor
eassosupervisor

副指导教师。

```
\heusetup{
  cassosupervisor={副导师~副教授},
  eassosupervisor={2nd Boss}
}
```

联合导师

ccosupervisor
ecosupervisor

硕士生联合指导教师，博士生联合导师。

```
\heusetup{
  ccosupervisor={联合导师~教授},
  ecosupervisor={3rd Boss}
}
```

3.4.8 日期

日期

cdate
csubmitdate
coralexdate
edate

日期内容包括封面日期、论文提交日期和论文答辩日期等。其中论文封面日期默认为当前时间，也可以自己指定。论文提交日期和论文答辩日期需要手工设置。

```
\heusetup{
  cdate={中文日期},
  csubmitdate={论文提交日期},
  coralexdate={论文答辩日期},
  edate={English Date}
}
```

3.4.9 学生类型

学生类型

profdegree
researchtype
cstudenttype
estudenttype

非全日制或专业学位申请者（profdegree=true），如：同等学力人员、工程硕士、工商管理硕士、高级管理人员工商管理硕士、公共管理硕士、中职教师、高校教师等，具体要求按照入学的培养计划中学科自行填写，具体名称不是本文档范畴。同时根据专业学位论文扉页格式要求，还需要通过 researchtype 参数设置论文研究类型。profdegree 和 researchtype 通过模板选项设置。cstudenttype 和 estudenttype 在封面定义文件 `back/cover.tex` 文件中通过 `\heusetup` 设置。

```
\heusetup{
  cstudenttype={工程硕士},
  estudenttype={Master of Art},
}
```

3.4.10 学号

学号

cstudentid

学号，具体要求按照入学的培养计划中学科自行填写。

```
\heusetup{
  cstudentid={9527},
}
```

3.4.11 原创性与授权声明

\authorization

原创性与授权声明页中的签名和日期是需要手写。模板中提供了未签名的空白论文原创性与授权声明页 auscan(empty).pdf和示例签名后的文件 auscan.pdf，可以先打印空白页后进行签字，并使用签字后的扫描版文件对 auscan.pdf 进行替换，并使用带参数的 \authorization 指令插入该扫描文件。

```
\authorization[auscan.pdf]
```

3.4.12 摘要

摘要正文

cabstract (*env.*)

eabstract (*env.*)

说明：摘要正文只能用环境命令的形式，不支持 *\heusetup*。

```
\begin{cabstract}
 摘要请写在这里...
\end{cabstract}

\begin{eabstract}
 Here comes the abstract in English...
\end{eabstract}
```

关键词

ckeywords

ekeywords

关键词用英文逗号分割写入相应的命令中，模板会解析各关键词并生成符合不同论文格式要求的关键词格式。

```
\heusetup{
  ckeywords={关键词 1, 关键词 2},
  ekeywords={keyword 1, keyword 2}
}
```

3.4.13 符号对照表

denotation (*env.*)

主要符号表环境，单独在文件 front/denotation.tex 中。跟据《哈尔滨工程大学研究生学位论文撰写规范》示例中要求，我校符号表是 table 环境，示例文件如下，由于我校要求博士论文图表标题是双语，所以任何对单个标题的全局格式调整都会影响到双语标题，所以这里使用 \vspace，具体见实现细节中的描述。

```
\begin{denotation}
\caption{国际单位制中具有专门名称的导出单位}
\vspace{0.5em}\centering\wuhao
\begin{tabular}{ccccc}
\toprule[1.5pt]
量的名称&单位名称&单位符号&其它表示实例\\
\midrule[1pt]
频率&赫[兹]&Hz&s-1\\
\bottomrule[1.5pt]
\end{tabular}
\end{table}
\end{denotation}
```

3.4.14 目录

目录不需要用户干预，自动生成，具体命令已经写在 `main.tex`中。

3.5 正文

3.5.1 图和表

哈尔滨工程大学博士毕业论文要求使用中英双语图题、表题，这增加了维护难度。因为现有唯一的方法是在已有的图题或表题的基础上再添加一行英语图题或表题。两个题之间的距离具体多少不在《哈尔滨工程大学研究生学位论文撰写规范》 中要求。目前的方法是用户手动调节该距离。关于图题《哈尔滨工程大学研究生学位论文撰写规范》和《哈尔滨工程大学本科生毕业论文撰写规范》 只规定了居中，并没有规定居中对其。然而评审老师很多喜欢居中且居中对齐。模板默认选项是居中且居中对齐，如果不喜欢居中对齐，那么需要在 `main.tex` 的文档类选项中设置选项 capcenterlast=false。详细方法见前文的介绍。 《哈尔滨工程大学研究生学位论文撰写规范》,《哈尔滨工程大学本科生毕业论文撰写规范》规定："每个图均应有图题（由图序和图名组成），图题不宜有标点符号，图名在图序之后空 1 个半角字符排写。图序按章编排，如第 1 章第一个插图的图号为"图 1.1"。图题置于图下，硕士论文只用中文，博士论文用中、英两种文字，居中书写，中文在上，要求中文用宋体 5 号字，英文用 Times New Roman 5 号字。有图注或其它说明时应置于图题之上。引用图应注明出处，在图题右上角加引用文献号。图中若有分图时，分图题置于分图之下或图题之下，可以只用中文书写，分图号用 (a)、(b) 等表示。图中各部分说明应采用中文（引用的外文图除外）或数字符号，各项文字说明置于图题之上（有分图时，置于分图题之上）。图中文字用宋体、Times New Roman 字体，字号尽量采用 5 号字（当字数较多时可用小 5 号字，以清晰表达为原则，但在一个插图内字号要统一）。同一图内使用文字应统一。图表中物理量、符号用斜体。" 单双语图题的方法如下，注释中说明。

```
\begin{figure}[htpb]
\centering
\includegraphics[width = 0.4\textwidth]{golfer}
\bicaption[golfer1]{}{注意图中文字尽量用五号字
}{Fig.$\!$}{The person playing golf}
\end{figure}
```

单张单图题的格式如下，

```
\begin{figure}[h]
\centering
\includegraphics[width = 0.4\textwidth]{golfer}
\caption{注意图中文字字号尽量用五号字}
\end{figure}
```

并排图例。

```
\begin{figure}[htbp]
\centering
\begin{minipage}{0.4\textwidth}
\centering
\includegraphics[width=\textwidth]{golfer}
\bicaption[golfer2]{}{打高尔夫球的人}{Fig.$\!$}{The person playing golf}
\end{minipage}
\begin{minipage}{0.4\textwidth}
\centering
\includegraphics[width=\textwidth]{golfer}
\bicaption[golfer3]{}{打高尔夫球的人}{Fig.$\!$}{The person playing golf}
\end{minipage}
\end{figure}
```

子图图例。

```
\begin{figure}[htbp]
\centering
\subfigure{\label{golfer41}}\addtocounter{subfigure}{-2}
\subfigure[The person playing golf]{\subfigure[打高尔夫球的人~1]{\
includegraphics[width=0.4\textwidth]{golfer}}}
\subfigure{\label{golfer42}}\addtocounter{subfigure}{-2}
\subfigure[The person playing golf]{\subfigure[打高尔夫球的人~2]{\
includegraphics[width=0.4\textwidth]{golfer}}}
\bicaption[golfer4]{}{打高尔夫球的人}{Fig.$\!$}{The person playing golf}
\end{figure}
```

表格示例，表格中的字体是可以自行调整的。

```
\begin{table}[htbp]
\bicaption[table1]{}{符合研究生院绘图规范的表格}{Table$\!$}{Table in agreement of
the standard from graduate school}
\vspace{0.5em}\centering\wuhao
\begin{tabular}{ccccc}
\toprule[1.5pt]
$D$(in) & $P_u$(lbs) & $u_u$(in) & $\beta$ & $G_f$(psi.in)\\
\midrule[1pt]
 5 & 269.8 & 0.000674 & 1.79 & 0.04089\\
10 & 421.0 & 0.001035 & 3.59 & 0.04089\\
20 & 640.2 & 0.001565 & 7.18 & 0.04089\\
\bottomrule[1.5pt]
\end{tabular}
\end{table}
```

因为长表格不是浮动体，不会自动调整位置、也不会自动调整字体大小，一切都要手动设置。设置起来特别繁琐，建议尽量不要使用长表格。

```
\ltfontsize{\dawu[1.667]} %设置表格内字体行间距
\dawu[1.667]\begin{longtable}{ccc} % 注意此处设置的是表格线距离
\longbionenumcaption{}{{\wuhao 中国省级行政单位一览 %此处要添加字体设置
}\label{table2}}{Table$\!$}{}{{\wuhao Overview of the provincial
administrative
unit of China}}{-0.5em}{3.15bp}\\ %注意后两个参数分别是中英标题间距、标题和表格的间
距。
%\caption{\wuhao 中国省级行政单位一览}\\[1em] %注意此处是标题和表格间距，这行
%是单语标题
\toprule[1.5pt] 名称 & 简称 & 省会或首府  \\ \midrule[1pt]
\endfirsthead
\multicolumn{3}{r}{表~\thetable（续表）}\vspace{0.5em}\\
\toprule[1.5pt] 名称 & 简称 & 省会或首府  \\ \midrule[1pt]
\endhead
\bottomrule[1.5pt]
\endfoot
北京市 & 京 & 北京\\
天津市 & 津 & 天津\\
河北省 & 冀 & 石家庄市\\
山西省 & 晋 & 太原市\\
内蒙古自治区 & 蒙 & 呼和浩特市\\
辽宁省 & 辽 & 沈阳市\\
吉林省 & 吉 & 长春市\\
黑龙江省 & 黑 & 哈尔滨市\\
上海市 & 沪/申 & 上海\\
江苏省 & 苏 & 南京市\\
浙江省 & 浙 & 杭州市\\
安徽省 & 皖 & 合肥市\\
福建省 & 闽 & 福州市\\
江西省 & 赣 & 南昌市\\
山东省 & 鲁 & 济南市\\
河南省 & 豫 & 郑州市\\
湖北省 & 鄂 & 武汉市\\
湖南省 & 湘 & 长沙市\\
广东省 & 粤 & 广州市\\
广西壮族自治区 & 桂 & 南宁市\\
海南省 & 琼 & 海口市\\
重庆市 & 渝 & 重庆\\
四川省 & 川/蜀 & 成都市\\
贵州省 & 黔/贵 & 贵阳市\\
云南省 & 云/滇 & 昆明市\\
西藏自治区 & 藏 & 拉萨市\\
陕西省 & 陕/秦 & 西安市\\
甘肃省 & 甘/陇 & 兰州市\\
青海省 & 青 & 西宁市\\
宁夏回族自治区 & 宁 & 银川市\\
新疆维吾尔自治区 & 新 & 乌鲁木齐市\\
香港特别行政区 & 港 & 香港\\
澳门特别行政区 & 澳 & 澳门\\
台湾省 & 台 & 台北市\\
```

```
\end{longtable}\normalsize %注意这里要恢复正常字体
```

3.5.2 公式

公式不做介绍，与正常用法一致。

3.5.3 数学环境

HeuThesis 定义了常用的数学环境：

axiom 公理	theorem 定理	definition 定义	proposition 命题	lemma 引理	conjecture 猜想	
proof 证明	corollary 推论	example 例子	exercise 练习	assumption 假设	remark 注释	problem 问题

比如：

```
\begin{definition}
  道千乘之国，敬事而信，节用而爱人，使民以时。
\end{definition}
```

产生（自动编号）：

定义 1.1 道千乘之国，敬事而信，节用而爱人，使民以时。

列举出来的数学环境毕竟是有限的，如果想用胡说这样的数学环境，那么可以定义：

```
\newtheorem{nonsense}{胡说}[chapter]
```

然后这样使用：

```
\begin{nonsense}
  契丹武士要来中原夺武林秘笈。—— 慕容博
\end{nonsense}
```

产生（自动编号）：

胡说 1.1 契丹武士要来中原夺武林秘笈。——慕容博

3.5.4 算法

我校算法不在规范中要求，且每个人都有自己的算法格式喜好，大家最好采用大家普遍接收的方式，做到美观、协调就可以。

3.5.5 引用参考文献

\inlinecite

学校要求的参考文献引用有两种模式：(1) 上标模式。比如“同样的工作有很多 [1,2]…”。(2) 正文模式。比如“文 [3] 中详细说明了…”。其中上标模式使用远比正文模式频繁，所以为了符合使用习惯，上标模式仍然用常规的 `\cite{⟨key⟩}`，而 `\inlinecite{⟨key⟩}` 则用来生成正文模式。

关于参考文献模板推荐使用 BibTeX，关于中文参考文献需要额外增加一个 Entry: `language`，将其设置为 `zh` 用来指示此参考文献为中文，以便 `heuthesis.bst` 处理。如：

```
@INPROCEEDINGS{cnproceed,
  author    = {王重阳 and 黄药师 and 欧阳峰 and 洪七公 and 段皇帝},
  title     = {武林高手从入门到精通},
  booktitle = {第~$N$~次华山论剑},
  year      = 2006,
  address   = {西安, 中国},
  month     = sep,
  language  = "zh",
}

@ARTICLE{cnarticle,
  AUTHOR  = "贾宝玉 and 林黛玉 and 薛宝钗 and 贾探春",
  TITLE   = "论刘姥姥食量大如牛之现实意义",
  JOURNAL = "红楼梦杂谈",
  PAGES   = "260--266",
  VOLUME  = "224",
  YEAR    = "1800",
  LANGUAGE= "zh",
}
```

注意如果不需要引用参考文献，请删除 main.tex 中 \bibliography 开头的两行，以避免可能的编译错误。

3.5.6 列表环境

itemize (*env.*)
enumerate (*env.*)
description (*env.*)

为了适合中文习惯，模板将这三个常用的列表环境用 enumitem 进行了纵向间距压缩。一方面清除了多余空间，另一方面用户可以自己指定列表环境的样式（如标签符号，缩进等）。细节请参看 enumitem 文档，此处不再赘述。

3.6 后文

3.6.1 结论

conclusion (*env.*)

结论之后为后文内容。

```
\begin{conclusions}

学位论文的结论作为论文正文的最后一章单独排写，但不加章标题序号。

结论应是作者在学位论文研究过程中所取得的创新性成果的概要总结，不能与摘要混为一
谈。博士学位论文结论应包括论文的主要结果、创新点、展望三部分，在结论中应概括论
文的核心观点，明确、客观地指出本研究内容的创新性成果（含新见解、新观点、方法创
新、技术创新、理论创新），并指出今后进一步在本研究方向进行研究工作的展望与设想
。对所取得的创新性成果应注意从定性和定量两方面给出科学、准确的评价，分（1）、
（2）、（3）…条列出，宜用“提出了”、“建立了”等词叙述。

\end{conclusions}
```

3.6.2 参考文献

在后文中的参考文献是自动生成的，不需要用户干预，具体命令在 main.tex中有示例。

3.6.3 附录

appendix (*env.*) 所有的附录都插到这里来。因为附录会更改默认的 chapter 属性，而后面的**个人简历**又需要恢复，所以实现为环境可以保证全局的属性不受影响。

```
\begin{appendix}
\input{back/appA.tex}
\end{appendix}
```

3.6.4 所发表文章

publication (*env.*) 虽然在《哈尔滨工程大学研究生学位论文撰写规范》《哈尔滨工程大学本科生毕业论文撰写规范》中都没有明确规定此处的格式，此处基本按照参考文献的要求进行设置。

```
\begin{publication}
\noindent\textbf{（一）发表的学术论文}
\begin{publist}
\item XXX, XXX. Static Oxidation Model of Al-Mg/C Dissipation Thermal
Protection Materials[J]. Rare Metal Materials and Engineering, 2010, 39(Suppl
. 1): 520-524.（SCI~收录，IDS号为~669JS，IF=0.16）
\item XXX, XXX. 精密超声振动切削单晶铜的计算机仿真研究[J]. 系统仿真学报，2007，19（4）：
738-741，753.（EI~收录号：20071310514841）
\item XXX, XXX. 局部多孔质气体静压轴向轴承静态特性的数值求解[J]. 摩擦学学报，2007（1）：
68-72.（EI~收录号：20071510544816）
\item XXX, XXX. 硬脆光学晶体材料超精密切削理论研究综述[J]. 机械工程学报，2003，39（8）：
15-22.（EI~收录号：2004088028875）
\item XXX, XXX. 基于遗传算法的超精密切削加工表面粗糙度预测模型的参数辨识以及切削参数优化[J
]. 机械工程学报，2005，41（11）：158-162.（EI~收录号：2006039650087）
\item XXX, XXX. Discrete Sliding Mode Cintrok with Fuzzy Adaptive Reaching Law
 on 6-PEES Parallel Robot[C]. Intelligent System Design and Applications,
Jinan, 2006: 649-652.（EI~收录号：20073210746529）
\end{publist}

\noindent\textbf{（二）申请及已获得的专利（无专利时此项不必列出）}
\begin{publist}
\item XXX, XXX. 一种温热外敷药制备方案：中国，88105607.3[P]. 1989-07-26.
\end{publist}

\noindent\textbf{（三）参与的科研项目及获奖情况}
\begin{publist}
\item XXX, XXX. XX~气体静压轴承技术研究，XX~省自然科学基金项目.课题编号：XXXX.
\item XXX, XXX. XX~静载下预应力混凝土房屋结构设计统一理论. 黑江省科学技术二等奖，2007.
\end{publist}
%\vfill
%\hangafter=1\hangindent=2em\noindent
%\setlength{\parindent}{2em}
\end{publication}
```

3.6.5 索引

ceindex (*env.*) 我校要求中英文双语索引。后文中的自动索引实际上不需要用户干预。

```
\begin{ceindex}
  %如果想要手动加索引，注释掉以下这一样，用wordlist环境
\printsubindex*
\end{ceindex}
```

手工添加索引的方法不推荐，模板中将去除该功能。

3.6.6 致谢声明

acknowledgement (*env.*) 把致谢做成一个环境更好一些，直接往里面写感谢的话就可以啦！

```
\begin{acknowledgement}
  ...
  感谢\heu\LaTeX\ 论文模板\heuthesis\ !
\end{acknowledgement}
```

3.6.7 简历

resume (*env.*) 个人简历。实际上，致谢和个人简历是自由发挥的地区，字体，文体，格式，内容，完全自己决定。

```
\begin{resume}
XXXX~年~XX~月~XX~日出生于~XXXX。

XXXX~年~XX~月考入~XX~大学~XX~院（系）XX~专业，XXXX~年~XX~月本科毕业并获得~XX~学学士学位。

XXXX~年~XX~月------XXXX~年~XX~月在~XX~大学~XX~院（系）XX~学科学习并获得~XX~学硕士学位。

XXXX~年~XX~月------XXXX~年~XX~月在~XX~大学~XX~院（系）XX~学科学习并获得~XX~学博士学位。

获奖情况：如获三好学生、优秀团干部、X~奖学金等（不含科研学术获奖）。

工作经历：
\end{resume}
```

3.7 其它

模板的配置文件 heuthesisbook.cfg 中定义了很多固定词汇，一般无须修改。如果有特殊需求，推荐在导言区使用 \renewcommand。

4 实现细节

4.1 基本信息

```
⟨*bst⟩
ENTRY                                   % class Entry {
{                                       % public:
  author                                %   String author;
  editor                                %   String editor;
  translator                            %   String translator;
  title                                 %   String title;
  edition                               %   String edition;
  address                               %   String address;
  publisher                             %   String publisher;
  pages                                 %   String pages;
  year                                  %   String year;
  date                                  %   String date;
  modifydate                            %   String modifydate;
  citedate                              %   String citedate;
  url                                   %   String url;
  doi                                   %   String doi;
  language                              %   String language;
  booktitle                             %   String booktitle;
  journal                               %   String journal;
  chapter                               %   String chapter;
  series                                %   String series;
  volume                                %   String volume;
  number                                %   String number;
  version                               %   String version;
  month                                 %   String month;
  school                                %   String school;
  institution                           %   String institution;
  organization                          %   String organization;
  type                                  %   String type;
  howpublished                          %   String howpublished;
  eid                                   %   String eid;
  key                                   %   String key;
  country                               %   String country;
  patentid                              %   String patentid;
  media                                 %   String media;
} {                                     %   //  declare integer variables
  required                              %   int required;  // withther the bi
} {                                     %   //  declare String variables
  label                                 %   String label;           //  label
  mark                                  %   String mark;            //  mark
                                        %   //  there is ahidden entry variab
                                        %   String sort_key;
}                                       % }
                                        %
%%%
```

```
                                                    %
INTEGERS {                                          % //  declare global int variables
  entry.count                                       % static int entry_count;          //
  longest.label.width                               % static int longest_label_width;  //
  i                                                 % static int i;
  j                                                 % static int j;
  k                                                 % static int k;
}                                                   %
                                                    %
%%%
```

```
% Predefined latex command used to format the style of bibitems
FUNCTION {env.bibbegin}                 { "\begin{thebibliography}" }
FUNCTION {env.bibend}                   { "\end{thebibliography}" }
FUNCTION {cmd.bibauthor}                { "\providecommand{\bibauthor}[1]{#1}" }
FUNCTION {cmd.bibeditor}                { "\providecommand{\bibeditor}[1]{#1}" }
FUNCTION {cmd.bibtranslator}            { "\providecommand{\bibtranslator}[1]{#1}" }
FUNCTION {cmd.bibtitle}                 { "\providecommand{\bibtitle}[1]{#1}" }
FUNCTION {cmd.bibbooktitle}             { "\providecommand{\bibbooktitle}[1]{#1}" }
FUNCTION {cmd.bibjournal}               { "\providecommand{\bibjournal}[1]{#1}" }
FUNCTION {cmd.bibmark}                  { "\providecommand{\bibmark}[1]{\mbox{#1}}" }
FUNCTION {cmd.bibcountry}               { "\providecommand{\bibcountry}[1]{#1}" }
FUNCTION {cmd.bibpatentid}              { "\providecommand{\bibpatentid}[1]{#1}" }
FUNCTION {cmd.bibedition}               { "\providecommand{\bibedition}[1]{#1}" }
FUNCTION {cmd.biborganization}          { "\providecommand{\biborganization}[1]{#1}" }
FUNCTION {cmd.bibaddress}               { "\providecommand{\bibaddress}[1]{#1}" }
FUNCTION {cmd.bibpublisher}             { "\providecommand{\bibpublisher}[1]{#1}" }
FUNCTION {cmd.bibinstitution}           { "\providecommand{\bibinstitution}[1]{#1}" }
FUNCTION {cmd.bibschool}                { "\providecommand{\bibschool}[1]{#1}" }
FUNCTION {cmd.bibvolume}                { "\providecommand{\bibvolume}[1]{#1}" }
FUNCTION {cmd.bibnumber}                { "\providecommand{\bibnumber}[1]{#1}" }
FUNCTION {cmd.bibversion}               { "\providecommand{\bibversion}[1]{#1}" }
FUNCTION {cmd.bibpages}                 { "\providecommand{\bibpages}[1]{#1}" }
FUNCTION {cmd.bibmodifydate}            { "\providecommand{\bibmodifydate}[1]{#1}" }
FUNCTION {cmd.bibcitedate}              { "\providecommand{\bibcitedate}[1]{#1}" }
FUNCTION {cmd.bibyear}                  { "\providecommand{\bibyear}[1]{#1}" }
FUNCTION {cmd.bibdate}                  { "\providecommand{\bibdate}[1]{#1}" }
FUNCTION {cmd.biburl}                   { "\providecommand{\biburl}[1]{\newline\url{#1}

%%%
```

```
  } if$                                                   %   }
}                                                         % }
                                                          %
%%%
```

```
        false                                   %         stop = false;
      } if$                                     %       }
    } while$                                    %     }
    pop$ length.i #1 -                          %     return length_i - 1;
  } if$                                         %   }
}                                               % }
                                                %
%%%
```

```
  } while$                                          %   }
  length -                                          %   int n = n - length(suffix);
  #1 swap$ substring$                               %   return substring(str, 1, n);
}                                                   % }
                                                    %
%%%
```

```
FUNCTION {remove.dots} {                          % String Entry::remove_dots(String st
  "" 'remove.dots.result :=                       %   remove_dots_result = "";
  { duplicate$ empty$ not } {                     %   while (! empty(str)) {
    duplicate$ #1 #2 substring$                   %     String tmp = substring(str, 1,
    "\." = {                                      %     if (tmp == "\.") {
      #3 global.max$ substring$                   %       str = substring(str, 3, globa
    } {                                           %     } else {
      duplicate$ #1 #1 substring$                 %       tmp = substring(str, 1, 1);
      duplicate$ "." = {                          %       if (tmp == ".") {
        pop$ #2 global.max$ substring$            %         str = substring(str, 2, glo
      } {                                         %       } else {
        remove.dots.result swap$ *                %         tmp = remove_dots_result +
        'remove.dots.result :=                    %         remove_dots_result = tmp;
        #2 global.max$ substring$                 %         str = substring(str, 2, glo
      } if$                                       %       }
    } if$                                         %     }
  } while$                                        %   }
  pop$ remove.dots.result                         %   return remove_dots_result;
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {add.brace} {                            % String Entry::add_brace(String str)
  "{" swap$ * "}" *                               %   return "{" + str + "}";
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {add.bracket} {                          % String Entry::bracket(String str) {
  "(" swap$ * ")" *                               %   return "(" + str + ")";
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {add.squarebracket} {                    % String Entry::add_squarebracket(Str
  "[" swap$ * "]" *                               %   return "[" + str + "]";
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {add.textit} {                           % String Entry::add_textit(String str
  "\textit{" swap$ * "}" *                        %   return "\textit{" + str + "}";
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {add.textbf} {                           % String Entry::add_textbf(String str
  "\textbf{" swap$ * "}" *                        %   return "\textbf{" + str + "}";
}                                                 % }
                                                  %
```

```
%%%
```

```
  } {                                                   %   } else {
    swap$ pop$ #2 global.max$ substring$                %     return substring(suffix, 2, glo
  } if$                                                 %   }
}                                                       % }
                                                        %
%%%
```

```
447   newline$                                          %   writeln();
448 }                                                   % }
449                                                     %
450 %%%%
```

```
                                                    %
%%%
```

```
                                                    %
%%%
```

```
    namelist nameindex "{ll}" format.name$        %     tmp = format_name(namelist, nam
    'lastname :=                                  %     lastname = change_case(tmp, "u'
    jrname empty$ not {                           %     if (! empty(jrname)) {
      jrname * " " *                              %       result = result + jrname + "
    } 'skip$ if$                                  %     }
    vonname empty$ not {                          %     if (! empty(vonname)) {
      vonname * " " *                             %       result = result + vonname + '
    } 'skip$ if$                                  %     }
    lastname empty$ not {                         %     if (! empty(lastname)) {
      lastname * " " *                            %       result = result + lastname +
    } 'skip$ if$                                  %     }
    firstname empty$ not {                        %     if (! empty(firstname)) {
      firstname * " " *                           %       result = result + firstname +
    } 'skip$ if$                                  %     }
    trim.end                                      %     result = trim_end(result);
  } while$                                        %   }
  nameindex namecount < {                         %   if (nameindex < namecount) {
    cap.et.al *                                   %     result = result + cap_et_al();
  } 'skip$ if$                                    %   }
}                                                 % }
                                                  %
                                                  %
%%%
```

```
  } {                                                   %   } else {
    format.names.en                                     %     editors = format_names_en(edito
  } if$                                                 %   }
  "editor" format.bibinfo                               %   return format_bibinfo(editors, "e
}                                                       % }
                                                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                        %
FUNCTION {format.translator} {                          % String Emtry::format_translator(Str
  is.in.chinese {                                       %   if (is_in_chinese) {
    duplicate$                                          %     String names = translators;
    format.names.zh                                     %     translators = format_names_zh(t
    swap$ num.names$ #3 > {                             %     if (num_names(names) > 3) {
      cap.translate.zh *                                %       translators = translators + c
    } {                                                 %     } else {
      cap.comma * cap.translate.zh *                    %       translators = translators + c
    } if$                                               %     }
  } {                                                   %   } else {
    duplicate$                                          %     String names = translators;
    format.names.en                                     %     translators = format_names_en(t
    swap$ num.names$ #3 > {                             %     if (num_names(names) > 3) {
      cap.translate.en *                                %       translators = translators + c
    } {                                                 %     } else {
      cap.comma * cap.translate.en *                    %       translators = translators + c
    } if$                                               %     }
  } if$                                                 %   }
  "translator" format.bibinfo                           %   return format_bibinfo(translator,
}                                                       % }
                                                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                        %
FUNCTION {format.title} {                               % String Emtry::format_title(String t
  "title" format.bibinfo                                %   return format_bibinfo(title, "tit
}                                                       % }
                                                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                        %
FUNCTION {format.booktitle} {                           % String Emtry::format_booktitle(Stri
  "booktitle" format.bibinfo                            %   return format_bibinfo(booktitle,
}                                                       % }
                                                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                        %
FUNCTION {format.mark} {                                % String Emtry::format_mark(String ma
  "[" swap$ * "]" *                                     %   mark = "[" + mark + "]";
  "mark" format.bibinfo                                 %   return format_bibinfo(mark, "mark
}                                                       % }
                                                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                        %
```

```
FUNCTION {format.country} {                          % String Emtry::format_country(String
  "country" format.bibinfo                           %   return format_bibinfo(country, "c
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                     %
FUNCTION {format.patentid} {                         % String Emtry::format_patentid(Strin
  "patentid" format.bibinfo                          %   return format_bibinfo(patentid, '
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                     %
FUNCTION {format.edition} {                          % String Emtry::format_edition(String
  duplicate$ is.number {                             %   if (is_number(edition)) {
    is.in.chinese {                                  %     if (is_in_chinese()) {
      cap.edition.zh *                               %       edition = edition + cap_editi
    } {                                              %     } else {
      num.to.ordinal.en cap.edition.en *             %       edition = num_to_ordinal_en(e
    } if$                                            %     }
  } 'skip$ if$                                       %   }
                                                     %   //  use a \mbox{} to prevent line
  "\mbox{" swap$ * "}" *                             %   edition = "\mbox{" + edition + "}
  "edition" format.bibinfo                           %   return format_bibinfo(edition, "e
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                     %
FUNCTION {format.organization} {                     % String Emtry::format_organization(S
  "organization" format.bibinfo                      %   return format_bibinfo(organizatio
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                     %
FUNCTION {format.address} {                          % String Emtry::format_address(String
  "address" format.bibinfo                           %   return format_bibinfo(address, "a
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                     %
FUNCTION {format.publisher} {                        % String Emtry::format_publisher(Stri
  "publisher" format.bibinfo                         %   return format_bibinfo(publisher,
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                     %
FUNCTION {format.institution} {                      % String Emtry::format_institution(St
  "institution" format.bibinfo                       %   return format_bibinfo(institution
}                                                    % }
                                                     %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
                                                    %
FUNCTION {format.school} {                          % String Emtry::format_school(String
  "school" format.bibinfo                           %   return format_bibinfo(school, "sc
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.year} {                            % String Emtry::format_year(String ye
  "year" format.bibinfo                             %   return format_bibinfo(year, "year
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.date} {                            % String Emtry::format_date(String da
  "date" format.bibinfo                             %   return format_bibinfo(date, "date
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.journal} {                         % String Emtry::format_journal(String
  "journal" format.bibinfo                          %   return format_bibinfo(journal, "j
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.volume} {                          % String Emtry::format_volume(String
  "volume" format.bibinfo                           %   return format_bibinfo(volume, "vo
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.number} {                          % String Emtry::format_number(String
  add.bracket                                       %   number = add_bracket(number);
  "number" format.bibinfo                           %   return format_bibinfo(number, "nu
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.report.number} {                   % String Emtry::format_report_number(
  "number" format.bibinfo                           %   return format_bibinfo(number, "nu
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {format.version} {                         % String Emtry::format_version(String
  "version" format.bibinfo                          %   return format_bibinfo(version, "v
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
```

```
FUNCTION {format.pages} {                          % String Emtry::format_pages(String p
  "pages" format.bibinfo                           %   return format_bibinfo(pages, "pag
}                                                  % }
                                                   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                   %
FUNCTION {format.modifydate} {                     % String Emtry::format_modifydate(Str
  add.bracket                                      %   modifydate = add_bracket(modifyda
  "modifydate" format.bibinfo                      %   return format_bibinfo(modifydate,
}                                                  % }
                                                   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                   %
FUNCTION {format.citedate} {                       % String Emtry::format_citedate(Strin
  add.squarebracket                                %   citedate = add_squarebracket(cite
  "citedate" format.bibinfo                        %   return format_bibinfo(citedate, '
}                                                  % }
                                                   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                   %
                                                   % // NOTE: do not use the format_bibi
                                                   % // since if the URL contains specia
                                                   % // as '%', the \biburl{} will be br
FUNCTION {format.url} {                            % String Emtry::format_url(String url
  "\url{" swap$ * "}" *                            %   return "\url{" + url + "}";
}                                                  % }
                                                   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                   %
FUNCTION {get.full.title} {                        % String Entry::get_full_title() {
  series empty$ {                                  %     if (empty(this.series)) {
    volume empty$ {                                %       if (empty(this.volume)) {
      title                                        %         return this.title;
    } {                                            %       } else {
      title cap.colon * cap.volume * volume *      %         return this.title + cap_col
    } if$                                          %       }
  } {                                              %     } else {
    volume empty$ {                                %       if (empty(this.volume)) {
      series cap.colon * title *                   %         return this.series + cap_co
    } {                                            %       } else {
      series cap.comma * cap.volume * volume *     %         String str = this.series +
      cap.colon * title *                          %         return str + cap_colon + th
    } if$                                          %       }
  } if$                                            %     }
}                                                  % }
                                                   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                   %
FUNCTION {get.full.booktitle} {                    % String Entry::get_full_booktitle()
  series empty$ {                                  %     if (empty(this.series)) {
```

```
    volume empty$ {                                  %       if (empty(this.volume)) {
      booktitle                                      %         return this.booktitle;
    } {                                              %       } else {
      booktitle cap.colon * cap.volume * volume *    %         return this.booktitle + cap
    } if$                                            %       }
  } {                                                %     } else {
    volume empty$ {                                  %       if (empty(this.volume)) {
      series cap.colon * booktitle *                 %         return this.series + cap_co
    } {                                              %       } else {
      series cap.comma * cap.volume * volume *       %         String str = this.series +
      cap.colon * booktitle *                        %         return str + cap_colon + th
    } if$                                            %       }
  } if$                                              %     }
}                                                    % }
                                                     %
%%%
```

```
FUNCTION {output.author} {                          % void Entry::output_author(int requi
  'required :=                                      %   this.required = required;
  author empty$ not {                               %   if (! empty(this.author)) {
    author format.author write$                     %     write(format_author(this.author
    cap.period write$                               %     write(cap_period);
  } {                                               %   } else {
    required {                                      %     if (required == 1) {
      "Require author: " cite$ * warning$           %       warning("Require author: " +
      cap.anonymous format.author write$            %       write(format_author(cap_anony
      cap.period write$                             %       write(cap_period);
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%
```

```
    required {                                      %     if (required == 1) {
      "Require title: " cite$ * warning$            %       warning("Require title: " + t
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%
```

```
  translator empty$ not {                          %   if (! empty(this.translator)) {
    cap.period write$                              %     write(cap_period);
    translator format.translator write$            %     write(format_translator(this.tr
  } {                                              %   } else {
    required {                                     %     if (required == 1) {
      "Require translator: " cite$ * warning$      %       warning("Require translator:
    } 'skip$ if$                                   %     }
  } if$                                            %   }
}                                                  % }
                                                   %
%%%
```

```
      cap.colon write$                                  %       write(cap_colon);
      cap.no.publisher format.publisher write$          %       write(format_publisher(cap_no
    } 'skip$ if$                                        %     }
  } if$                                                 %   }
}                                                       % }
                                                        %
%%%
```

```
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%
```

```
    cap.space write$                                  %     write(cap_space);
    citedate format.citedate write$                   %     write(format_citedate(this.cite
  } {                                                 %   } else {
    required is.online or {                           %     if ((required == 1) || (is_onli
      "Require citedate: " cite$ * warning$           %       warning("Require citedate: "
    } 'skip$ if$                                      %     }
  } if$                                               %   }
}                                                     % }
                                                      %
%%%
```

```
                                                    %
FUNCTION {output.report.number} {                   % void Entry::output_report_number(in
  'required :=                                      %   this.required = required;
  number empty$ not {                               %   if (! empty(this.number)) {
    cap.colon write$                                %     write(cap_colon);
    number format.report.number write$              %     write(format_report_number(this
  } {                                               %   } else {
    required {                                      %     if (required == 1) {
      "Require number: " cite$ * warning$           %       warning("Require number: " +
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {output.country} {                         % void Entry::output_country(int requ
  'required :=                                      %   this.required = required;
  country empty$ not {                              %   if (! empty(this.country)) {
    cap.colon write$                                %     write(cap_colon);
    country format.country write$                   %     write(format_country(this.count
  } {                                               %   } else {
    required {                                      %     if (required == 1) {
      "Require country: " cite$ * warning$          %       warning("Require country: " +
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {output.patentid} {                        % void Entry::output_patentid(int rec
  'required :=                                      %   this.required = required;
  patentid empty$ not {                             %   if (! empty(this.patentid)) {
    cap.comma write$                                %     write(cap_comma);
    patentid format.patentid write$                 %     write(format_patentid(this.pate
  } {                                               %   } else {
    required {                                      %     if (required == 1) {
      "Require patentid: " cite$ * warning$         %       warning("Require patentid: "
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                    %
FUNCTION {output.start.year} {                      % void Entry::output_start_year(int r
  'required :=                                      %   this.required = required;
  year empty$ not {                                 %   if (! empty(this.year)) {
    year extract.before.first.dash                  %     String str = extract_before_fir
    format.year write$                              %     write(format_year(str));
  } {                                               %   } else {
    required {                                      %     if (required == 1) {
```

```
      "Require year: " cite$ * warning$          %       warning("Require year: " + th
    } 'skip$ if$                                 %     }
  } if$                                          %   }
}                                                % }
                                                 %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                 %
FUNCTION {output.start.volume} {                 % void Entry::output_start_volume(int
  'required :=                                   %   this.required = required;
  volume empty$ not {                            %   if (! empty(this.volume)) {
    cap.comma write$                             %     write(cap_comma);
    volume extract.before.first.dash             %     String str = extract_before_fir
    format.volume write$                         %     write(format_volume(str));
  } {                                            %   } else {
    required {                                   %     if (required == 1) {
      "Require volume: " cite$ * warning$        %       warning("Require volume: " +
    } 'skip$ if$                                 %     }
  } if$                                          %   }
}                                                % }
                                                 %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                 %
FUNCTION {output.start.number} {                 % void Entry::output_start_number(int
  'required :=                                   %   this.required = required;
  number empty$ not {                            %   if (! empty(this.number)) {
    number extract.before.first.dash             %     String str = extract_before_fir
    format.number write$                         %     write(format_number(str));
  } {                                            %   } else {
    required {                                   %     if (required == 1) {
      "Require number: " cite$ * warning$        %       warning("Require number: " +
    } 'skip$ if$                                 %     }
  } if$                                          %   }
}                                                % }
                                                 %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                 %
FUNCTION {output.end.year} {                     % void Entry::output_end_year(int rec
  'required :=                                   %   this.required = required;
  year empty$ not {                              %   if (! empty(this.year)) {
    year extract.after.last.dash                 %     String str = extract_after_last
    format.year write$                           %     write(format_year(str));
  } {                                            %   } else {
    required {                                   %     if (required == 1) {
      "Require year: " cite$ * warning$          %       warning("Require year: " + th
    } 'skip$ if$                                 %     }
  } if$                                          %   }
}                                                % }
                                                 %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                 %
```

```
FUNCTION {output.end.volume} {                    % void Entry::output_end_volume(int r
  'required :=                                    %   this.required = required;
  volume empty$ not {                             %   if (! empty(this.volume)) {
    cap.comma write$                              %     write(cap_comma);
    volume extract.after.last.dash                %     String str = extract_after_last
    format.volume write$                          %     write(format_volume(str));
  } {                                             %   } else {
    required {                                    %     if (required == 1) {
      "Require volume: " cite$ * warning$         %       warning("Require volume: " +
    } 'skip$ if$                                  %     }
  } if$                                           %   }
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {output.end.number} {                    % void Entry::output_end_number(int r
  'required :=                                    %   this.required = required;
  number empty$ not {                             %   if (! empty(this.number)) {
    number extract.after.last.dash                %     String str = extract_after_last
    format.number write$                          %     write(format_number(str));
  } {                                             %   } else {
    required {                                    %     if (required == 1) {
      "Require number: " cite$ * warning$         %       warning("Require number: " +
    } 'skip$ if$                                  %     }
  } if$                                           %   }
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {output.url.or.doi} {                    % void Entry::output_url_or_doi(int r
  'required :=                                    %   this.required = required;
  url empty$ not {                                %   if (! empty(this.url)) {
    cap.period write$                             %     write(cap_period);
    url format.url write$                         %     write(format_url(this.url));
  } {                                             %   } else {
    doi empty$ not {                              %     if (! empty(this.doi)) {
      cap.period write$                           %       write(cap_period);
      cap.doi.url doi * format.url write$         %       write(format_url(cap_doi_url
    } {                                           %     } else {
      required {                                  %       if (required == 1) {
        "Require URL or DOI: " cite$ * warning$   %         warning("Require URL or DOI
      } 'skip$ if$                                %       }
    } if$                                         %     }
  } if$                                           %   }
}                                                 % }
                                                  %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
                                                  %
FUNCTION {output.url} {                           % void Entry::output_url(int required
  'required :=                                    %   this.required = required;
```

```
  url empty$ not {                                  %   if (! empty(this.url)) {
    cap.period write$                               %     write(cap_period);
    url format.url write$                           %     write(format_url(this.url));
  } {                                               %   } else {
    required {                                      %     if (required == 1) {
      "Require URL: " cite$ * warning$              %        warning("Require URL: " + th
    } 'skip$ if$                                    %     }
  } if$                                             %   }
}                                                   % }
                                                    %
%%%
```

```
  "M" set.mark                                    %   set_mark("M");
  book.impl                                       %   book_impl();
}                                                 % }
                                                  %
%%%
```

```
  true output.school                                  %   output_school(true);
  cap.comma write$                                    %   write(cap_comma);
  true output.year                                    %   output_year(true);
  false output.pages                                  %   output_pages(false);
  false output.citedate                               %   output_citedate(false);
  false output.url.or.doi                             %   output_url_or_doi(false);
  end.bibitem                                         %   end_bibitem();
}                                                     % }
                                                      %
%%%
```

```
  true output.address                          %     output_address(true);
  true output.institution                      %     output_institution(true);
  cap.comma write$                             %     write(cap_comma);
} {                                            %   } else {
  cap.period write$                            %     write(cap_period);
} if$                                          %   }
true output.year                               %   output_year(true);
false output.pages                             %   output_pages(false);
false output.citedate                          %   output_citedate(false);
false output.url.or.doi                        %   output_url_or_doi(false);
end.bibitem                                    %   end_bibitem();
}                                              % }
                                               %
%%%
```

```
    cap.comma write$                                  %     write(cap_comma);
  } {                                                 %   } else {
    cap.period write$                                 %     write(cap_period);
  } if$                                               %   }
  true output.year                                    %   output_year(true);
  false output.pages                                  %   output_pages(false);
  false output.citedate                               %   output_citedate(false);
  false output.url.or.doi                             %   output_url_or_doi(false);
  end.bibitem                                         %   end_bibitem();
}                                                     % }
                                                      %
%%%
```

```
                                                    %
%%%
```

```
  cap.period write$                          %     write(cap_period);
} if$                                        %   }
true output.year                             %   output_year(true);
false output.modifydate                      %   output_modifydate(false);
true output.citedate                         %   output_citedate(true);
true output.url                              %   output_url(true);
end.bibitem                                  %   end_bibitem();
}                                            % }
                                             %
%%%
```

```
                                                    %
FUNCTION {unpublished} {                            % void Entry::unpublished() {
  "H" set.mark                                      %   set_mark("H");
  start.bibitem                                     %   start_bibitem();
  true output.author                                %   output_author(true);
  true output.title                                 %   output_title(true);
  true output.mark                                  %   output_mark(true);
  cap.period write$                                 %   write(cap_period);
  true output.year                                  %   output_year(true);
  false output.citedate                             %   output_citedate(false);
  false output.url.or.doi                           %   output_url_or_doi(false);
  end.bibitem                                       %   end_bibitem();
}                                                   % }
                                                    %
%%%
```

```
  inbook.impl                                   %   inbook_impl();
}                                               % }
                                                %
%%%
```

```
  end.bibitem                                          %   end_bibitem();
}                                                      % }
                                                       %
%%%
```

```
  newline$                                    %   writeln();
  cmd.bibschool write$                        %   write(cmd_bibschool);
  newline$                                    %   writeln();
  cmd.bibvolume write$                        %   write(cmd_bibvolume);
  newline$                                    %   writeln();
  cmd.bibnumber write$                        %   write(cmd_bibnumber);
  newline$                                    %   writeln();
  cmd.bibversion write$                       %   write(cmd_bibversion);
  newline$                                    %   writeln();
  cmd.bibpages write$                         %   write(cmd_bibpages);
  newline$                                    %   writeln();
  cmd.bibmodifydate write$                    %   write(cmd_bibmodifydate);
  newline$                                    %   writeln();
  cmd.bibcitedate write$                      %   write(cmd_bibcitedate);
  newline$                                    %   writeln();
  cmd.bibyear write$                          %   write(cmd_bibyear);
  newline$                                    %   writeln();
  cmd.bibdate write$                          %   write(cmd_bibdate);
  newline$                                    %   writeln();
  cmd.biburl write$                           %   write(cmd_biburl);
  newline$                                    %   writeln();
}                                             % }
                                              %
%%%
```

```
}                                                    % }
                                                     %
%%%
```

```
  setkeys=\kvsetkeys}

\newif\ifheu@bachelor
\newif\ifheu@master
\newif\ifheu@doctor
\define@key{heu}{type}{%
  \heu@bachelorfalse
  \heu@masterfalse
  \heu@doctorfalse
  \expandafter\csname heu@#1true\endcsname}

\newif\ifheu@productdevelopment
\newif\ifheu@projectdplanning
\newif\ifheu@projectdesign
\newif\ifheu@applicationresearch
\define@key{heu}{researchtype}{%
  \heu@productdevelopmentfalse
  \heu@projectdplanningfalse
  \heu@projectdesignfalse
  \heu@applicationresearchfalse
  \expandafter\csname heu@#1true\endcsname}
\ifheu@productdevelopment\relax\else
\ifheu@projectdplanning\relax\else
\ifheu@projectdesign\relax\else
\heu@applicationresearchtrue
\fi
\fi
\fi

\newif\ifheu@qingdao
\newif\ifheu@yantai
\newif\ifheu@harbin
\define@key{heu}{campus}{%
  \heu@qingdaofalse
  \heu@yantaifalse
  \heu@harbinfalse
  \expandafter\csname heu@#1true\endcsname}
\ifheu@yantai\relax\else
\ifheu@qingdao\relax\else
\heu@harbintrue
\fi
\fi

\newif\ifheu@opening
\newif\ifheu@midterm
\define@key{heu}{stage}{%
  \heu@openingfalse
  \heu@midtermfalse
  \expandafter\csname heu@#1true\endcsname}

```

```
\DeclareBoolOption[true]{raggedbottom}
\DeclareBoolOption[false]{pifootnote}
\DeclareBoolOption[false]{debug}
\DeclareBoolOption[true]{toc}
\DeclareBoolOption[true]{newtxmath}

\DeclareStringOption{fontset}

\DeclareDefaultOption{\PassOptionsToClass{\CurrentOption}{ctexart}}

\ProcessKeyvalOptions*
\PassOptionsToPackage{no-math}{fontspec}

\ifheu@bachelor\relax\else
\ifheu@master\relax\else
\ifheu@doctor\relax\else
\ClassError{heuthesisart}%
{
  \MessageBreak Please specify thesis type in option:
  \MessageBreak type=[bachelor | master | doctor]
}
\fi
\fi
\fi

\ifheu@productdevelopment\relax\else
\ifheu@projectdplanning\relax\else
\ifheu@projectdesign\relax\else
\ifheu@applicationresearch\relax\else
\ClassError{heuthesisartplus}%
{
  \MessageBreak Please specify researchtype in option:
  \MessageBreak researchtype=[productdevelopment | projectdplanning | projectdesign | a
}
\fi
\fi
\fi
\fi

\ifheu@opening\relax\else
\ifheu@midterm\relax\else
\ClassError{heuthesisart}%
{
  \MessageBreak Please specify stage in option:
  \MessageBreak stage=<opening|midterm>
}
\fi
\fi

\ifthenelse%
```

```
{\equal{\heu@fontset}{}}%
{%
  \PassOptionsToPackage{AutoFakeBold=2}{xeCJK}
}%
{%
  \ifthenelse%
  {\equal{\heu@fontset}{siyuan}}%
  {\relax}%
  {%
    \PassOptionsToPackage{AutoFakeBold=2}{xeCJK}
  }%
  \PassOptionsToClass{fontset=\heu@fontset}{ctexart}
}%
\LoadClass[a4paper,UTF8,zihao=-4,scheme=plain]{ctexart}

\RequirePackage{etoolbox}
\RequirePackage{ifxetex}
\ifxetex
\else
  \ClassError{heuthesis}%
     {Please use: \MessageBreak
     xelatex}{}
\fi
\RequirePackage{xparse}
\RequirePackage{amsmath}
\RequirePackage[amsmath,thmmarks,hyperref]{ntheorem}
\RequirePackage{amssymb}
\ifheu@newtxmath
\RequirePackage{newtxmath}
\fi
\RequirePackage{graphicx}
\RequirePackage{pdfpages}
\includepdfset{fitpaper=true}
\RequirePackage{enumitem}       % 使用 enumitem 宏包, 改变列表项的格式
\RequirePackage{environ}
\ifheu@raggedbottom
  \RequirePackage[bottom,perpage,hang]{footmisc}
  \raggedbottom
\else
  \RequirePackage[perpage,hang]{footmisc}
\fi
\RequirePackage{bbding}
\ifheu@pifootnote
  \RequirePackage{pifont}
\fi
\RequirePackage{xeCJKfntef}
\RequirePackage{diagbox}
\RequirePackage{longtable}
\RequirePackage{booktabs}
\RequirePackage[sort&compress,numbers]{natbib}
```

```
\RequirePackage{hyperref}
\hypersetup{%
  CJKbookmarks=true,
  linktoc=all,
  bookmarksnumbered=true,
  bookmarksopen=true,
  bookmarksopenlevel=1,
  breaklinks=true,
  colorlinks=false,
  plainpages=false,
  pdfborder=0 0 0}
\urlstyle{same}

\RequirePackage{geometry}
\geometry{
  a4paper,
  hcentering,
  ignoreall,
  nomarginpar,
  centering,
  text={150true mm,236true mm},
  left=30true mm,
  head=5true mm,
  headsep=2true mm,
  footskip=0true mm,
  foot=5.2true mm
}

\ifheu@debug%
\RequirePackage{layout}
\RequirePackage{layouts}
\RequirePackage{lineno}
\fi

\RequirePackage{fancyhdr}
\RequirePackage{tabularx}
\RequirePackage{varwidth}
\RequirePackage{changepage}
\RequirePackage{multicol}
\RequirePackage[below]{placeins}% 允许上一个 section 的浮动图形出现在下一个 section 的开始部分，还提供\FloatBarrier 命令，使所有未处理的浮动图形立即被处理
\RequirePackage{flafter}         % 使得所有浮动体不能被放置在其浮动环境之前，以免浮动体在引述它的文本之前出现.
\RequirePackage{multirow}        % 使用 Multirow 宏包，使得表格可以合并多个 row 格
\RequirePackage{subfigure}       % 支持子图 %centerlast 设置最后一行是否居中
\RequirePackage[subfigure]{ccaption} % 支持双语标题
\RequirePackage{xltxtra}

\renewcommand\normalsize{%
  \@setfontsize\normalsize{12bp}{20.50398bp}%
```

```
  \abovedisplayskip=8pt
  \abovedisplayshortskip=8pt
  \belowdisplayskip=\abovedisplayskip
  \belowdisplayshortskip=\abovedisplayshortskip}

\def\heu@def@fontsize#1#2{%
  \expandafter\newcommand\csname #1\endcsname[1][1.3]{%
    \fontsize{#2}{##1\dimexpr #2}\selectfont}}
\heu@def@fontsize{dachu}{58.0bp}
\heu@def@fontsize{chuhao}{42.2bp}
\heu@def@fontsize{xiaochu}{36.1bp}
\heu@def@fontsize{yihao}{26.1bp}
\heu@def@fontsize{xiaoyi}{24.1bp}
\heu@def@fontsize{erhao}{22.1bp}
\heu@def@fontsize{xiaoer}{18.1bp}
\heu@def@fontsize{sanhao}{16.1bp}
\heu@def@fontsize{xiaosan}{15.1bp}
\heu@def@fontsize{sihao}{14.1bp}
\heu@def@fontsize{banxiaosi}{13.0bp}
\heu@def@fontsize{xiaosi}{12.1bp}
\heu@def@fontsize{dawu}{11.0bp}
\heu@def@fontsize{wuhao}{10.5bp}
\heu@def@fontsize{xiaowu}{9.0bp}
\heu@def@fontsize{liuhao}{7.5bp}
\heu@def@fontsize{xiaoliu}{6.5bp}
\heu@def@fontsize{qihao}{5.5bp}
\heu@def@fontsize{bahao}{5.0bp}

%-- 设置字号 --%
\ctexset{%
  section={
    afterindent=true,
    beforeskip={7.5bp},% 上下空 0.5 行
    afterskip={7.5bp},
    format={\heiti\xiaosan[1.25]},
    aftername=\enspace,
    fixskip=true,
    break={},
  },
  subsection={
    afterindent=true,
    beforeskip={7bp},
    afterskip={7bp},
    format={\heiti\sihao[1.25]},
    aftername=\enspace,
    fixskip=true,
    break={},
  },
  subsubsection={
    afterindent=true,
```

```
    beforeskip={3bp},
    afterskip={3bp},
    format={\heiti\normalsize},
    aftername=\enspace,
    fixskip=true,
    break={},
  },
  paragraph/afterindent=true,
  subparagraph/afterindent=true
}

\def\heu@def@term#1{%
  \define@key{heu}{#1}{\csname #1\endcsname{##1}}
  \expandafter\gdef\csname #1\endcsname##1{%
    \expandafter\gdef\csname heu@#1\endcsname{##1}}
  \csname #1\endcsname{}}

\heu@def@term{ctitlecover} % 中文标题封面
\heu@def@term{csubject}
\heu@def@term{cauthor}
\heu@def@term{cstudentid}
\heu@def@term{cclassid}
\heu@def@term{caffil}
\heu@def@term{csupervisor}
\heu@def@term{csubmitdate}
\heu@def@term{coralexdate}
\heu@def@term{cdate}

\def\heu@parse@keywords#1{
  \define@key{heu}{#1}{\csname #1\endcsname{##1}}
  \expandafter\gdef\csname heu@#1\endcsname{}
  \expandafter\gdef\csname #1\endcsname##1{
    \@for\reserved@a:=##1\do{
      \expandafter\ifx\csname heu@#1\endcsname\@empty\else
      \expandafter\g@addto@macro\csname heu@#1\endcsname{%
        \ignorespaces\csname heu@#1@separator\endcsname}
      \fi
      \expandafter\expandafter\expandafter\g@addto@macro%
      \expandafter\csname heu@#1\expandafter\endcsname\expandafter{\reserved@a}}}}

\def\heusetup{\kvsetkeys{heu}}
```

开题报告封面

```
\newcommand{\heu@report@titlepage@graduate}{
  \ifthenelse%
  {\equal{\heu@fontset}{siyuan}}%
  {\xiaosi[1]\vspace*{0.65em}}%
  {\xiaosi[1]\textcolor[rgb]{1,1,1}{\songti{\heu@motto}}}%
  \vspace*{10mm}
```

```
  \begin{center}
    \kaishu\xiaoer\textbf{\heu@cschoolname\ifheu@qingdao\heu@qingdaocampus\fi\ifheu@yan
  \end{center}
  \vspace{1mm}
  \begin{center}
    \songti\erhao\textbf{\heu@cxuewei\heu@cthesisname
      \ifheu@opening
      \heu@stage@opening
      \else
      \ifheu@midterm
      \heu@stage@midterm
      \fi
      \fi
      \heu@stage@doctype
    }
  \end{center}
   \vspace{10mm}
   \parbox[t][3cm][t]{\textwidth}{
     \begin{center}
       \songti\xiaoer\textbf{\heu@cthesistitleprefix\heu@title@csep\heu@ctitlecover}
     \end{center}
   }
     \parbox[b][3cm][t]{\textwidth}{
       \begin{center}\songti\sanhao
       \renewcommand{\arraystretch}{1.8}
       \begin{tabular}{l@{\ \ }c}
         \textbf{\heu@graduate@caffiltitle:} & \underline{\makebox[6.1cm]{\textbf{\heu@
         \textbf{\heu@graduate@cmajortitle:} & \underline{\makebox[6.1cm]{\textbf{\heu@
         \textbf{\heu@graduate@supervisor:} & \underline{\makebox[6.1cm]{\textbf{\heu@c
         \textbf{\heu@graduate@studenttitle:} & \underline{\makebox[6.1cm]{\textbf{\heu
         \textbf{\heu@graduate@studentid:} & \underline{\makebox[6.1cm]{\textbf{\heu@cs
         \textbf{\heu@graduate@datetitle:} & \underline{\makebox[6.1cm]{\textbf{\heu@cd
       \end{tabular}\renewcommand{\arraystretch}{1}
     \end{center}
     }
   \vfill
   \ifheu@harbin
     \heu@harbin@schoolbottommark
   \else
   \ifheu@qingdao
     \heu@qingdao@schoolbottommark
   \else
   \ifheu@yantai
     \heu@yantai@schoolbottommark
   \fi
   \fi
}

\newcommand{\heu@report@titlepage@bachelor}{
   \ifthenelse%
```

```
  {\equal{\heu@fontset}{siyuan}}%
  {\xiaosi[1]\vspace*{0.65em}}%
  {\xiaosi[1]\textcolor[rgb]{1,1,1}{\songti{\heu@motto}}}%
  \vspace*{10mm}
  \begin{center}
    \includegraphics[width=6.2cm]{heulogo}
  \end{center}
  \begin{center}
    \songti\xiaoyi\textbf{\heu@bachelor@cthesisname
      \ifheu@opening
        \heu@stage@opening
      \else
        \ifheu@midterm
          \heu@stage@midterm
        \fi
      \fi
      \heu@stage@doctype
    }
  \end{center}
  \vspace{15mm}
  \parbox[t][4.0cm][t]{\textwidth}{
    \begin{center}
      \songti\xiaoer\textbf{\heu@cthesistitleprefix\heu@title@csep\heu@ctitlecover}
    \end{center}
  }
    \parbox[b][6cm][t]{\textwidth}{
      \begin{center}\songti\sanhao
      \renewcommand{\arraystretch}{2.1}
      \begin{tabular}{l@{\ \ }c}
        \textbf{\heu@bachelor@cmajortitle} & \underline{\makebox[6.1cm]{\textbf{\heu@c
        \textbf{\heu@bachelor@cstudenttitle} & \underline{\makebox[6.1cm]{\textbf{\heu
        \textbf{\heu@bachelor@cstudentidtitle} & \underline{\makebox[6.1cm]{\textbf{\h
        \textbf{\heu@bachelor@csupervisortitle} & \underline{\makebox[6.1cm]{\textbf{\
        \textbf{\heu@bachelor@cdatetitle} & \underline{\makebox[6.1cm]{\textbf{\heu@co
      \end{tabular}\renewcommand{\arraystretch}{1}
    \end{center}
    }
  \vfill
  \ifheu@yantai
    \relax
  \else
    \heu@harbin@bachelor@schoolbottommark
  \fi
}

\newcommand{\heu@report@backpage@bachelor}{
  \thispagestyle{empty}
  \noindent\parbox[t][6.5cm][t]{\textwidth}{\heu@bachelor@teachercomment}
  \noindent\parbox[b][6cm][t]{\textwidth}{\heu@bachelor@teachersign\underline{\makebox[
}
```

```

\renewcommand\tableofcontents{%
  \thispagestyle{empty}
  \normalsize\@starttoc{toc}
}

\def\makecover{
  \begin{titlepage}
    \ifheu@bachelor
      \heu@report@titlepage@bachelor
    \else
      \heu@report@titlepage@graduate
    \fi
    \clearpage
    \ifheu@toc
      \tableofcontents
      \clearpage
    \fi
  \end{titlepage}
}
\def\makebackcover{
    \clearpage
    \heu@report@backpage@bachelor
}

\newcommand\bibstyle@numerical{\bibpunct{[}{]}{,}{s}{,}{\textsuperscript{,}}}
\newcommand\bibstyle@inline{\bibpunct{[}{]}{,}{n}{,}{\heu@inline@sep}}

\citestyle{numerical}
\DeclareRobustCommand\inlinecite{\@inlinecite}
\def\@inlinecite#1{\begingroup\citestyle{inline}\let\@cite\NAT@citenum\citep{#1}\endgro
\let\onlinecite\inlinecite
\newcommand{\lcite}{\inlinecite}
\renewenvironment{thebibliography}[1]{%
  \list{\@biblabel{\@arabic\c@enumiv}}%
  {\renewcommand{\makelabel}[1]{##1\hfill}
    \settowidth{\labelwidth}{\@biblabel{#1}}
    \setlength{\labelsep}{0.5em}
    \setlength{\itemindent}{0pt}
    \setlength{\leftmargin}{\labelsep+\labelwidth}
    \addtolength{\itemsep}{-0.8em}
    \usecounter{enumiv}%
    \let\p@enumiv\@empty
  \renewcommand\theenumiv{\@arabic\c@enumiv}}%
  \sloppy\frenchspacing
  \flushbottom
  \clubpenalty0
  \@clubpenalty \clubpenalty
  \widowpenalty0%
  \interlinepenalty-50%
```

```
\sfcode`\.\@m}
{\def\@noitemerr
  {\@latex@warning{Empty `thebibliography' environment}}%
\endlist\frenchspacing}
\AtEndOfClass{\input{heuthesisart.cfg}}
\AtEndOfClass{\sloppy}
⟨/artcls⟩
```

4.2 定义选项

⟨*bookcls⟩

此处添加 book 控制项，用于 cfg 中定理等控制

```
\RequirePackage{ifthen}
\RequirePackage{kvoptions}
\SetupKeyvalOptions{
  family=heu,
  prefix=heu@,
  setkeys=\kvsetkeys}
\newif\ifheu@bachelor
\newif\ifheu@master
\newif\ifheu@doctor
\newif\ifheu@postdoc
\define@key{heu}{type}{%
  \heu@bachelorfalse
  \heu@masterfalse
  \heu@doctorfalse
  \heu@postdocfalse
  \expandafter\csname heu@#1true\endcsname}
```

此处设置专业学位论文研究类型，没有明确给出时默认为**应用研究**型。

```
\newif\ifheu@productdevelopment
\newif\ifheu@projectdplanning
\newif\ifheu@projectdesign
\newif\ifheu@applicationresearch
\define@key{heu}{researchtype}{%
  \heu@productdevelopmentfalse
  \heu@projectdplanningfalse
  \heu@projectdesignfalse
  \heu@applicationresearchfalse
  \expandafter\csname heu@#1true\endcsname}
\ifheu@productdevelopment\relax\else
\ifheu@projectdplanning\relax\else
\ifheu@projectdesign\relax\else
\heu@applicationresearchtrue
\fi
\fi
\fi
```

此处设置校区，没有明确给出烟台或者青岛时候，默认为哈尔滨校区。

```
\newif\ifheu@qingdao
\newif\ifheu@yantai
\newif\ifheu@harbin
```

```
\define@key{heu}{campus}{%
  \heu@qingdaofalse
  \heu@yantaifalse
  \heu@harbinfalse
  \expandafter\csname heu@#1true\endcsname}
\ifheu@yantai\relax\else
\ifheu@qingdao\relax\else
\heu@harbintrue
\fi
\fi
```

设置版芯。

```
\newif\ifheu@english
\newif\ifheu@chinese
\define@key{heu}{language}{%
  \heu@englishfalse
  \heu@chinesefalse
  \expandafter\csname heu@#1true\endcsname}
\ifheu@english\relax\else
\heu@chinesetrue
\fi
\newif\ifheu@geometrynewone
\newif\ifheu@geometrynewtwo
\newif\ifheu@geometrynewno
\define@key{heu}{newgeometry}{%
  \heu@geometrynewonefalse
  \heu@geometrynewtwofalse
  \heu@geometrynewnofalse
  \expandafter\csname heu@geometrynew#1true\endcsname}
```

设置字距选项，因为 word 和规范不一致

```
\newif\ifheu@zijvregu
\newif\ifheu@zijvword
\define@key{heu}{zijv}{%
  \heu@zijvregufalse
  \heu@zijvwordfalse
  \expandafter\csname heu@zijv#1true\endcsname}
```

设置中文参考文献中相邻两个标号使用逗号“,”还是短线“-”。

```
\newif\ifheu@citetwocomma
\newif\ifheu@citetwoendash
\heu@citetwoendashtrue
\define@key{heu}{citetwo}{%
  \heu@citetwocommafalse
  \heu@citetwoendashfalse
  \expandafter\csname heu@citetwo#1true\endcsname
}
```

目录中英文是否用 Arial 字体（默认关闭）。

```
\DeclareBoolOption[false]{arialtoc}
```

章节标题中的英文是否用 Arial 字体（默认打开）。

```
\DeclareBoolOption[false]{arialtitle}
```

raggedbottom 选项（默认开启）。如果不开启这个选项，会出现一页中尽量上下对齐，段的间距大。如果开启，尽量使段间距保持一致，页面底部出现空白。

```
\DeclareBoolOption[true]{raggedbottom}
```

在脚注标记中使用 pifont 的带圈数字（默认使用）。

```
\DeclareBoolOption[true]{pifootnote}
```

字体间距设置（默认关闭）。

```
\DeclareBoolOption[false]{glue}
```

文科生四级目录设置（默认关闭）。

```
\DeclareBoolOption[false]{tocfour}
```

章标题是否悬挂居中（默认开启）

```
\DeclareBoolOption[true]{chapterhang}
```

是否是专业学位（默认否）。

```
\DeclareBoolOption[false]{profdegree}
```

是否有子标题（默认是）。

```
\DeclareBoolOption[false]{subtitle}
```

是否开启 debug 模式（默认否）。如果开启，载入显示行号等的包，只为开发调试用。

```
\DeclareBoolOption[false]{debug}
```

是否使用右开页（默认否）。

```
\DeclareBoolOption[false]{openright}
```

空白页是否无页眉页脚（默认否）。

```
\DeclareBoolOption[false]{pageempty}
```

是否为提交图书馆电子版。

```
\DeclareBoolOption[false]{library}
```

图题和标题最后一行是否居中对其（默认是，非规范要求）。

```
\DeclareBoolOption[false]{capcenterlast}
```

子图图题和标题最后一行是否居中对其（默认是，非规范要求）。

```
\DeclareBoolOption[false]{subcapcenterlast}
```

中文目录中 Abstract 是否均为大写

```
\DeclareBoolOption[false]{absupper}
```

此处添加控制本科论文的页码横线选项

```
\DeclareBoolOption[false]{bsmainpagenumberline}
\DeclareBoolOption[false]{bsfrontpagenumberline}
\DeclareBoolOption[true]{bsheadrule}
```

数学字体是否使用新罗马

```
\DeclareBoolOption[true]{newtxmath}
```

添加参考文献分割开关

```
\DeclareBoolOption[false]{splitbibitem}
```

此处添加英文目录开关

```
\DeclareBoolOption[false]{engtoc}
```

声明字体选项。

```
\DeclareStringOption{fontset}
```

将其余选项默认传递给 ctexbook。

```
\DeclareDefaultOption{\PassOptionsToClass{\CurrentOption}{ctexbook}}
```

解析用户传递过来的选项，并加载 ctexbook。

```
\ProcessKeyvalOptions*
```

使用 XeTeX 引擎时，fontspec 宏包会被 xeCJK 自动调用。传递给 fontspec 宏包 no-math 选项，避免部分数学符号字体自动调整为 CMR。其他引擎下没有这个问题，这一行会被无视。

```
\PassOptionsToPackage{no-math}{fontspec}
```

载入单双面打印设置，默认本、硕、博均采用双面打印。

```
\ifheu@bachelor
\PassOptionsToClass{twoside}{book}
\fi
\ifheu@master
\PassOptionsToClass{twoside}{book}
\fi
\ifheu@doctor
\PassOptionsToClass{twoside}{book}
\fi
```

设置字体。由于宋体没有粗体，且我校模板的标题要求使用粗宋体，于是面临 CTeX 的经典的伪粗体 bug：“首次出现伪粗体字体之后的正常字体无法复制”。但如果使用自带宋体的思源字体，那么不必使用伪粗体。模板只给出了新 windows 字体的思源字体设置，且思源字体版本为 Adobe 版。

```
\ifthenelse%
{\equal{\heu@fontset}{}}%
{%
  \PassOptionsToPackage{AutoFakeBold=2}{xeCJK}
}%
{%
  \ifthenelse%
  {\equal{\heu@fontset}{siyuan}}%
  {\relax}%
  {%
    \PassOptionsToPackage{AutoFakeBold=2}{xeCJK}
  }%
  \PassOptionsToClass{fontset=\heu@fontset}{ctexbook}
}%
```

使用 ctexbook 类，优于调用 ctex 宏包。

```
\LoadClass[a4paper,openany,UTF8,zihao=-4,scheme=plain]{ctexbook}
```

用户至少要提供一个选项，指定论文类型。

```
\ifheu@bachelor\relax\else
  \ifheu@master\relax\else
    \ifheu@doctor\relax\else
      \ifheu@postdoc\relax\else
         \ClassError{heuthesis}%
            {Please specify thesis type in option: \MessageBreak
             type=[bachelor | master | doctor | postdoc]}{}
      \fi
    \fi
  \fi
\fi
```

设置字距默认选项

```
\ifheu@zijvregu\relax\else
\ifheu@zijvword\relax\else
\heu@zijvregutrue
\fi
\fi
```

重新设置 geometry 的默认选项

```
\ifheu@geometrynewone\relax\else
\ifheu@geometrynewtwo\relax\else
\ifheu@geometrynewno\relax\else
\heu@geometrynewtwotrue
\fi
\fi
\fi
% 设置专业学位论文类型
%     \begin{macrocode}
\ifheu@profdegree{
\ifheu@productdevelopment\relax\else
  \ifheu@projectdplanning\relax\else
    \ifheu@projectdesign\relax\else
      \ifheu@applicationresearch\relax\else
        \ClassError{heuthesisartplus}%
        { Please specify researchtype in option:\MessageBreak
          researchtype=[productdevelopment | projectdplanning | projectdesign | applica
        }{}
      \fi
    \fi
  \fi
\fi}
\fi
```

4.3 装载宏包

引用的宏包和相应的定义。

```
\RequirePackage{etoolbox}
\RequirePackage{ifxetex}
\ifxetex
\else
  \ClassError{heuthesis}%
    {Please use: \MessageBreak
      xelatex}{}
\fi
\RequirePackage{xparse}
```

$\mathcal{A}\mathcal{M}\mathcal{S}$-TEX 宏包，用来排出更加漂亮的公式。

```
\RequirePackage{amsmath}
```

定理类环境宏包，其中 amsmath 选项用来兼容 $\mathcal{A}\mathcal{M}\mathcal{S}$-TEX 的宏包

```
\RequirePackage[amsmath,thmmarks,hyperref]{ntheorem}
\RequirePackage{amssymb}
```

设置默认英文字体.

```
\setmainfont{Times New Roman}
\setsansfont{Arial}
\setmonofont{Courier New}
```

单独使用 newtxmath 宏包改变数学字体时，会使 operators，\mathrm，\mathit，\mathbf 命令使用 CM 字体，根据 SJTUThesis 的字体设置 进行调整

```
\ifheu@newtxmath
  \let\oldencodingdefault\encodingdefault
  \let\oldrmdefault\rmdefault
  \let\oldsfdefault\sfdefault
  \let\oldttdefault\ttdefault
  \def\encodingdefault{T1}
  \renewcommand{\rmdefault}{ntxtlf}
  \renewcommand{\sfdefault}{qhv}
  \renewcommand{\ttdefault}{ntxtt}
  \RequirePackage{newtxmath}
  \let\encodingdefault\oldencodingdefault
  \let\rmdefault\oldrmdefault
  \let\sfdefault\oldsfdefault
  \let\ttdefault\oldttdefault
\fi
```

图形支持宏包。

```
\RequirePackage{graphicx}
```

pdfpages 宏包便于我们插入扫描后的授权页和声明页 PDF 文档。

```
\RequirePackage{pdfpages}
\includepdfset{fitpaper=true}
```

更好的列表环境。

```
\RequirePackage{enumitem}        % 使用 enumitem 宏包，改变列表项的格式
\RequirePackage{environ}
```

禁止 LaTeX 自动调整多余的页面底部空白，并保持脚注仍然在底部。脚注按页编号。

```
\ifheu@raggedbottom
  \RequirePackage[bottom,perpage,hang]{footmisc}
  \raggedbottom
\else
  \RequirePackage[perpage,hang]{footmisc}
\fi
```

扩展符号字符宏包。

```
\RequirePackage{bbding}
\ifheu@pifootnote
  \RequirePackage{pifont}
\fi
```

利用 xeCJKfntef 实现汉字的下划线和盒子内两段对齐，并可以避免 \makebox[⟨*width*⟩][⟨*s*⟩] 可能产生的 underful boxes。

```
\RequirePackage{xeCJKfntef}
```

表格控制

```
\RequirePackage{diagbox}
\RequirePackage{longtable}
```

使用三线表：\toprule，\midrule，\bottomrule。

```
\RequirePackage{booktabs}
```

重定义 booktabs 宏包的默认线宽，绘制三线表时不再需要显式设置，以实现更好的内容与格式分离

```
\heavyrulewidth=1.5pt
\lightrulewidth=1pt
\cmidrulewidth=0.5pt
```

参考文献引用宏包。

```
% \RequirePackage[sort&compress]{natbib}
```

生成有书签的 pdf 及其开关，请结合 gbk2uni 避免书签乱码。

```
\RequirePackage{hyperref}
\hypersetup{%
  CJKbookmarks=true,
  linktoc=all,
  bookmarksnumbered=true,
  bookmarksopen=true,
  bookmarksopenlevel=1,
  breaklinks=true,
  colorlinks=false,
  plainpages=false,
  pdfborder=0 0 0}
```

设置 url 样式，与上下文一致

```
\urlstyle{same}
```

引用文献标准 GBT2015

```
\RequirePackage[sort&compress]{gbt7714}
\ifheu@citetwocomma
  \patchcmd{\NAT@citexnum}{\def@NAT@last@yr{-\NAT@penalty}}
  {%
    \ifx\NAT@last@yr\relax
    \def@NAT@last@yr{\@citea}%
    \else
    \def@NAT@last@yr{-\NAT@penalty}%
    \fi
  }%
  {\typeout{Use comma between neighboring citations}}%
  {\ClassError{heuhesisbook}{fail to patch \NAT@citexnum}{}}%
\fi
```

4.4 页面设置

这部分是根据 哈尔滨工程大学研院函〔2010〕26 号《关于调整研究生学位论文版面大小的通知》的规定："版面一律改成 A4 纸页边距上、下设置为 28mm，左、右设置为 25mm，页眉、页脚设置为 20mm。装订时上、下、右各切除 3mm，学位论文成品版面大小为 207mm*291mm"

要求设置。论文尺寸为 207x291mm，版芯尺寸 140x220mm，版心距左边 25mm，距右边 20mm，距上下边各 20mm。这里只设置版心尺寸和版心距左边距尺寸，其他尺寸由装订裁切进行控制。

```
\ifheu@debug\RequirePackage[showframe]{geometry}\else\RequirePackage{geometry}\fi
\geometry{% 版芯定义
  a4paper, % 210 * 297mm
  hcentering,
  ignoreall,
  nomarginpar,
  centering,
  text={160true mm,241true mm},
  left=25true mm,
  head=20true mm,
  headsep=2true mm,
  footskip=0true mm,
  foot=6true mm
}
```

载入显示行号的包。

```
\ifheu@debug%
\RequirePackage{layout}
\RequirePackage{layouts}
\RequirePackage{lineno}
\fi
```

利用 fancyhdr 设置页眉页脚。

```
\RequirePackage{fancyhdr}
```

其他包，表格、数学符号包子图最后一行图题是否居中选项

```
\RequirePackage{tabularx}
\RequirePackage{varwidth}
```

此处 changepage 环境用来控制索引页面的左右边距，规范中给出的示例的边距要大于正文。

```
\RequirePackage{changepage}
\RequirePackage{multicol}
\RequirePackage[below]{placeins}% 允许上一个 section 的浮动图形出现在下一个 section 的开始部分，还提供\FloatBarrier 命令，使所有未处理的浮动图形立即被处理
\RequirePackage{flafter}        % 使得所有浮动体不能被放置在其浮动环境之前，以免浮动体在引述它的文本之前出现.
\RequirePackage{multirow}       % 使用 Multirow 宏包，使得表格可以合并多个 row 格
\ifheu@subcapcenterlast
\PassOptionsToPackage{centerlast}{subfigure}
\fi
\RequirePackage{subfigure}% 支持子图 %centerlast 设置最后一行是否居中
\RequirePackage[subfigure]{ccaption} % 支持双语标题
```

中英文索引包。

```
\RequirePackage[makeindex]{splitidx}
\newindex[]{china}
\newindex[]{english}
⟨/bookcls⟩
```

索引格式。

```
⟨*ist⟩
headings_flag 1
heading_prefix "\{\\vskip -\\baselineskip\\centering\\normalsize\\textbf\{"
heading_suffix "\}\\par\}\\nopagebreak\\wuhao\n"
delim_0 "\\hspace*{\\fill}"
delim_1 "\\hspace*{\\fill}"
⟨/ist⟩
```

排版 logo。

```
⟨bookcls⟩\RequirePackage{xltxtra}
```

4.5 主文档格式

4.5.1 页面选项

\cleardoublepage 对于 *openright* 选项，必须保证章首页右开。如果 *library* 为真，则强制设置 *openright* 为真。

```
⟨*bookcls⟩
\ifheu@english
\RequirePackage[xindy,order=letter]{glossaries-extra}
\newglossarystyle{engstyle}{%
  \setglossarystyle{long}%
  \renewenvironment{theglossary}%
  {\vspace{-\intextsep}\centering\begin{longtable}{@{\hskip 1cm}p{3.5cm}p{\glsdescwidth
     {\end{longtable}}}
\setabbreviationstyle[acronym]{long-short}
\else
\RequirePackage{glossaries}
\setacronymstyle{short-long}
\fi

\renewcommand*{\genacrfullformat}[2]{%
  \glsentrylong{#1}%
}
\ifheu@english
\makenoidxglossaries
\else
\makeglossaries
\fi
\ifheu@library\heu@openrightfalse\else\relax\fi
\ifheu@pageempty
  \let\heu@cleardoublepage\cleardoublepage
  \newcommand{\heu@clearemptydoublepage}{%
    \clearpage{\pagestyle{heu@empty}\heu@cleardoublepage}
  }
  \let\cleardoublepage\heu@clearemptydoublepage
\fi
```

\frontmatter 我们的单面和双面模式与常规的不太一样。

```
\renewcommand\frontmatter{%
```

```
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \@mainmatterfalse
  \pagenumbering{Roman}
  \pagestyle{heu@empty}
```

设置字距

```
\ifheu@openright\cleardoublepage\else\clearpage\fi
\ifheu@zijvword
\ziju{0.05416667}
\else
\ifheu@zijvregu
\ziju{0.05580808}
\fi
\fi
}
```

\mainmatter 根据打印店操作，\mainmatter 命令的逻辑是，双面打印时第一章必须在奇数页设置第一章必须在奇数页

```
\renewcommand\mainmatter{%
```

添加英文版限制

```
  \ifheu@doctor%
    \ifheu@library\clearpage\else\cleardoublepage\fi
    \else%
    \clearpage
  \fi%
  \@mainmattertrue
  \pagenumbering{arabic}
  \pagestyle{heu@headings}
}
```

\backmatter 后文开页选项，根据 *openright* 选项决定分页方式。

```
\renewcommand\backmatter{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \@mainmattertrue
  \fancyhf{}
  \pagestyle{heu@titleheadings}
 }
⟨/bookcls⟩
```

4.5.2 字体

\normalsize 根据我校规定，正文小四号 (12bp) 字，但对行距没有给出明确要求。

```
⟨*bookcls⟩
```

1bp = 1/72 inch, 1 inch = 25.4 mm

```
\renewcommand\normalsize{%
  \@setfontsize\normalsize{12bp}{\ifheu@glue 20.50394bp \@plus 2.834646bp \@minus 0bp\e
  \abovedisplayskip=8pt
  \abovedisplayshortskip=8pt
  \belowdisplayskip=\abovedisplayskip
```

```
\belowdisplayshortskip=\abovedisplayshortskip}
```

WORD 中的字号对应该关系如下（1bp = 72.27/72 pt）：

初号	42.2bp	14.82mm	42.1575pt
小初	36.1bp	12.70mm	36.135 pt
一号	26.1bp	9.17mm	26.0975pt
小一	24.1bp	8.47mm	24.09pt
二号	22.1bp	7.76mm	22.0825pt
小二	18.1bp	6.35mm	18.0675pt
三号	16.1bp	5.64mm	16.06pt
小三	15.1bp	5.29mm	15.05625pt
四号	14.1bp	4.94mm	14.0525pt
小四	12.1bp	4.23mm	12.045pt
五号	10.5bp	3.70mm	10.59375pt
小五	9.0bp	3.18mm	9.03375pt
六号	7.5bp	2.56mm	
小六	6.5bp	2.29mm	
七号	5.5bp	1.94mm	
八号	5.0bp	1.76mm	

\heu@def@fontsize 根据习惯定义字号。用法：`\heu@def@fontsize{⟨字号名称⟩}{⟨磅数⟩}` 避免了字号选择和行距的紧耦合。所有字号定义时为单倍行距，并提供选项指定行距倍数。

```
\def\heu@def@fontsize#1#2{%
  \expandafter\newcommand\csname #1\endcsname[1][1.3]{%
    \fontsize{#2}{##1\dimexpr #2}\selectfont}}
```

\dachu 一组字号定义。可以在论文中使用 `\fontsize` 格式指定，如 `\sihao` 。

\chuhao
\xiaochu
\yihao
\xiaoyi
\erhao
\xiaoer
\sanhao
\xiaosan
\sihao
\banxiaosi
\xiaosi
\dawu
\wuhao
\xiaowu
\liuhao
\xiaoliu
\qihao
\bahao

```
\heu@def@fontsize{dachu}{58.0bp}
\heu@def@fontsize{chuhao}{42.2bp}
\heu@def@fontsize{xiaochu}{36.1bp}
\heu@def@fontsize{yihao}{26.1bp}
\heu@def@fontsize{xiaoyi}{24.1bp}
\heu@def@fontsize{erhao}{22.1bp}
\heu@def@fontsize{xiaoer}{18.1bp}
\heu@def@fontsize{sanhao}{16.1bp}
\heu@def@fontsize{xiaosan}{15.1bp}
\heu@def@fontsize{sihao}{14.1bp}
\heu@def@fontsize{banxiaosi}{13.0bp}
\heu@def@fontsize{xiaosi}{12.1bp}
\heu@def@fontsize{dawu}{11.0bp}
\heu@def@fontsize{wuhao}{10.5bp}
\heu@def@fontsize{xiaowu}{9.0bp}
\heu@def@fontsize{liuhao}{7.5bp}
\heu@def@fontsize{xiaoliu}{6.5bp}
\heu@def@fontsize{qihao}{5.5bp}
\heu@def@fontsize{bahao}{5.0bp}
⟨/bookcls⟩
```

4.5.3 页眉页脚

\heu@empty
\heu@plain
\heu@headings
\heu@titleheadings

定义四种页眉页脚格式：

- heu@empty：页眉页脚都没有
- heu@plain:只显示页脚的页码。\chapter 自动调用 \thispagestyle{heu@plain}。
- heu@headings：页眉页脚同时显示，奇数页显示章节
- heu@titleheadings：页眉页脚同时显示，奇数页显示论文题目

```
⟨*bookcls⟩
\let\heu@headrule\headrule
```

空白页面页眉页脚格式控制

```
\fancypagestyle{heu@empty}{%
  \fancyhf{}
  \let\headrule\heu@headrule%
  \renewcommand{\headrulewidth}{0pt}
  \renewcommand{\footrulewidth}{0pt}
}
```

正文页眉页脚格式控制

```
\fancypagestyle{heu@headings}{%
  \fancyhf{}
  \ifheu@english
```

英文版页眉格式控制

```
    \fancyhead[C]{\wuhao[0]\leftmark}
    \fancyfoot[C]{\wuhao\thepage}
    \renewcommand{\headrule}{
      \vskip 0.6mm
      \hrule\@height0.5mm\@width\headwidth
      \vskip 0.3mm
      \hrule\@height0.8mm\@width\headwidth
    }
  \else
    \ifheu@postdoc
```

博后页眉控制

```
      \fancyhead[CO]{\songti\wuhao[0]\leftmark}
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@postdoc@documenttitlenospace}
    \fi
    \ifheu@doctor
      \fancyhead[CO]{\songti\wuhao[0]\leftmark}
      \ifheu@harbin
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@cdegree\heu@cthesisname}
      \fi
      \ifheu@qingdao
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@qingdaocampus\heu@cdegree\h
      \fi
      \ifheu@yantai
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@yantaicampus\heu@cdegree\he
      \fi
    \fi
```

```
    \ifheu@master
      \fancyhead[CO]{\songti\wuhao[0]\leftmark}
      \ifheu@harbin
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@cdegree\heu@cthesisname}
      \fi
      \ifheu@qingdao
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@qingdaocampus\heu@cthesisna
      \fi
      \ifheu@yantai
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@yantaicampus\heu@cdegree\he
      \fi
    \fi
    \ifheu@bachelor
      \fancyhead[CO]{\songti\wuhao[0]\leftmark}
      \ifheu@harbin
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@bachelor@cxuewei\heu@bachel
      \fi
      \ifheu@qingdao
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@qingdaocampus\heu@bachelor@
      \fi
      \ifheu@yantai
        \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@yantaicampus\heu@bachelor@c
      \fi
    \fi
    \fancyfoot[C]{\wuhao\thepage}
    \renewcommand{\headrule}{
    \vskip 0.6mm
    \hrule\@height0.5mm\@width\headwidth
    \vskip 0.3mm
    \hrule\@height0.8mm\@width\headwidth
    }
  \fi
  \renewcommand{\footrulewidth}{0pt}
}
```

前文页眉页脚格式控制（奇数页页眉为论文题目）

```
%
\fancypagestyle{heu@titleheadings}%
{%
  \fancyhead{} %
  \fancyhead[CO]{\songti\wuhao[0]\heu@ctitle}
  \ifheu@postdoc
    \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@postdoc@documenttitlenospace}
  \fi
  \ifheu@doctor
    \ifheu@harbin
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@cdegree\heu@cthesisname}
    \fi
    \ifheu@qingdao
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@qingdaocampus\heu@cdegree\heu
```

```
    \fi
    \ifheu@yantai
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@yantaicampus\heu@cdegree\heu@
    \fi
  \fi
  \ifheu@master
    \ifheu@harbin
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@cdegree\heu@cthesisname}
    \fi
    \ifheu@qingdao
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@qingdaocampus\heu@cthesisname
    \fi
    \ifheu@yantai
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@yantaicampus\heu@cdegree\heu@
    \fi
  \fi
  \ifheu@bachelor
    \ifheu@harbin
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@bachelor@cxuewei\heu@bachelor
    \fi
    \ifheu@qingdao
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@qingdaocampus\heu@bachelor@cx
    \fi
    \ifheu@yantai
      \fancyhead[CE]{\songti\wuhao[0]\heu@cschoolname\heu@yantaicampus\heu@bachelor@cxu
    \fi
  \fi
  \fancyfoot[C]{\wuhao\thepage}       % 只显示号码
  \renewcommand{\headrulewidth}{0pt}  % 页眉线粗细
  \renewcommand{\footrulewidth}{0pt}  % 页脚线粗细
  \renewcommand{\headrule}{
    \vskip 0.6mm
    \hrule\@height0.5mm\@width\headwidth
    \vskip 0.3mm
    \hrule\@height0.8mm\@width\headwidth
  }
}
\AtBeginDocument{%
  \pagestyle{heu@empty}
  \renewcommand{\chaptermark}[1]{\@mkboth{\CTEXthechapter\enspace#1}{}}}
⟨/bookcls⟩
```

4.5.4 段落

全文首行缩进 2 字符，标点符号用全角

```
⟨*bookcls⟩
\ctexset{%
  punct=quanjiao,
  space=auto,
  autoindent=true}
```

利用 enumitem 命令调整默认列表环境间的距离，以符合中文习惯。

```
\setlist{nosep}
⟨/bookcls⟩
```

4.5.5 脚注

脚注符合中文习惯，数字带圈。

```
⟨*bookcls⟩
\def\heu@textcircled#1{%
  \ifnum\value{#1} >9
    \ClassError{heuthesis}%
      {Too many footnotes in this page.}{Keep footnote less than 10.}
  \fi
  \ifheu@pifootnote%
    \ding{\the\numexpr\value{#1}+171\relax}%
  \else%
    \textcircled{\xiaoliu\arabic{#1}}%
  \fi}
\renewcommand{\thefootnote}{\heu@textcircled{footnote}}
\renewcommand{\thempfootnote}{\heu@textcircled{mpfootnote}}
```

定义脚注分割线，字号（宋体小五），以及悬挂缩进（1.5 字符）。

```
\def\footnoterule{\vskip-3\p@\hrule\@width0.3\textwidth\@height0.4\p@\vskip2.6\p@}
\let\heu@footnotesize\footnotesize
\renewcommand\footnotesize{\heu@footnotesize\xiaowu[1.5]}
\footnotemargin1.5em\relax
```

\@makefnmark 默认是上标样式，而在脚注部分要求为正文大小。利用\patchcmd 动态调整 \@makefnmark 的定义。

```
\let\heu@makefnmark\@makefnmark
\def\heu@@makefnmark{\hbox{{\normalfont\@thefnmark}}}
\pretocmd{\@makefntext}{\let\@makefnmark\heu@@makefnmark}{}{}
\apptocmd{\@makefntext}{\let\@makefnmark\heu@makefnmark}{}{}
⟨/bookcls⟩
```

4.5.6 数学相关

允许太长的公式断行、分页等。

```
⟨*bookcls⟩
\allowdisplaybreaks[4]
\predisplaypenalty=0  % 公式之前可以换页，公式出现在页面顶部
\postdisplaypenalty=0
```

设置公式编号方式为 X-n

```
\renewcommand\theequation{\ifnum \c@chapter>\z@ \thechapter-\fi\@arabic\c@equation}
```

设置公式前后随意断页公式距前后文的距离由 4 个参数控制，参见 \normalsize 的定义。同时为了让 amsmath 的 \tag* 命令得到正确的格式，我们必须修改这些代码。\make@df@tag 是定义 \tag* 和 \tag 内部命令的。\make@df@tag@@ 处理 \tag*，我们就改它！

```
\def\make@df@tag{\@ifstar\make@df@tag@@\make@df@tag@@@}
\def\make@df@tag@@#1{%
  \gdef\df@tag{\maketag@@@{#1}\def\@currentlabel{#1}}}
```

```
\def\make@df@tag{\@ifstar\heu@make@df@tag@@\make@df@tag@@@}
\def\heu@make@df@tag@@#1{\gdef\df@tag{\heu@maketag{#1}\def\@currentlabel{#1}}}
\iffalse
\ifheu@bachelor
  \def\heu@maketag#1{\maketag@@@{%
    (\ignorespaces\text{\equationname\hskip0.5em}#1\unskip\@@italiccorr)}}
  \def\tagform@#1{\maketag@@@{%
    (\ignorespaces\text{\equationname\hskip0.5em}#1\unskip\@@italiccorr)\equcaption{#1}
\fi
\fi
\def\heu@maketag#1{\maketag@@@{(\ignorespaces #1\unskip\@@italiccorr)}}
\def\tagform@#1{\maketag@@@{(\ignorespaces #1\unskip\@@italiccorr)\equcaption{#1}}}
```

修改 \tagform 会影响 \eqref。

```
\renewcommand{\eqref}[1]{\textup{(\ref{#1})}}
```

⟨/bookcls⟩

定理标题使用黑体，正文使用宋体，冒号隔开。

⟨*bookcfg⟩

```
\theorembodyfont{\normalfont}
\theoremsymbol{\ensuremath{\square}}
\ifheu@english
\theoremheaderfont{\normalfont\bf}
\newtheorem*{proof}{Proof}
\else
\theoremheaderfont{\normalfont\heiti}
\newtheorem*{proof}{证明}
\fi
```

英文版黑体加粗

```
\theoremstyle{plain}
\theoremsymbol{}
\theoremseparator{}
\ifheu@english
\newtheorem{assumption}{Assumption}[chapter]
\newtheorem{definition}{Definition}[chapter]
\newtheorem{proposition}{Proposition}[chapter]
\newtheorem{lemma}{Lemma}[chapter]
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{axiom}{Axiom}[chapter]
\newtheorem{corollary}{Corollary}[chapter]
\newtheorem{exercise}{exercise}[chapter]
\newtheorem{example}{Example}[chapter]
\newtheorem{remark}{Remark}[chapter]
\newtheorem{problem}{Problem}[chapter]
\newtheorem{conjecture}{Conjecture}[chapter]
\newtheorem{fact}{Fact}[chapter]
```

```
\else
\newtheorem{assumption}{假设}[chapter]
\newtheorem{definition}{定义}[chapter]
\newtheorem{proposition}{命题}[chapter]
\newtheorem{lemma}{引理}[chapter]
\newtheorem{theorem}{定理}[chapter]
\newtheorem{axiom}{公理}[chapter]
\newtheorem{corollary}{推论}[chapter]
\newtheorem{exercise}{练习}[chapter]
\newtheorem{example}{例}[chapter]
\newtheorem{remark}{注释}[chapter]
\newtheorem{problem}{问题}[chapter]
\newtheorem{conjecture}{猜想}[chapter]
\newtheorem{fact}{事实}[chapter]
\fi
⟨/bookcfg⟩
⟨*artcfg⟩
\theorembodyfont{\normalfont}
\theoremheaderfont{\normalfont\heiti}
\theoremsymbol{\ensuremath{\square}}
\newtheorem*{proof}{证明}
\theoremstyle{plain}
\theoremsymbol{}
\theoremseparator{}
\newtheorem{assumption}{假设}
\newtheorem{definition}{定义}
\newtheorem{proposition}{命题}
\newtheorem{lemma}{引理}
\newtheorem{theorem}{定理}
\newtheorem{axiom}{公理}
\newtheorem{corollary}{推论}
\newtheorem{exercise}{练习}
\newtheorem{example}{例}
\newtheorem{remark}{注释}
\newtheorem{problem}{问题}
\newtheorem{conjecture}{猜想}
\ctexset{%
  contentsname={目\hspace{\ccwd} 录},
  figurename= 图,
  tablename= 表
}
\let\CJK@todaysave=\today
\def\CJK@todaysmall@short{\the\year 年 \the\month 月}
\def\CJK@todaysmall{\the\year 年 \the\month 月 \the\day 日}
\def\CJK@todaybig@short{\zhdigits{\the\year} 年\zhnumber{\the\month} 月}
\def\CJK@todaybig{\zhdigits{\the\year} 年\zhnumber{\the\month} 月\zhnumber{\the\day}
日}
\def\CJK@today{\CJK@todaysmall}
\renewcommand\today{\CJK@today}
\newcommand\CJKtoday[1][1]{%
```

```
  \ifcase#1\def\CJK@today{\CJK@todaysave}
    \or\def\CJK@today{\CJK@todaysmall}
    \or\def\CJK@today{\CJK@todaybig}
  \fi}
\cdate{\ifheu@bachelor\CJK@todaysmall\else\CJK@todaysmall@short\fi}
\ifheu@doctor
\gdef\heu@cxueweishort{博}
\gdef\heu@cxuewei{\heu@cxueweishort 士}
\gdef\heu@cdegree{\heu@cxueke\heu@cxuewei}
\def\heu@cauthortitle{\heu@cxueweishort 士研究生}
\fi
\ifheu@master
\gdef\heu@cxueweishort{硕}
\gdef\heu@cxuewei{\heu@cxueweishort 士}
\gdef\heu@cdegree{\heu@cxueke\heu@cxuewei}
\def\heu@cauthortitle{\heu@cxueweishort 士研究生}
\fi
\ifheu@bachelor
\gdef\heu@cxuewei{学士}
\fi
\def\heu@stage@opening{开题}
\def\heu@stage@midterm{中期}
\def\heu@stage@doctype{报告}
\def\heu@bachelor@cxuewei{本科生}
\def\heu@bachelor@cthesisname{毕业设计（论文）}
\def\heu@bachelor@cmajortitle{专\hspace{2\ccwd} 业}
\def\heu@bachelor@cstudenttitle{学\hspace{2\ccwd} 生}
\def\heu@bachelor@cstudentidtitle{学\hspace{2\ccwd} 号}
\def\heu@bachelor@cclass{班\hspace{2\ccwd} 号}
\def\heu@bachelor@caffiltitle{学院（系）名称}
\def\heu@bachelor@caffiltitleyt{烟台学院}
\def\heu@bachelor@caffiltitleqd{青岛学院}
\def\heu@bachelor@csupervisortitle{指导教师}
\def\heu@bachelor@cdatetitle{日\hspace{2\ccwd} 期}
\newcommand{\heu@bachelor@teachercomment}{指导教师评语：}
\newcommand{\heu@bachelor@teachersign}{指导教师签字：}
\newcommand{\heu@bachelor@checkdate}{检查日期：}
\def\heu@cthesistitleprefix{题\hspace{\ccwd} 目}
\def\heu@graduate@caffiltitle{院\hspace{3\ccwd}（系）}
\def\heu@graduate@cmajortitle{学\hspace{4\ccwd} 科}
\def\heu@graduate@supervisor{导\hspace{4\ccwd} 师}
\def\heu@graduate@studenttitle{研\hspace{1.5\ccwd} 究\hspace{1.5\ccwd} 生}
\def\heu@graduate@studentid{学\hspace{4\ccwd} 号}
\def\heu@graduate@datetitle{\ifheu@opening\heu@stage@opening
\else\ifheu@midterm\heu@stage@midterm\fi\fi\heu@stage@doctype 日期}
\def\heu@graduate@enrolldate{入\hspace{0.6666666\ccwd} 学\hspace{0.6666666\ccwd}
  时\hspace{0.6666666\ccwd} 间}
\def\heu@graduate@thesistitle{论\hspace{0.6666666\ccwd} 文\hspace{0.6666666\ccwd}
  题\hspace{0.6666666\ccwd} 目}
\def\heu@graduate@cafflimajor{学\hspace{0.6666666\ccwd} 科\hspace{0.6666666\ccwd}
```

```
专\hspace{0.6666666\ccwd} 业}
\def\heu@cschoolname{哈尔滨工程大学}
\def\heu@qingdaocampus{（青岛）}
\def\heu@cthesisname{学位论文}
\def\heu@yantaicampus{（烟台）}
\def\heu@cthesismidtermcheck{\heu@cxueweishort\hspace{.125\ccwd} 士\hspace{.125\ccwd} 学\hspace{.125\ccwd} 位\hspace{.125\ccwd} 论\hspace{.125\ccwd} 文\hspace{.125\ccwd} 中\hspace{.125\ccwd} 期\hspace{.125\ccwd} 检\hspace{.125\ccwd} 查\hspace{.125\ccwd} 报\hspace{.125\ccwd} 告}
\def\heu@cschoolnametitle{学位授予单位}
\def\heu@csubmitdatatitle{论文提交日期}
\def\heu@coralexdatetitle{论文答辩日期}
\def\heu@caffiltitle{所在单位}
\def\heu@csubjecttitle{学科、专业}
\def\heu@cdegreetitle{申请学位}
\def\heu@csupervisortitle{指导教师}
\def\heu@cassosupervisortitle{副导师}
\def\heu@ccosupervisortitle{联合导师}
\def\heu@creviewertitle{学位论文主审人}
\def\heu@title@csep{：}
\def\heu@natclassifiedindextitle{分类号}
\def\heu@internatclassifiedindextitle{ＵＤＣ}
\def\heu@secretlevel{密级}
\def\heu@cnumbertitle{编号}
\newcommand{\heu@authorsig}{作者签名：}
\newcommand{\heu@teachersig}{导师签名：}
\newcommand{\heu@frontdate}{日期：}
\newcommand{\heu@authorizationtitle}{学位论文使用权限}
\newcommand{\heu@qingdao@schoolbottommark}{
  \begin{center}
    \songti\sanhao\textbf{哈工程（青岛）制}
  \end{center}
}
\newcommand{\heu@yantai@schoolbottommark}{
   \begin{center}
       \songti\sanhao\textbf{哈工程（烟台）制}
   \end{center}
}
\newcommand{\heu@harbin@schoolbottommark}{
  \begin{center}
    \songti\sanhao\textbf{研究生院制}
  \end{center}
}
\newcommand{\heu@harbin@bachelor@schoolbottommark}{
  \begin{center}
    \songti\xiaoer\textbf{哈尔滨工程大学本科生院制}
  \end{center}
}
\newcommand{\heu@doctor@midterm@note}{
\thispagestyle{empty}
```

```
{\begin{center}
\songti\sanhao\textbf{说\hspace{4\ccwd} 明}
\end{center}
}
\vspace{1cm}\songti\sihao[2]
\begin{enumerate}[itemsep=-4bp]
\item 博士研究生学位论文中期报告一般在入学后的第五学期末完成。
\item 此报告由学生本人填写，请导师签字后提交一份给检查小组。答辩结束后由检查组长签署意见。
\item 报告经中期报告检查组长签字同意后，由学科部保存，以备论文答辩时参考。研究生院学生培养处将对研究生的学位论文中期报告进行抽查。
\item 报告统一用 A4 纸打印。
\end{enumerate}
}
\newcommand{\heu@datefill}{\hspace{2.5em} 年\hspace{1.5em} 月\hspace{1.5em} 日}
\def\heu@motto{大工至善，大学至臻}
\def\heuthesis{\textsc{heu}\-\textsc{Thesis}}
\def\heu{哈尔滨工程大学}
⟨/artcfg⟩
```

4.5.7 浮动对象以及表格

设置浮动对象和文字之间的距离，由于规范中没有明确规定，根据经验，设置成正文汉字高度。

```
⟨*bookcls⟩
\ifheu@postdoc
\setlength{\intextsep}{10.5bp}
\setlength{\textfloatsep}{10.5bp}
\setlength{\floatsep}{10.5bp}
```

添加博后的浮动间距设置

```
\else
\setlength{\intextsep}{\ifheu@glue 8.50398bp \@plus 2.83465bp \@minus 0bp\else 8.50398b
\setlength{\textfloatsep}{\ifheu@glue 8.50398bp \@plus 2.83465bp \@minus 0bp\else 8.503
\setlength{\floatsep}{\ifheu@glue 12bp \@plus 2.83465bp \@minus 0bp\else 12bp\fi}
\fi
```

此处设置 float 在 p 选项时间隔，此处不设置\@fptop 和\@fpbot 以确保居中。下面这组命令使浮动对象的缺省值稍微宽松一点，从而防止幅度对象占据过多的文本页面，也可以防止在很大空白的浮动页上放置很小的图形。

```
\g@addto@macro\appendix{\renewcommand*{\thefigure}{\thechapter.\arabic{figure}}}
\g@addto@macro\appendix{\renewcommand*{\thetable}{\thechapter.\arabic{table}}}
\g@addto@macro\appendix{\renewcommand*{\theequation}{\thechapter.\arabic{equation}}}
\renewcommand{\textfraction}{0.15}
\renewcommand{\topfraction}{0.85}
\renewcommand{\bottomfraction}{0.65}
\renewcommand{\floatpagefraction}{0.60}
```

\@makecaption 根据我校规范，本科和硕博的图题序号之后的空格不一样。《哈尔滨工程大学研究生学位论文撰写规范》2.13.1 中规定：“每个图均应有图题（由图序和图名组成），图题不宜有标点符号，图名在图序之后空 1 个半角字符排写。” 《哈尔滨工程大学本科生毕业论文撰写规范》2.13.1

中规定："每个图均应有图题（由图序和图名组成），图题不宜有标点符号，图名在图序之后空 1 个字符排写。" 我校规范中没有明确规定是否标题是否居中对齐，这里给出一个居中选项自行调整。注意，我校只规定："居中书写"。此处不额外添加悬挂处理。

```
\long\def\@makecaption#1#2{%
  \vskip\abovecaptionskip
  \wuhao\sbox\@tempboxa{#1\ifheu@bachelor\hskip\ccwd\else\enskip\fi#2}%
  \ifdim \wd\@tempboxa >\hsize
    \ifheu@capcenterlast%
      \vskip 6.3bp%
        {\setbox0=\vbox{#1\ifheu@bachelor\hskip\ccwd\else\enskip\fi#2}
           \setbox1=\vbox{%
             \unvbox0
             \setbox2=\lastbox
             \hbox to \textwidth{\hfill\unhcopy2 \unskip\unskip\hfill}
            }
          \unvbox1}
      \else%
        #1\ifheu@bachelor\hskip\ccwd\else\enskip\fi#2%
      \fi%
    \par
  \else
  \global \@minipagefalse
  \hb@xt@\hsize{\hfil\box\@tempboxa\hfil}%
  \fi
\vskip\belowcaptionskip}
```

\LT@makecaption 有关 longtable 的 caption 支持

```
\long\def\LT@makecaption#1#2#3{%
  \LT@mcol\LT@cols c{\hbox to\z@{\hss\parbox[t]\LTcapwidth{%
    \wuhao\sbox\@tempboxa{#1{#2\ifheu@bachelor\hskip\ccwd\else\enskip\fi }#3}%
    \ifdim\wd\@tempboxa>\hsize
      \ifheu@capcenterlast%
        \vskip 6.3bp%
        {\setbox0=\vbox{#1{#2\ifheu@bachelor\hskip\ccwd\else\enskip\fi }#3}
          \setbox1=\vbox{%
             \unvbox0
             \setbox2=\lastbox
             \hbox to \textwidth{\hfill\unhcopy2 \unskip\unskip\hfill}
          }
        \unvbox1}
      \else%
        #1{#2\ifheu@bachelor\hskip\ccwd\else\enskip\fi }#3%
      \fi%
      \par
    \else
      \global \@minipagefalse
      \hbox to\hsize{\hfil\box\@tempboxa\hfil}
    \fi
    \endgraf\vskip\belowcaptionskip}%
```

```
\hss}}}
```

\longbionenumcaption 长表格的双语标题是一个坑. 因为第一不能用浮动格式，只能用 longtable 包中的 tabular ，这样表题只能使用表格中前两行来写。这样出现了一个问题是，中英表题的间距，标题和表第一行间距，表格内部间距等多个变量的协调问题。这个问题只要使用 tabular 的形式，就是无解的。唯一的方法就是把这些参数都给用户列出来。以下，第 2，5 参数为中英双语标题内容，1，4 为标题参数。6 为中英标题间距，7 为表题和表格间距。

```
\renewcommand*{\longbionenumcaption}[7]{%
\@if@contemptyarg{#1}{\caption{#2}}{\caption[#1]{#2}}%
\global\let\@cont@oldtablename\tablename
\gdef\tablename{#3}
\global\let\LT@c@ption\@cont@LT@nonumintoc
\\[#6]
\@if@contemptyarg{#4}{\caption{#5}}{\caption[#4]{#5}}%
\global\let\tablename\@cont@oldtablename
\global\let\LT@c@ption\@cont@oldLT@c@ption
\vspace{#7}}
```

\ltfontsize 我们采用 longtable 来处理跨页的表格。同样我们需要设置其默认字体为五号，行距设置为 1.3 倍行距。此处还需要提供一个设置长表格内部字体的命令。

```
\let\heu@LT@array\LT@array
\def\LT@array{\wuhao\heu@LT@array} % set default font size
\newcommand{\ltfontsize}[1]{\def\LT@array{#1\heu@LT@array}}
```

图表名称及格式。

```
\renewcommand{\thesubtable}{(\alph{subtable})}
\renewcommand{\thefigure}{\arabic{chapter}.\arabic{figure}}% 使图编号为 7.1 的格式 %\protect{~}
\renewcommand{\thesubfigure}{(\alph{subfigure})}% 使子图编号为 (a) 的格式
\renewcommand{\p@subfigure}{\thefigure~} % 使子图引用为 7.1(a) 的格式，母图编号和子图编号之间用 ~ 加一个空格
\renewcommand{\thetable}{\arabic{chapter}.\arabic{table}}% 使表编号为 7.1 的格式
```

调整罗列环境、浮动格式、间距。

```
\setitemize{leftmargin=0em,itemsep=0em,partopsep=0em,parsep=0em,topsep=0em,itemindent=3
\setenumerate{leftmargin=0em,itemsep=0em,partopsep=0em,parsep=0em,topsep=0em,itemindent
\newcommand{\citeup}[1]{\textsuperscript{\cite{#1}}}
\captionnamefont{\wuhao}
\captiontitlefont{\wuhao}
\renewcommand{\subcapsize}{\wuhao}
\setlength{\abovecaptionskip}{0pt}% 为了双标题之间的间距，不能设置
\setlength{\belowcaptionskip}{0pt}
% 自定义项目列表标签及格式 \begin{publist} 列表项 \end{publist}
\newcounter{pubctr} % 自定义新计数器
\newenvironment{publist}{%% 定义新环境
\begin{list}{[\arabic{pubctr}]} %% 标签格式
    {
     \usecounter{pubctr}
     \setlength{\leftmargin}{1.7em}   % 左边界 \leftmargin =\itemindent + \labelwidth +
```

```
    \setlength{\itemindent}{0em}      % 标号缩进量
    \setlength{\labelsep}{0.5em}      % 标号和列表项之间的距离，默认 0.5em
    \setlength{\rightmargin}{0em}     % 右边界
    \setlength{\topsep}{0ex}          % 列表到上下文的垂直距离
    \setlength{\parsep}{0ex}          % 段落间距
    \setlength{\itemsep}{0ex}         % 标签间距
    \setlength{\listparindent}{0pt} % 段落缩进量
   }}
{\end{list}}
```

设置定理定义格式

```
\renewtheoremstyle{plain}
{\item[\hskip\labelsep \theorem@headerfont ##1\ ##2\theorem@separator]}
{\item[\hskip\labelsep \theorem@headerfont ##1\ ##2\ ##3\theorem@separator]}
\theorembodyfont{\songti\rmfamily}
\theoremheaderfont{\heiti\rmfamily}
\theoremsymbol{$\square$}
\setlength{\theorempreskipamount}{0pt}
\setlength{\theorempostskipamount}{-2pt}
\setlength{\parindent}{2em}
\arraycolsep=1.6pt
⟨/bookcls⟩
```

4.5.8 章节标题

```
⟨*bookcfg⟩
\ifheu@english
\ctexset{%
  chapter/name={Chapter\enskip,},
  appendixname={Appendix},
  contentsname={Contents},
  listfigurename={List of Figures},
  listtablename={List of Tables},
  figurename={Figure},
  tablename={Table},
  bibname={References},
}
\else
\ctexset{%
  chapter/name={第，章},
  appendixname= 附录,
  contentsname={目\hspace{\ccwd} 录},
  listfigurename= 插图索引,
  listtablename= 表格索引,
  figurename= 图,
  tablename= 表,
  bibname= 参考文献,
  indexname= 索引,
}
\fi
\ifheu@postdoc
```

```
\ctexset{
  contentsname={目录},
}
\fi
```

添加 Article 和 English 版本以及 postdoc 版本标题名称设置

```
\ifheu@english
\newcommand\equationname{Equation}
\else
\newcommand\equationname{公式}
\fi
\newcommand\listfigureename{Index of figure}
\newcommand\listtableename{Index of table}
\newcommand\listequationename{Index of equation}
\newcommand\listequationname{公式索引}
```

设置英文版公式名称

```
\newcommand{\cabstractename}{Abstract (In Chinese)}
```

此处删除冗余选项

```
\newcommand{\eabstractcname}{Abstract}
\newcommand{\eabstractename}{Abstract (In English)}
\newcommand{\cabstractcname}{摘要}

```

添加博后摘要

```
\newcommand{\heu@ckeywords@title}{关键词：}
\def\heu@ckeywords@separator{；}
\def\heu@ekeywords@separator{,}
\let\CJK@todaysave=\today
\def\CJK@todaysmall@short{\the\year 年 \the\month 月}
\def\CJK@todaysmall{\the\year 年 \the\month 月 \the\day 日}
\def\CJK@todaybig@short{\zhdigits{\the\year} 年\zhnumber{\the\month} 月}
\def\CJK@todaybig{\zhdigits{\the\year} 年\zhnumber{\the\month} 月\zhnumber{\the\day}
日}
\def\CJK@today{\CJK@todaysmall}
\renewcommand\today{\CJK@today}
\newcommand\CJKtoday[1][1]{%
  \ifcase#1\def\CJK@today{\CJK@todaysave}
    \or\def\CJK@today{\CJK@todaysmall}
    \or\def\CJK@today{\CJK@todaybig}
  \fi}
```

按照 word 示范要求，此处使用阿拉伯数字

```
\cdate{\ifheu@bachelor\CJK@todaysmall\else\CJK@todaysmall@short\fi}
\edate{\ifcase \month \or January\or February\or March\or April\or May%
       \or June\or July \or August\or September\or October\or November
       \or December\fi\unskip,\ \ \the\year}
```

⟨/bookcfg⟩

按照我校要求，页面中标题之下不少于一行。

⟨*bookcls⟩

```
\def\heu@title@font{%
  \ifheu@arialtitle\sffamily\else\heiti\fi}
```

```
\newcommand\heu@chapter@titleformat[1]{% 开启悬挂缩进选项
  \ifheu@english
  \MakeUppercase{#1}
  \else
    \ifthenelse%
    {\equal{#1}{\eabstractcname}}%
    {\bfseries #1}%
    % 实现章标题的居中加悬挂缩进,注意,此处一定是\CTEX@chaptername\CTEX@chapter@aftername, 否则是英文标题长度
    {\ifheu@chapterhang\settowidth{\hangindent}{\CTEX@chaptername\CTEX@chapter@afternam
  \fi
}
```

添加英文版目录 chapter title 控制

```
\newcommand\heu@chapter@engformat[1]{% 开启悬挂缩进选项
  \begin{center}
    \heu@title@font
    \if@mainmatter
    \xiaoer[1.57481]
    \else
    \sanhao[1.57481]
    \fi\bfseries
    #1
  \end{center}}
\newcommand\heu@chapter@engnameformat[1]{% 开启悬挂缩进选项
  \begin{center}
    \MakeUppercase{#1}
  \end{center}}
\newcommand\heu@chapter@postdocformat[1]{\centering\heu@title@font\xiaoer[1.57481]\text
```

添加博后标题格式

```
\renewcommand\@afterheading{%
  \@nobreaktrue
  \everypar{%
    \if@nobreak
      \@nobreakfalse
      \clubpenalty 1
      \if@afterindent \else
        {\setbox\z@\lastbox}%
      \fi
    \else
      \clubpenalty 1
      \everypar{}%
    \fi}}
```

设置一到四级标题、目录、书签格式。

按照规范中规定行距的 mm 数定义各级行距

```
\ifheu@english
  \ctexset{%
    chapter={
      afterindent=true,
```

```
      pagestyle={heu@headings},
      beforeskip={28.34646bp},% 一个空行 1.57481 × 18
      afterskip={28.74646bp},%0.8 应该不计算间距 0.8 × 18 + 0.57481×18
      aftername={\par},
      format={\heu@chapter@engformat},
      nameformat={\heu@chapter@engnameformat},%\center 会影响之后全局
      numberformat=\relax,
      titleformat={\heu@chapter@titleformat},
      fixskip=true, % 添加这一行去除默认间距
      %hang=true,
    },
    section={
      afterindent=true,
      beforeskip={\ifheu@glue 19.84252bp \@plus 2.834646bp \@minus 0bp \else 19.84252bp
上下空 0.5 行
      afterskip={\ifheu@glue 19.84252bp \@plus 2.834646bp \@minus 0bp \else 19.84252bp
      format={\heu@title@font\ifheu@glue\fontsize{15bp}{21bp \@plus 1.677267bp \@minus
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    subsection={
      afterindent=true,
      beforeskip={\ifheu@glue 17.00787bp \@plus 2.834646bp \@minus 0bp \else 17.00787bp
      afterskip={\ifheu@glue 17.00787bp \@plus 2.834646bp \@minus 0bp \else 17.00787bp
      format={\heu@title@font\ifheu@glue\fontsize{14bp}{18bp \@plus 1.842609bp \@minus
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    subsubsection={
      afterindent=true,
      beforeskip={\ifheu@glue 8.503937bp \@plus 2.834646bp \@minus 0bp \else 8.503937bp
      afterskip={\ifheu@glue 8.503937bp \@plus 2.834646bp \@minus 0bp \else 8.503937bp
      format={\heu@title@font\normalsize},
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    paragraph/afterindent=true,
    subparagraph/afterindent=true
  }
\else
\ifheu@postdoc
  \ctexset{%
    chapter={
      afterindent=true,
      pagestyle={heu@headings},
      beforeskip={28.34646bp},% 一个空行 1.57481 × 18
      afterskip={28.74646bp},%0.8 应该不计算间距 0.8 × 18 + 0.57481×18
```

```
      aftername=\enspace,
      format={\heu@chapter@postdocformat},%\center 会影响之后全局
      nameformat=\relax,
      numberformat=\relax,
      titleformat=\heu@chapter@titleformat,
      fixskip=true, % 添加这一行去除默认间距
      %hang=true,
    },
    section={
      afterindent=true,
      beforeskip={\ifheu@glue 19.84252bp \@plus 2.834646bp \@minus 0bp \else 19.84252bp
上下空 0.5 行
      afterskip={\ifheu@glue 19.84252bp \@plus 2.834646bp \@minus 0bp \else 19.84252bp
      format={\heu@title@font\ifheu@glue\fontsize{15bp}{21bp \@plus 1.677267bp \@minus
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    subsection={
      afterindent=true,
      beforeskip={\ifheu@glue 17.00787bp \@plus 2.834646bp \@minus 0bp \else 17.00787bp
      afterskip={\ifheu@glue 17.00787bp \@plus 2.834646bp \@minus 0bp \else 17.00787bp
      format={\heu@title@font\ifheu@glue\fontsize{14bp}{18bp \@plus 1.842609bp \@minus
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    subsubsection={
      afterindent=true,
      beforeskip={\ifheu@glue 8.503937bp \@plus 2.834646bp \@minus 0bp \else 8.503937bp
      afterskip={\ifheu@glue 8.503937bp \@plus 2.834646bp \@minus 0bp \else 8.503937bp
      format={\heu@title@font\normalsize},
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    paragraph/afterindent=true,
    subparagraph/afterindent=true
  }
\else
  \ctexset{%
    chapter={
      afterindent=true,
      pagestyle={heu@headings},
      beforeskip={28.34646bp},% 一个空行 1.57481 × 18
      afterskip={28.74646bp},%0.8 应该不计算间距 0.8 × 18 + 0.57481×18
      aftername=\enspace,
      format={\centering\heu@title@font\xiaoer[1.57481]},%\center 会影响之后全
局
      nameformat=\relax,
```

```
      numberformat=\relax,
      titleformat=\heu@chapter@titleformat,
      fixskip=true, % 添加这一行去除默认间距
      %hang=true,
    },
    section={
      afterindent=true,
      beforeskip={\ifheu@glue 19.84252bp \@plus 2.834646bp \@minus 0bp \else 19.84252bp
上下空 0.5 行
      afterskip={\ifheu@glue 19.84252bp \@plus 2.834646bp \@minus 0bp \else 19.84252bp
      format={\heu@title@font\ifheu@glue\fontsize{15bp}{21bp \@plus 1.677267bp \@minus
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    subsection={
      afterindent=true,
      beforeskip={\ifheu@glue 17.00787bp \@plus 2.834646bp \@minus 0bp \else 17.00787bp
      afterskip={\ifheu@glue 17.00787bp \@plus 2.834646bp \@minus 0bp \else 17.00787bp
      format={\heu@title@font\ifheu@glue\fontsize{14bp}{18bp \@plus 1.842609bp \@minus
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    subsubsection={
      afterindent=true,
      beforeskip={\ifheu@glue 8.503937bp \@plus 2.834646bp \@minus 0bp \else 8.503937bp
      afterskip={\ifheu@glue 8.503937bp \@plus 2.834646bp \@minus 0bp \else 8.503937bp
      format={\heu@title@font\normalsize},
      aftername=\enspace,
      fixskip=true,
      break={},
    },
    paragraph/afterindent=true,
    subparagraph/afterindent=true
  }
\fi
\fi
```

此处添加英文版和博后的 title 设置设置附表、附录格式。

```
\ifheu@english
\NewDocumentCommand{\heu@appendix@chapter}{s o m o}{%
  \IfBooleanT{#1}%
  {
    \phantomsection
    \IfValueTF{#2}{
      \markboth{#2}{#2}
    }{
      \markboth{#3}{#3}
    }
```

```
    \ifthenelse%
    {\equal{#3}{\cabstractcname}}%
    {\relax}
    {\addcontentsline{toc}{chapter}{\texorpdfstring{\ifheu@arialtitle\sffamily\heiti\el

    \IfValueT{#4}{\addcontentsline{toe}{chapter}{\texorpdfstring{\bfseries #4}{#4}}}
    \heu@chapter*{#3}
  }
}
\else
```

添加英文版附表格式

```
\NewDocumentCommand{\heu@appendix@chapter}{s o m o}{%
  \IfBooleanT{#1}%
  {
    \phantomsection
    \IfValueTF{#2}{
      \markboth{#2}{#2}
    }{
      \markboth{#3}{#3}
    }
    \ifthenelse%
      {\equal{#3}{\eabstractcname}}%
      {\addcontentsline{toc}{chapter}{\texorpdfstring{\ifheu@arialtitle\sffamily\heiti\
      {\addcontentsline{toc}{chapter}{\texorpdfstring{\ifheu@arialtitle\sffamily\heiti\
    \IfValueT{#4}{\addcontentsline{toe}{chapter}{\texorpdfstring{\bfseries #4}{#4}}}
    \heu@chapter*{#3}
  }
}
\fi
\newcommand{\BiAppChapter}[2]     % 该附录命令适用于有章节的完整附录
{\phantomsection
 \chapter{#1}
```

此处添加保护选项添加`\texorpdfstring`命令去除书签中带有格式时的警告

```
 \addcontentsline{toe}{chapter}{\texorpdfstring{\bfseries \xiaosi Appendix \thechapter~
}
```

设置章节命令。s: 星号，表示在目录中出不出现序号。m: 必须要有的选项，中文章节名称也即目录中名称，页眉中名称，书签中的名称。o: 可选内容，没有就默认是正文章节，如果有，则是英文目录中显示的内容。

\chapter

```
\let\heu@chapter\chapter
\RenewDocumentCommand{\chapter}{s o m o}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi\phantomsection%
  \IfBooleanTF{#1}%
  {%    if \chapter*
    \IfNoValueTF{#2}%
    {\heu@chapter*{#3}}%
    {\heu@chapter*[#2]{#3}}%
```

```
    \IfValueT{#4}{%
```

此处添加保护选项

```
            \addcontentsline{toe}{chapter}{\texorpdfstring{\bfseries #4}{#4}}
    }
  }%
  {%    if \chapter
    \IfNoValueTF{#2}%
    {\heu@chapter{#3}}%
    {\heu@chapter[#2]{#3}}%
    \IfValueT{#4}{%
```

此处需删除章节的空白此处添加保护选项

```
            \addcontentsline{toe}{chapter}{\texorpdfstring{\bfseries\relax Chapter \the
    }
  }
}
```

\section

```
\let\heu@section\section
\RenewDocumentCommand\section{s o m o}{
  \IfBooleanTF{#1}%
  {%    if \section*
    \IfNoValueTF{#2}%
    {\heu@section*{#3}}%
    {\heu@section*[#2]{#3}}%
    \IfValueT{#4}{%
      \addcontentsline{toe}{section}{#4}
    }
  }%
  {%    if \section
    \IfNoValueTF{#2}%
    {\heu@section{#3}}%
    {\heu@section[#2]{#3}}%
    \IfValueT{#4}{%
```

此处需删除章节的空白

```
    \addcontentsline{toe}{section}{\protect\numberline{\csname thesection\endcsname}\ig
    }
  }
}
```

\subsection

```
\let\heu@subsection\subsection
\RenewDocumentCommand\subsection{s o m o}{
  \IfBooleanTF{#1}%
  {%    if \subsection*
    \IfNoValueTF{#2}%
    {\heu@subsection*{#3}}%
    {\heu@subsection*[#2]{#3}}%
    \IfValueT{#4}{%
```

```
      \addcontentsline{toe}{subsection}{#4}
    }
  }%
  {%    if \subsection
    \IfNoValueTF{#2}%
    {\heu@subsection{#3}}%
    {\heu@subsection[#2]{#3}}%
    \IfValueT{#4}{%
```

此处需删除章节的空白添加\ignorespaces 选项，矫正英文目录多出一个空白而无法对其的 bug

```
    \addcontentsline{toe}{subsection}{\protect\numberline{\csname thesubsection\endcsna
    }
  }
}
```

\subsubsection

```
\let\heu@subsubsection\subsubsection
\RenewDocumentCommand\subsubsection{s o m o}{
  \IfBooleanTF{#1}%
  {%    if \subsubsection*
    \IfNoValueTF{#2}%
    {\heu@subsubsection*{#3}}%
    {\heu@subsubsection*[#2]{#3}}%
    \IfValueT{#4}{%
      \addcontentsline{toe}{subsubsection}{#4}
    }
  }%
  {%    if \subsubsection
    \IfNoValueTF{#2}%
    {\heu@subsubsection{#3}}%
    {\heu@subsubsection[#2]{#3}}%
    \IfValueT{#4}{%
```

此处需删除章节的空白添加\ignorespaces 选项，矫正英文目录多出一个空白而无法对其的 bug

```
    \addcontentsline{toe}{subsubsection}{\protect\numberline{\csname thesubsubsection\e
    }
  }
}
```

4.5.9 定义封面

封面信息。

```
\def\heu@def@term#1{%
  \define@key{heu}{#1}{\csname #1\endcsname{##1}}
  \expandafter\gdef\csname #1\endcsname##1{%
    \expandafter\gdef\csname heu@#1\endcsname{##1}}
  \csname #1\endcsname{}}

```

```
\heu@def@term{statesecrets} % 密级
\heu@def@term{cnumber} % 编号
\heu@def@term{natclassifiedindex}  % 国内图书分类号
\heu@def@term{intclassifiedindex}  % 国际图书分类号
\heu@def@term{ctitlecover} % 中文标题封面
\heu@def@term{ctitle} % 中文标题
\heu@def@term{csubtitle} % 中文副标题
\heu@def@term{cxueke} % 中文学科
\heu@def@term{cauthor} % 中文作者
\heu@def@term{csupervisor} % 中文导师
\heu@def@term{cprofessional}% 职称
\heu@def@term{csupervisorcaffil}% 导师单位
\heu@def@term{cassosupervisor} % 中文副导师
\heu@def@term{ccosupervisor}% 中文联合导师
\heu@def@term{creviewer}% 论文主审人
\heu@def@term{caffil}% 中文院系
\heu@def@term{csubject}% 中文专业
\heu@def@term{csubmitdate}% 论文提交日期
\heu@def@term{coralexdate}% 论文答辩日期
\heu@def@term{cdate}% 论文封面日期
```

添加博后封皮选项

```
\heu@def@term{cpositionname}% 博士后工作站名称
\heu@def@term{cfinishdate}% 完成日期
\heu@def@term{cstartdate}% 开始日期
\heu@def@term{cenddate}% 博士后期满时间

\heu@def@term{cstudentid}% 学号
\heu@def@term{cstudenttype}% 学生类型
\heu@def@term{ctitleone}% 第一行标题
\heu@def@term{ctitletwo}% 第二行标题

\heu@def@term{etitle} % 英文标题
\heu@def@term{esubtitle} % 英文标题
\heu@def@term{exueke} % 英文学科
\heu@def@term{eauthor} % 英文作者
\heu@def@term{esupervisor} % 英文导师
\heu@def@term{eassosupervisor} % 英文副导师
\heu@def@term{ecosupervisor} % 英文联合导师
\heu@def@term{eaffil}% 英文院系
\heu@def@term{esubject}% 英文学科
\heu@def@term{esubmitdate}% 英文论文提交日期
\heu@def@term{eoralexdate}% 英文论文答辩日期
\heu@def@term{edate}% 英文日期
\heu@def@term{estudenttype}% 英文学生类型

\newcommand{\heu@@cabstract}[1]{\long\gdef\heu@cabstract{#1}}
\newenvironment{cabstract}{\Collect@Body\heu@@cabstract}{}
\newcommand{\heu@@eabstract}[1]{\long\gdef\heu@eabstract{#1}}
\newenvironment{eabstract}{\Collect@Body\heu@@eabstract}{}
```

```
\def\heu@parse@keywords#1{
  \define@key{heu}{#1}{\csname #1\endcsname{##1}}
  \expandafter\gdef\csname heu@#1\endcsname{}
  \expandafter\gdef\csname #1\endcsname##1{
    \@for\reserved@a:=##1\do{
      \expandafter\ifx\csname heu@#1\endcsname\@empty\else
        \expandafter\g@addto@macro\csname heu@#1\endcsname{%
          \ignorespaces\csname heu@#1@separator\endcsname}
      \fi
      \expandafter\expandafter\expandafter\g@addto@macro%
        \expandafter\csname heu@#1\expandafter\endcsname\expandafter{\reserved@a}}}}
\heu@parse@keywords{ckeywords}
\heu@parse@keywords{ekeywords}
\def\heusetup{\kvsetkeys{heu}}
⟨/bookcls⟩
```

定义封面中用到的词汇。

```
⟨*bookcfg⟩
\ifheu@postdoc
\gdef\heu@cxueweishort{博}
\gdef\heu@exuewei{Doctor}
\gdef\heu@exueweier{D.Eng}% 英文学位名称缩写
\gdef\heu@cxuewei{\heu@cxueweishort 士}
\gdef\heu@cdegree{\heu@cxueke\heu@cxuewei}
\gdef\heu@edegree{\heu@exuewei \ of \heu@exueke}
\def\heu@cauthortitle{\heu@cxueweishort 士研究生}
\fi

```

添加博后选项

```

\ifheu@doctor
\gdef\heu@cxueweishort{博}
\gdef\heu@exuewei{Doctor}
\gdef\heu@exueweier{D.Eng}% 英文学位名称缩写
\gdef\heu@cxuewei{\heu@cxueweishort 士}
\gdef\heu@cdegree{\heu@cxueke\heu@cxuewei}
\gdef\heu@edegree{\heu@exuewei \ of \heu@exueke}
\def\heu@cauthortitle{\heu@cxueweishort 士研究生}
\fi

\ifheu@master
\gdef\heu@cxueweishort{硕}
\gdef\heu@exuewei{Master}
\gdef\heu@exueweier{M.Eng}% 英文学位名称缩写
\gdef\heu@cxuewei{\heu@cxueweishort 士}
\gdef\heu@cdegree{\heu@cxueke\heu@cxuewei}
\gdef\heu@edegree{\heu@exuewei \ of \heu@exueke}
\def\heu@cauthortitle{\heu@cxueweishort 士研究生}
\fi

```

```
\ifheu@bachelor
\gdef\heu@cxuewei{学士}
\fi

\def\heu@postdoc@classifiedindex{分类号}
\def\heu@postdoc@secretlevel{密级}
\def\heu@postdoc@number{编号}
\def\heu@postdoc@UDC{U\hspace{0.4\ccwd}D\hspace{0.4\ccwd}C：}
\def\heu@stage@opening{开题}
\def\heu@stage@midterm{中期}
\def\heu@stage@doctype{报告}

```

调整博后封皮中字间距

```
\def\heu@postdoc@documenttitle{博\hspace{0.58\ccwd} 士\hspace{0.58\ccwd} 后\hspace{0.58\
    研\hspace{0.58\ccwd} 究\hspace{0.58\ccwd} 工\hspace{0.58\ccwd} 作\hspace{0.58\ccwd}
    报\hspace{0.58\ccwd} 告}
\def\heu@postdoc@documenttitlenospace{博士后研究工作报告}
\def\heu@postdoc@finishdate{工作完成日期}
\def\heu@postdoc@submitdate{报告提交日期}
\def\heu@postdoc@authorname{博士后姓名}
\def\heu@postdoc@supervisor{合作导师}
\def\heu@postdoc@positionname{流动站名称}
\def\heu@postdoc@major{专业名称}
\def\heu@postdoc@startdate{研究工作起始时间}
\def\heu@postdoc@enddate{研究工作期满时间}
\def\heu@postdoc@separator{：}
```

添加本科开题封皮中 title 定义

```
\def\heu@bachelor@cxuewei{本科生}
\def\heu@bachelor@cthesisname{毕业论文}
\def\heu@bachelor@cstudenttitle{学生姓名}
\def\heu@bachelor@cstudentidtitle{学\hspace{2\ccwd} 号}
\def\heu@bachelor@secretlevel{密\hspace{2\ccwd} 级}
\def\heu@bachelor@ccolegetitle{所在学院}
\def\heu@bachelor@cmajortitlecov{专业名称}
\def\heu@bachelor@cmajortitlesec{所在专业}
\def\heu@bachelor@csupervisortitle{指导教师}
\def\heu@bachelor@cprofessionaltitle{职称}
\def\heu@bachelor@cthesistitle{题目}
\def\heu@bachelor@cdatetitle{日\hspace{2\ccwd} 期}
\def\heu@bachelor@caffiltitle{学院（系）名称}
\def\heu@bachelor@cupervisoraffiltitle{所在单位}
\def\heu@bachelor@caffiltitleqd{青岛学院}
\def\heu@bachelor@caffiltitleya{烟台学院}
⟨/bookcfg⟩
```

此处添加青岛校区设置

```
⟨*bookcfg⟩
\newcommand{\heu@bachelor@teachercomment}{指导教师评语：}
\newcommand{\heu@bachelor@teachersign}{指导教师签字：}
```

```
\newcommand{\heu@bachelor@checkdate}{检查日期：}

\def\heu@cthesistitleprefix{题\hspace{\ccwd} 目}
\def\heu@cthesisname{学位论文}
\def\heu@cschoolname{哈尔滨工程大学}
⟨/bookcfg⟩
```

此处添加校区设置

```
⟨*bookcfg⟩
\def\heu@qingdaocampus{（青岛）}
\def\heu@yantaicampus{（烟台）}
\def\heu@harbincampus{（哈尔滨）}

\def\heu@cschoolnametitle{学位授予单位}
\def\heu@csubmitdatatitle{论文提交日期}
\def\heu@coralexdatetitle{论文答辩日期}
\def\heu@caffiltitle{所在单位}
\def\heu@ccolegetitle{所在学院}
\def\heu@csubjecttitle{学科、专业}
\def\heu@cprodrgreetitle{学位类别}
\def\heu@cprosubjecttile{专业类别}
\def\heu@cdegreetitle{申请学位}
\def\heu@csupervisortitle{指导教师}
\def\heu@cassosupervisortitle{副导师}
\def\heu@ccosupervisortitle{校外导师}
\def\heu@creviewertitle{学位论文主审人}
\def\heu@title@csep{：}
\def\heu@eauthortitle{Candidate}
\def\heu@esupervisortitle{Supervisor}
\def\heu@eassosupervisortitle{Associate Supervisor}
\def\heu@ecosupervisortitle{Co Supervisor}
\def\heu@edegreetitle{Academic Degree Applied for}
\def\heu@esubjecttitle{Specialty}
\def\heu@eaffiltitle{College}
\def\heu@ecategorytitle{Professional category}
\def\heu@esubmitdatetitle{Date of Submission}
\def\heu@eoralexdatetitle{Date of Oral Examination}
\def\heu@eschoolnametitle{University}
\def\heu@eschoolname{Harbin Engineering University}
\def\heu@title@esep{: }
\def\heu@natclassifiedindextitle{分类号}
\def\heu@internatclassifiedindextitle{ＵＤＣ}
\def\heu@secretlevel{密级}
\def\heu@cnumbertitle{编号}
```

专业学位论文扉页研究类型选项

```
\ifheu@productdevelopment
  \def\heu@searcetype{（{\wuhao\SquareSolid} 产品设计\hspace{0.5\ccwd}{\wuhao\Square}
  工程规划\hspace{0.5\ccwd}{\wuhao\Square} 工程设计\hspace{0.5\ccwd}{\wuhao\Square}
  应用研究）}
\fi
```

```
\ifheu@projectdplanning
  \def\heu@searcetype{（{\wuhao\Square} 产品设计\hspace{0.5\ccwd}{\wuhao\SquareSolid} 工程规划\hspace{0.5\ccwd}{\wuhao\Square} 工程设计\hspace{0.5\ccwd}{\wuhao\Square} 应用研究）}
\fi
\ifheu@projectdesign
  \def\heu@searcetype{（{\wuhao\Square} 产品设计\hspace{0.5\ccwd}{\wuhao\Square} 工程规划\hspace{0.5\ccwd}{\wuhao\SquareSolid} 工程设计\hspace{0.5\ccwd}{\wuhao\Square} 应用研究）}
\fi
\ifheu@applicationresearch
  \def\heu@searcetype{（{\wuhao\Square} 产品设计\hspace{0.5\ccwd}{\wuhao\Square} 工程规划\hspace{0.5\ccwd}{\wuhao\Square} 工程设计\hspace{0.5\ccwd}{\wuhao\SquareSolid} 应用研究）}
\fi

\def\heu@conclusion@ctitle{结论}

\def\heu@conclusion@etitle{Conclusions}
\def\heu@bibname@etitle{References}

\def\heu@appendix@ctitle{附录}
\def\heu@appendix@etitle{Appendix}

\def\heu@acknowledgement@ctitle{致谢}
\def\heu@acknowledgement@etitle{Acknowledgements}

\ifheu@postdoc
\def\heu@resume@ctitle{博士后个人简历}
\else
\def\heu@resume@ctitle{个人简历}
\fi
```

添加博后封皮 title 的选项

```
\def\heu@correspondingaddr@ctitle{永久通讯地址}

\def\heu@resume@etitle{Resume}
\def\heu@authorization@ctitle{学位论文原创性声明}
\def\heu@authorization@etitle{Statement of copyright and Letter of authorization}

\newcommand{\heu@authorsig}{作者（签字）：}
\newcommand{\heu@teachersig}{导师（签字）：}
\newcommand{\heu@frontdate}{日期：}
\newcommand{\heu@denotation@ctitle}{物理量名称及符号表}
\newcommand{\heu@denotation@etitle}{List of physical quantity and symbol}
\newcommand{\heu@authorizationtitle}{学位论文授权使用声明}

\newcommand{\heu@qingdao@schoolbottommark}{
  \begin{center}
```

```
      \songti\sanhao\textbf{哈工程（青岛）制}
    \end{center}
    \begin{center}
      \songti\sanhao\textbf{二〇二〇年三月}
    \end{center}
}
\newcommand{\heu@yantai@schoolbottommark}{
     \begin{center}
         \songti\sanhao\textbf{哈工程（烟台）制}
     \end{center}
     \begin{center}
         \songti\sanhao\textbf{二〇二〇年三月}
     \end{center}
}
\newcommand{\heu@harbin@schoolbottommark}{
    \begin{center}
      \songti\sanhao\textbf{研究生院制}
    \end{center}
    \begin{center}
      \songti\sanhao\textbf{二〇一四年九月}
    \end{center}
}
\newcommand{\heu@harbin@bachelor@schoolbottommark}{
    \begin{center}
      \songti\xiaoer\textbf{哈尔滨工程大学本科生院制}
    \end{center}
}

\newcommand{\heu@doctor@midterm@note}{
\thispagestyle{empty}
{\begin{center}
\songti\sanhao\textbf{说\hspace{4\ccwd} 明}
\end{center}
}
\vspace{1cm}\songti\sihao[2]
\begin{enumerate}[itemsep=-4bp]
\item 博士研究生学位论文中期报告一般在入学后的第五学期末完成。
\item 此报告由学生本人填写，请导师签字后提交一份给检查小组。答辩结束后由检查组长签署意见。
\item 报告经中期报告检查组长签字同意后，由学科部保存，以备论文答辩时参考。研究生院学生培养处将对研究生的学位论文中期报告进行抽查。
\item 报告统一用 A4 纸打印。
\end{enumerate}
}
%  \end{macrocode}
%  \changes{v1.0.5}{2026/03/01}{根据 2026.02.28 哈尔滨工程大学最新要求修改原创型声明和授权使用声明}
%  \begin{macrocode}
\newcommand{\heu@authorizationtext}{%
{\wuhao 本人完全了解学校保护知识产权的有关规定，即研究生在校攻读学位期间论文工作的知识产权属于 \heu@cschoolname 。 \heu@cschoolname 有权保留并向国家有关部门或机构送交论文的复
```

```
印件。本人允许 \heu@cschoolname 将论文的部分或全部内容编入有关数据库进行检索，可采用影印、缩印或扫描等复制手段保存和汇编本学位论文，可以公布论文的全部内容。同时本人保证毕业后结合学位论文研究课题再撰写的论文一律注明作者第一署名单位为 \heu@cschoolname 。涉密学位论文待解密后适用本声明。}

{\wuhao 本论文（{\xiaoliu{\Square}} 在授予学位后即可 \hspace{0.1em} {\xiaoliu{\Square}} 在授予学位 24 个月（含）后 \hspace{0.1em} {\xiaoliu{\Square}} 全部内容在授予学位 \underline{\hspa
月后【限填 24 个月以上，须学校审批】\hspace{0.1em} {\xiaoliu{\Square}} 部分内容在授予学位 \underline{\hspace{1.5em}} 个月后【须学校审批】\hspace{0.1em} {\xiaoliu{\Square}} 解密后）由 \heu@cschoolname 送交有关部门进行保存汇编、公布使用等。}}

\newcommand{\heu@declarename@bachelor}{\heu@cschoolname 本科毕业设计（论文）原创性声明}
\newcommand{\heu@authorizationtext@bachelor}{%
{\wuhao 本人郑重声明:在\heu@cschoolname 攻读学士学位期间,所提交的毕业设计(论文)《\heu@ctitle》,是本人在导师指导下独立进行研究工作所取得的成果。对本文的研究工作做出重要贡献的个人和集体，均已在文中以明确方式注明，其它未注明部分不包含他人已发表或撰写过的研究成果，不存在购买、由他人代写、剽窃和伪造数据等作假行为。本人完全意识到本声明的法律结果由本人承担。}}
\newcommand{\heu@declarename}{学位论文原创性声明}
\newcommand{\heu@declaretext}{%
{\wuhao 本人郑重声明：本论文的所有工作，是在导师的指导下，由作者本人独立完成的。有关观点、方法、数据和文献的引用已在文中指出，并与参考文献相对应。本论文在撰写过程中，所有由人工智能技术辅助生成的内容，作者均已明确标注，并对其真实性、准确性及合规性承担最终责任。论文中所有研究设计、数据分析、核心观点、结论及创新性研究成果均为作者独立思考完成，不存在将人工智能技术或工具标注为作者或完成人的情况。除文中已注明引用的内容外，本论文不包含任何其他个人或集体已经公开发表的作品成果。对本文的研究做出重要贡献的个人和集体，均已在文中以明确方式标明。本人完全意识到本声明的法律结果由本人承担。}}

\ifheu@english
\newcommand{\heu@publication@ctitle}{Author's Publications}
\else
\ifheu@postdoc
\newcommand{\heu@publication@ctitle}{博士后期间的研究成果}
\else
\newcommand{\heu@publication@ctitle}{攻读\heu@cxuewei 学位期间发表的论文及其他成果}
\fi
\fi
\newcommand{\heu@doctorpublication@ctitle}{博士生期间的研究成果}
```

博后和英文版附录 title

```
\newcommand{\heu@datefill}{\hspace{3.0em} 年\hspace{1.5em} 月\hspace{1.5em} 日}
\newcommand{\heu@publication@etitle}{Papers published in the period of Ph.D. education}
\def\heu@index@etitle{Index}
\def\heu@motto{大工至善，大学至臻}
\def\heu@cbraceleft{（}
\def\heu@cbraceright{）}
\def\heu@ebraceleft{(}
\def\heu@ebraceright{)}
```

```
⟨/bookcfg⟩
```

中英文封面。

```
⟨*bookcls⟩
\newlength{\heu@title@width}
\newcommand{\heu@put@title}[2][\heu@title@width]{%
  \begin{CJKfilltwosides}[b]{#1}#2\end{CJKfilltwosides}}

\def\heu@first@titlepage{%
  \ifheu@postdoc\heu@first@titlepage@postdoc\else\ifheu@bachelor\heu@first@titlepage@ba

\def\heu@second@titlepage{%
  \ifheu@postdoc\heu@second@titlepage@postdoc\else\ifheu@bachelor\heu@second@titlepage@

```

博后封皮

```
\newcommand{\heu@first@titlepage@postdoc}{
  {\songti\sihao\renewcommand{\arraystretch}{2}\
    \begin{tabular}{@{}r@{}l@{\hspace{7\ccwd}}r@{}l@{\hspace{3em}}r}
      \heu@postdoc@classifiedindex & \underline{\makebox[6\ccwd]{\heu@natclassifiedinde
      \heu@postdoc@UDC             & \underline{\makebox[6\ccwd]{\heu@intclassifiedinde
    \end{tabular}\renewcommand{\arraystretch}{1}}
  \vspace*{0.5cm}
  \parbox[t][5cm][b]{\textwidth}{\begin{center}\heiti\xiaoer[2]\heu@cschoolname\par
      \heu@postdoc@documenttitle\end{center}}
  \parbox[b][4cm][b]{\textwidth}{
    \songti\xiaoer[2]\renewcommand{\arraystretch}{1.5}
    \begin{center}
      \begin{tabular}{c}
        \makebox[9cm]{\heu@ctitleone}\\
        \hline
        \makebox[9cm]{\heu@ctitletwo}\\
        \hline
      \end{tabular}\renewcommand{\arraystretch}{1}
    \end{center}}
    \begin{center}
      \songti\sihao[2]\centering\heu@cauthor
    \end{center}
  \parbox[b][4.5cm][b]{\textwidth}{
    \songti\sihao[2]\renewcommand{\arraystretch}{1.5}
    \begin{center}
      \begin{tabular}{rl}
        \heu@postdoc@finishdate & \underline{\makebox[6cm]{\heu@cfinishdate}} \\
        \heu@postdoc@submitdate & \underline{\makebox[6cm]{\heu@csubmitdate\hfill}} \\
      \end{tabular}\renewcommand{\arraystretch}{1}
    \end{center}
    }
    \vfill
    {\songti\sihao[2]
     \begin{center}
       \heu@cschoolname%
```

```
      %\ifheu@harbin
      %\heu@harbincampus
      %\fi
      \ifheu@yantai
      \heu@yantaicampus
      \fi
      \ifheu@qingdao
      \heu@qingdaocampus
      \fi
    \end{center}
    \begin{center}
      \heu@cdate
    \end{center}}
}

```

搏后封二

```
\newcommand{\heu@second@titlepage@postdoc}{
  \ifthenelse%
  {\equal{\heu@fontset}{siyuan}}%
  {\xiaosi[1]\vspace*{0.65em}}%
  {\xiaosi[1]\textcolor[rgb]{1,1,1}{\songti{\heu@motto}}}%
  \vspace*{1.2cm}
  \begin{center}
    \parbox[t][8cm][t]{\textwidth}{
      \begin{center}\songti\sanhao[2]\heu@ctitlecover\end{center}
      \vspace*{16bp}
      \begin{center}\sihao[2]\MakeUppercase{\heu@etitle}\end{center}
    }
    \parbox[t][5cm][t]{\textwidth}{
      \begin{center}\songti\sihao[2]
        \begin{tabularx}{0.6\linewidth}{l@{\heu@postdoc@separator}l}
          \makebox[9em][s]{\heu@postdoc@authorname}   & \heu@cauthor \\
          \makebox[9em][s]{\heu@postdoc@supervisor}   & \heu@csupervisor \\
          \makebox[9em][s]{\heu@postdoc@positionname} & \heu@cpositionname \\
          \makebox[9em][s]{\heu@postdoc@major}        & \heu@csubject \\
          \makebox[9em][s]{\heu@postdoc@startdate}    & \heu@cstartdate \\
          \makebox[9em][s]{\heu@postdoc@enddate}      & \heu@cenddate \\
        \end{tabularx}
      \end{center}
    }
  \end{center}
  \vfill
  {\songti\sihao[2]
    \begin{center}
      \heu@cschoolname%
      %\ifheu@harbin
      %\heu@harbincampus
      %\fi
      \ifheu@yantai
```

```
      \heu@yantaicampus
      \fi
      \ifheu@qingdao
      \heu@qingdaocampus
      \fi
    \end{center}
    \begin{center}
      \heu@cdate
    \end{center}}}

```

本科论文封面

```
\newcommand{\heu@first@titlepage@bachelor}{
\ifthenelse%
  {\equal{\heu@fontset}{siyuan}}%
  {\xiaosi[1]\textcolor[rgb]{1,1,1}{\songti{\heu@motto}}}%
  \vspace*{0.1cm}
  \begin{center}
  {\songti\xiaosi
    \begin{tabular}{@{\qquad\qquad}@{}r@{}l@{}l@{\qquad}}
     &  & \\
     &  &
    \end{tabular}}\hfill
  {\songti\xiaosi
    \begin{tabular}{@{}r@{: }l@{}l@{\qquad}}
    \heu@bachelor@cstudentidtitle & \underline{\makebox[2.5cm]{\heu@cstudentid}}   & \\
    \heu@bachelor@secretlevel     & \underline{\makebox[2.5cm]{\heu@statesecrets}} &
    \end{tabular}}
  \parbox[t][2.0cm][t]{\textwidth}{\begin{center} \end{center} }
  \parbox[t][1.2cm][t]{\textwidth}{
    \begin{center}\xiaoer[1]\songti{\heu@cschoolname\heu@bachelor@cxuewei\heu@bachelor@
  \parbox[t][1.2cm][t]{\textwidth}{}
  \parbox[t][7.8cm][t]{\textwidth}{%
    \begin{center}\erhao\heiti\heu@ctitlecover\end{center}
    \ifheu@subtitle\begin{center}\hspace{-4em}\xiaoer\heiti\pozhehao\heu@csubtitle\end{
  \parbox[t][7.4cm][t]{\textwidth}{
  {\xiaosan
    \setlength{\heu@title@width}{7em}
    \begin{center} \renewcommand{\arraystretch}{1.62} \songti
    \begin{tabular}{l@{\heu@title@csep}l}
    {\songti\heu@put@title{\heu@bachelor@caffiltitle}}      & \heu@caffil \\
    {\songti\heu@put@title{\heu@bachelor@cmajortitlecov}}   & \heu@csubject \\
    {\songti\heu@put@title{\heu@bachelor@cstudenttitle}}    & \heu@cauthor\\
    {\songti\heu@put@title{\heu@bachelor@csupervisortitle}} & \heu@csupervisor\quad\heu
    \end{tabular} \renewcommand{\arraystretch}{1}
    \end{center}}}
  \parbox[t][1.4cm][t]{\textwidth}{
  \begin{center}\kaishu\xiaoer{\heu@cschoolname}\end{center}}
  {\songti\xiaosan{\heu@cdate}}
  \end{center}
```

```
}

```

本科封二

```
\newcommand{\heu@second@titlepage@bachelor}{
  \vspace*{0.5cm}
  {\songti\xiaosi
    \begin{tabular}{@{\qquad\qquad}@{}r@{}l@{}l@{\qquad}}
    &  & \\
    &  &
    \end{tabular}}\hfill
  {\songti\xiaosi
    \begin{tabular}{@{}r@{: }l@{}l@{\qquad}}
    \heu@bachelor@cstudentidtitle & \underline{\makebox[2.5cm]{\heu@cstudentid}}   & \\
    \heu@bachelor@secretlevel     & \underline{\makebox[2.5cm]{\heu@statesecrets}} &
    \end{tabular}}
  \vspace*{1.5cm}
  \begin{center}
    \parbox[t][8.0cm][t]{\textwidth}
    {\heiti\xiaosan
        \begin{center}\erhao\heiti\heu@ctitlecover\end{center}
      \ifheu@subtitle\begin{center}\hspace{-4em}\xiaoer\heiti\pozhehao\heu@csubtitle\en
      \begin{center}\erhao{\heu@etitle}\end{center}
      \begin{center}\xiaoer\textbf{\ifheu@subtitle\heu@title@esep\heu@esubtitle\fi}\enc
    }
  \parbox[t][9.8cm][b]{\textwidth}
  {\heiti\xiaosan
    \setlength{\heu@title@width}{6em}
    \begin{center} \renewcommand{\arraystretch}{1.5} \songti
    \begin{tabular}{l@{\heu@title@csep}l}
    \textbf{\heu@put@title{\heu@bachelor@cstudenttitle}}        & \heu@cauthor\\
    \textbf{\heu@put@title{\heu@bachelor@ccolegetitle}}         & \heu@caffil\\
    \textbf{\heu@put@title{\heu@bachelor@cmajortitlesec}}       & \heu@csubject\\
    \textbf{\heu@put@title{\heu@bachelor@csupervisortitle}}     & \heu@csupervisor\\
    \textbf{\heu@put@title{\heu@bachelor@cprofessionaltitle}}   & \heu@cprofessional\\
    \textbf{\heu@put@title{\heu@bachelor@cupervisoraffiltitle}} & \heu@csupervisorcaffi
    \textbf{\heu@put@title{\heu@csubmitdatatitle}}              & \heu@csubmitdate\\
    \textbf{\heu@put@title{\heu@coralexdatetitle}}              & \heu@coralexdate\\
    \textbf{\heu@put@title{\heu@cschoolnametitle}}              & \heu@cschoolname
    \end{tabular} \renewcommand{\arraystretch}{1}
    \end{center}
  }
  \end{center}
}

```

硕博论文封面

```
\newcommand{\heu@first@titlepage@other}{
\ifthenelse%
{\equal{\heu@fontset}{siyuan}}%
{\xiaosi[1]\textcolor[rgb]{1,1,1}{\songti{\heu@motto}}}%
```

```
\vspace*{0.1cm}
\begin{center}
  {\songti\xiaosi
    \begin{tabular}{@{\qquad}@{}r@{: }l@{}l@{\qquad}}
    \heu@natclassifiedindextitle      & \underline{\makebox[2.5cm]{\heu@natclassifiedin
    \heu@internatclassifiedindextitle & \underline{\makebox[2.5cm]{\heu@intclassifiedin
    \end{tabular}}\hfill
  {\songti\xiaosi
    \begin{tabular}{@{}r@{: }l@{}l@{\qquad}}
    \heu@secretlevel  & \underline{\makebox[2.5cm]{\heu@statesecrets}} & \\
    \heu@cnumbertitle & \underline{\makebox[2.5cm]{\heu@cnumber}}      &
    \end{tabular}}

  \vspace*{2.0cm}
  \parbox[t][1.2cm][t]{\textwidth}{\xiaoer[1]
  \begin{center}\xiaoer[1]\songti{\heu@cdegree\heu@cthesisname}\end{center}}
% 修改第二行标题：学术类型
  \parbox[t][1.2cm][t]{\textwidth}{\quad}
    \parbox[t][6.0cm][t]{\textwidth}{%
  \begin{center}\erhao\heiti\heu@ctitlecover\end{center}
    \ifheu@subtitle\begin{center}\hspace{-4em}\xiaoer\heiti\pozhehao\heu@csubtitle\end{
    \parbox[t][7.4cm][t]{\textwidth}{
    {\xiaosan
    \setlength{\heu@title@width}{7em}
    \begin{center} \renewcommand{\arraystretch}{1.62} \songti
      \begin{tabular}{l@{\heu@title@csep}l}
      {\songti\heu@put@title{\heu@cauthortitle}}           & \heu@cauthor\\
      {\songti\heu@put@title{\heu@csupervisortitle}}       & \heu@csupervisor\\
      \ifx\heu@cassosupervisor\@empty\else%
        {\songti\heu@put@title{\heu@cassosupervisortitle}} & \heu@cassosupervisor\\
      \fi
      \ifheu@profdegree%
        {\songti\heu@put@title{\heu@ccosupervisortitle}}   & \heu@ccosupervisor \\
        {\songti\heu@put@title{\heu@cprodrgreetitle}}      & \heu@cstudenttype \\
      \else%
        {\songti\heu@put@title{\heu@csubjecttitle}}        & \heu@csubject \\
      \fi
      \ifx\heu@creviewer\@empty\else%
        {\songti\heu@put@title{\heu@creviewertitle}}       & \heu@creviewer
      \fi
      \end{tabular} \renewcommand{\arraystretch}{1}
    \end{center}}}
    \parbox[t][3.0cm][b]{\textwidth}{
  \begin{center}\kaishu\xiaoer{\heu@cschoolname}\end{center}}
  {\songti\xiaosan{\heu@cdate}}
  \end{center}
}

```

硕博论文内封

```
\newcommand{\heu@second@titlepage@other}{
  \begin{center}
    {\songti\xiaosi
      \begin{tabular}{@{\qquad}@{}r@{: }l@{}l@{\qquad}}
    \heu@natclassifiedindextitle      & \underline{\makebox[2.5cm]{\heu@natclassifiedin
    \heu@internatclassifiedindextitle & \underline{\makebox[2.5cm]{\heu@intclassifiedin
    \end{tabular}}\hfill
    {\songti\xiaosi
      \begin{tabular}{@{}r@{: }l@{}l@{\qquad}}
    \heu@secretlevel & \underline{\makebox[2.5cm]{\heu@statesecrets}} &   \\
    \heu@cnumbertitle & \underline{\makebox[2.5cm]{\heu@cnumber}} &
    \end{tabular}}
  \parbox[t][2.0cm][t]{\textwidth}{\begin{center} \end{center}}
    \parbox[t][1.2cm][t]{\textwidth}{\xiaoer[1]
  \begin{center}\songti\heu@cdegree\heu@cthesisname\end{center}}
% 修改第二行标题：学术类型
  \parbox[t][1.2cm][t]{\textwidth}{\quad}
    \parbox[t][5cm][t]{\textwidth}{\erhao
  \begin{center}\heiti\heu@ctitlecover\end{center}
    \ifheu@subtitle\hspace{-4em}\xiaoer\heiti\pozhehao\heu@csubtitle\fi
    \ifheu@profdegree{\begin{center}\sihao\songti\par\heu@searcetype\end{center}}\else{
    \parbox[t][9.8cm][b]{\textwidth}
    {\sihao
      \setlength{\heu@title@width}{6em}
      \begin{center} \renewcommand{\arraystretch}{1.5} \songti
    \begin{tabular}{l@{\heu@title@csep}l}
      \textbf{\heu@put@title{\heu@cauthortitle}}           & \heu@cauthor\\
      \textbf{\heu@put@title{\heu@csupervisortitle}}       & \heu@csupervisor\\
      \ifx\heu@cassosupervisor\@empty\else%
        \textbf{\heu@put@title{\heu@cassosupervisortitle}} & \heu@cassosupervisor\\
      \fi
      \ifheu@profdegree%
        \textbf{\heu@put@title{\heu@ccosupervisortitle}}   & \heu@ccosupervisor\\
        \textbf{\heu@put@title{\heu@cprosubjecttile}}      & \heu@csubject\\
        \textbf{\heu@put@title{\heu@ccolegetitle}}         & \heu@caffil\\
      \else%
        \textbf{\heu@put@title{\heu@cdegreetitle}}         & \heu@cdegree\\
        \textbf{\heu@put@title{\heu@csubjecttitle}}        & \heu@csubject\\
        \textbf{\heu@put@title{\heu@caffiltitle}}          & \heu@caffil\\
      \fi
      \textbf{\heu@put@title{\heu@csubmitdatatitle}}       & \heu@csubmitdate\\
      \textbf{\heu@put@title{\heu@coralexdatetitle}}       & \heu@coralexdate\\
      \textbf{\heu@put@title{\heu@cschoolnametitle}}       & \heu@cschoolname
    \end{tabular} \renewcommand{\arraystretch}{1}
    \end{center} }
  \end{center}
}
```

硕博论文英文内封

```
\newcommand{\emultiline}[2][c]{\renewcommand{\arraystretch}{1}\begin{tabular}[#1]{@{}l@
```

```
\newcommand{\heu@engcover}{
  {\songti\xiaosi
    \begin{tabular}{@{\qquad}l@{}l@{\qquad}}
    Classified Index:\ \heu@natclassifiedindex & \\
    U.D.C:\ \heu@intclassifiedindex &
    \end{tabular}}\hfill
  {\songti\xiaosi
    \begin{tabular}{@{}r@{ }l@{}l@{\qquad}}
     &  &   \\
     &  &
    \end{tabular}}
  \vspace*{1em}
  \begin{center}
% 工程硕士、非全日制学位论文英文封面第一行文字不同。
  \parbox[t][1.6cm][t]{\textwidth}{\begin{center} \end{center} }
    \parbox[t][1.2cm][t]{\textwidth}{\xiaoer[1]
      \begin{center}
        \ifheu@profdegree A Thesis for the Degree of {\heu@estudenttype}
        \else A Dissertation for the Degree of {\heu@exueweier}
        \fi
      \end{center}}
    \parbox[t][1.2cm][t]{\textwidth}{\xiaoer[1] \qquad}
    \parbox[t][7cm][t]{\textwidth}{%
    \begin{center}\erhao{\heu@etitle}\end{center}
    \begin{center}\xiaoer{\ifheu@subtitle\heu@title@esep\heu@esubtitle\fi}\end{center}}
    {\xiaosi\renewcommand{\arraystretch}{1.3}
      \begin{tabular}{@{}r@{~}l@{}}
        \textbf{\heu@eauthortitle\heu@title@esep}            & \heu@eauthor\\
        \textbf{\heu@esupervisortitle\heu@title@esep}        & \heu@esupervisor\\
        \ifx\heu@eassosupervisor\@empty\else%
          \textbf{\heu@put@title{\heu@eassosupervisortitle}} & \heu@eassosupervisor\\
        \fi
        \ifheu@profdegree%
          \textbf{\heu@eassosupervisortitle\heu@title@esep}  & \heu@eassosupervisor\\
          \textbf{\heu@ecategorytitle\heu@title@esep}        & \heu@esubject\\
          \textbf{\heu@eaffiltitle\heu@title@esep}           & \heu@eaffil\\
        \else%
            \textbf{\heu@edegreetitle\heu@title@esep}        & \heu@edegree\\
          \textbf{\heu@esubjecttitle\heu@title@esep}         & \heu@esubject\\
        \fi
        \textbf{\heu@esubmitdatetitle\heu@title@esep}        & \heu@esubmitdate\\
        \textbf{\heu@eoralexdatetitle\heu@title@esep}        & \heu@eoralexdate\\
        \textbf{\heu@eschoolnametitle\heu@title@esep}        & \heu@eschoolname
      \end{tabular}\renewcommand{\arraystretch}{1}}
  \end{center}
}

```

生成封面。

```
\def\makecover{
```

```
\phantomsection
\pdfbookmark[0]{\heu@ctitle}{ctitle}
\xiaosi[1]%
\begin{titlepage}
  \ifheu@yantai
    \ifheu@bachelor
      \relax
    \else
      \heu@first@titlepage
      \ifheu@library\clearpage\else\cleardoublepage\fi
    \fi
  \else
    \heu@first@titlepage
    \ifheu@library\clearpage\else\cleardoublepage\fi
  \fi
  \heu@second@titlepage
  \ifheu@library\clearpage\else\cleardoublepage\fi
  \ifheu@postdoc
  \relax
  \else
  \ifheu@bachelor
  \relax
  \else
  \phantomsection
  \pdfbookmark[0]{\heu@etitle}{etitle}
  \heu@engcover
  \ifheu@library\clearpage\else\cleardoublepage\fi
  \fi
  \fi
\end{titlepage}
\normalsize
\cleardoublepage}
```

生成独创性声明和授权签字页。

```
\def\heu@authorization@other{%
\ifheu@openright\cleardoublepage\else\clearpage\fi
%  \heu@appendix@chapter*{\heu@authorization@ctitle}[\heu@authorization@etitle]
%  \xiaosi[1.5]\vspace{\baselineskip}
\begin{center}
    \quad\\
    \xiaoer\heiti\heu@cschoolname\\%
    \vspace{0.25\baselineskip}%
    \heu@declarename
\end{center}
\vspace{0.25\baselineskip}
\par\heu@declaretext
\vspace{\baselineskip}
\par\hspace{23em}{\wuhao \heu@authorsig}
\vspace{0.25\baselineskip}
\par\hspace{23em}{\wuhao \heu@frontdate\heu@datefill}
```

```
  \vspace{\baselineskip}
  \begin{center}
      \xiaoer\heiti\heu@cschoolname\\%
      \vspace{0.25\baselineskip}%
      \heu@authorizationtitle
  \end{center}
  \vspace{0.25\baselineskip}
  \par\heu@authorizationtext
  \vspace{\baselineskip}
  \par\hspace{0em}{\wuhao \heu@authorsig} \hspace{17em} {\wuhao \heu@teachersig}
  \vspace{0.5\baselineskip}
  \par\hspace{0em}{\wuhao \heu@frontdate\heu@datefill} \hspace{12.5em}{\wuhao \heu@fron
\def\heu@authorization@bachelor{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \heu@appendix@chapter*{\heu@declarename@bachelor}
  \heu@authorizationtext@bachelor
  \vspace{\baselineskip}
  \par\hspace{6em}\heu@authorsig\hfill\heu@frontdate\heu@datefill}
\NewDocumentCommand{\authorization}{o}{%
  \IfNoValueTF{#1}{%
      \thispagestyle{empty}%
      % 本科不需要原创性和授权页，删掉
      %\ifheu@bachelor\heu@authorization@bachelor\else\heu@authorization@other\fi
      \heu@authorization@other
  }{%
      \ifheu@openright\cleardoublepage\else\clearpage\fi%
      \includepdf[fitpaper=true,pagecommand={%
          \thispagestyle{empty}%
          % 原创性和授权页不加到目录
          %\phantomsection\addcontentsline{toc}{chapter}{\ifheu@bachelor\heu@declarenam
          %\ifheu@doctor%
          %  \addcontentsline{toe}{chapter}{\texorpdfstring{\bfseries \heu@authorizatio
          %\fi%
      }]{#1}%
  }%
}
```

生成摘要和关键字。

```
\newbox\heu@kw
\newcommand\heu@put@keywords[2]{%
  \begingroup
    \setbox\heu@kw=\hbox{#1}
    \noindent\hangindent\wd\heu@kw\hangafter1%
    \box\heu@kw#2\par
  \endgroup}
\newcommand{\makeabstract}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \ifheu@english
  \addtocontents{toc}{~\hfill\textbf{Page}\par}
  \addtocontents{\ext@table}{{\noindent\sihao\textbf{Table}~\hfill\textbf{Page}}\par}%
```

```
  \addtocontents{\ext@figure}{{\noindent\sihao\textbf{Figure}~\hfill\textbf{Page}}\par}
  \fi
```

摘要不加入目录、不在页眉显示

```
%  \heu@appendix@chapter*{\cabstractcname}[\cabstractcname]
  \pagestyle{heu@titleheadings}
  \pagenumbering{Roman}
  \begin{center}
    ~\\
    \xiaoer\heiti\cabstractcname
  \end{center}\par
  \heu@cabstract
  \vskip12bp
  \heu@put@keywords{\ifheu@postdoc\bfseries\fi\heiti\heu@ckeywords@title}{\heu@ckeywords}
```

博后关键字加粗

```
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \clearpage
```

摘要不加入目录、不在页眉显示

```
%  \heu@appendix@chapter*{\eabstractcname}[\eabstractcname]
  \begin{center}
    ~\\
    \xiaoer\textbf{\eabstractcname}
  \end{center}\par
  \heu@eabstract
  \vskip12bp
  \heu@put@keywords{\textbf{Keywords:\enskip}}{\heu@ekeywords}}
```

定义符号表。

```
\newenvironment{denotation}[1][2.5cm]{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \heu@appendix@chapter*{\heu@denotation@ctitle}[\heu@denotation@etitle]
\setcounter{table}{0}
\renewcommand{\thetable}{\arabic{table}}% 使表编号为 1 的格式
  }{\renewcommand{\thetable}{\arabic{chapter}.\arabic{table}}% 使表编号为 7.1 的格式
\setcounter{table}{0}}%
```

定义索引、目录格式

```
\def\heu@starttoc#1{% #1: float type, prepend type name in \listof*** entry.
  \let\oldnumberline\numberline
  \def\numberline##1{\oldnumberline{\csname #1name\endcsname\hskip.4em ##1}}
  \@starttoc{\csname ext@#1\endcsname}
  \let\numberline\oldnumberline}
\def\heu@listof#1#2{% #1: float type
  \clearpage
%  \ifheu@openright\cleardoublepage\else\clearpage\fi
```

前文的图、表清单不列入目录

```
%  \heu@appendix@chapter*[\csname list#1name\endcsname]{\csname list#1name\endcsname}[#
  \begin{center}
```

```
    ~\\
    {\xiaoer\heiti{\csname list#1name\endcsname}}
  \end{center}\par
  \heu@starttoc{#1}}
```

只有英文版用到，且需要 clearpage

```

\renewcommand\listoffigures{\heu@listof{figure}{\listfigureename}}
\renewcommand*\l@figure{\addvspace{6bp}\@dottedtocline{1}{0em}{4em}}
\renewcommand\listoftables{\heu@listof{table}{\listtableename}}
\let\l@table\l@figure
\def\ext@equation{loe}
\def\equcaption#1{%
  \addcontentsline{\ext@equation}{equation}%
                   {\protect\numberline{#1}}}
\newcommand\listofequations{\heu@listof{equation}{\listequationename}}
\let\l@equation\l@figure
```

4.5.10 目录

默认目录是三级，如需要四级目录需要设置 tocfour=true。

```
\setcounter{secnumdepth}{3}
\setcounter{tocdepth}{2}
\ifheu@bachelor\ifheu@tocfour\setcounter{tocdepth}{3}\fi\fi
\renewcommand\tableofcontents{%
\ifheu@openright\cleardoublepage\else\clearpage\fi
\phantomsection
\markboth{\contentsname}{ccontent}
```

工程大学论文要求：目录中的目录位置是空白。

```
%\heu@chapter*{\contentsname}
\begin{center}
  ~\\
  \xiaoer\heiti\contentsname
\end{center}\par
\pdfbookmark[0]{\contentsname}{ccontent}
\normalsize\@starttoc{toc}
\ifheu@doctor
\tableofengcontents
```

有些硕士的范例中带有英文目录，这里添加手动开关 engtoc=true

```
\else
\ifheu@engtoc
\tableofengcontents
\fi
\fi
```

将英文目录嵌入中文目录命令中

```
\ifheu@english
\printnoidxglossary[sort=word,style=engstyle,title={Nomenclature},nopostdot,nonumberlis
\listoftables
```

```
\listoffigures
\fi
}
```

添加英文 toc 控制按照我校要求的目录格式。

```
\ifheu@arialtoc
  \def\heu@toc@font{\sffamily}
\fi
\def\@pnumwidth{4em}% 规定中的提前悬挂
\def\@tocrmarg{\@pnumwidth}
\def\@dotsep{1}
```

此处临时更改一下对齐方式。CTEX 似乎无法应对双语目录。将 leftskip 设置参数至于外侧，以便后续添加可以适应标题长度的`\contentsline` 方法

```
\setlength\@tempdima{4em}%
\patchcmd{\@dottedtocline}{#4}{\csname heu@toc@font\endcsname #4}{}{}
\patchcmd{\@dottedtocline}{\hb@xt@\@pnumwidth}{\hbox}{}{}
\renewcommand*\l@chapter[2]{%
  \ifnum \c@tocdepth >\m@ne
    \addpenalty{-\@highpenalty}%
    %\vskip 4bp \@plus\p@
    \begingroup
      \parindent \z@ \rightskip \@pnumwidth
      \parfillskip -\@pnumwidth
      \leavevmode
      \advance\leftskip\@tempdima
      \hskip -\leftskip
      % numberline is called here, and it uses \@tempdima
      {\ifheu@bachelor\rmfamily\else\csname
        heu@toc@font\endcsname\fi\heiti\ifheu@postdoc\bfseries\fi\ifheu@english\bfserie
```

博后和英文版需要加粗

```
      \leaders\hbox{$\m@th\mkern \@dotsep mu\hbox{.}\mkern \@dotsep mu$}\hfill
      \nobreak{\normalfont\normalcolor #2}\par
      \penalty\@highpenalty
    \endgroup
  \fi}
```

按工程大学标准, 缩小目录中各级标题之间的缩进，使它们相隔一个字符距离，也就是 12pt。

```
\renewcommand*\l@section{\@dottedtocline{1}{1em}{1.8em}}
\renewcommand*\l@subsection{\@dottedtocline{2}{2em}{2.5em}}
\renewcommand*\l@subsubsection{\@dottedtocline{3}{3\ccwd}{3.1em}}
```

英文目录格式。

```
\def\@dotsep{0.75}            % 定义英文目录的点间距
\setlength\leftmargini {0pt}
\setlength\leftmarginii {0pt}
\setlength\leftmarginiii {0pt}
\setlength\leftmarginiv {0pt}
\setlength\leftmarginv {0pt}
\setlength\leftmarginvi {0pt}
```

```
\def\engcontentsname{\bfseries Contents}
\newcommand\tableofengcontents{
```

此处添加英文目录的章标题格式，默认细点

```
  \def\l@chapter{\@dottedtocline{0}{0em}{5em}}% 控制英文目录：细点\@dottedtocline  粗点\@dottedtoclinebold
  \@restonecolfalse
  \chapter*{\engcontentsname  %chapter* 上移一行，避免在 toc 中出现。
  \pdfbookmark[0]{Contents}{econtent}
    \@mkboth{%
  \engcontentsname}{\engcontentsname}}
```

此处临时更改一下对齐方式。CTEX 似乎无法应对双语目录。

```
  \@starttoc{toe}%
  \if@restonecol\twocolumn\fi}
\def\@dotsep{0.75}            % 定义英文目录的点间距
```

目录中附录的章号格式。

```
\ctexset{%
  appendix/number=\ifheu@bachelor\arabic{chapter}\else\Alph{chapter}\fi,
}
```

设置附录、结论、参考文献等格式。

```
\let\heu@appendix\appendix
\renewenvironment{appendix}[2]{%
\heu@appendix
\ifheu@bachelor\renewcommand{\thechapter}{\arabic{chapter}}\fi
\ifheu@openright\cleardoublepage\else\clearpage\fi
\renewcommand{\thechapter}{#1} % 修改附录序号
\ifheu@english
\addcontentsline{toc}{chapter}{\heu@appendix@etitle #1 ~ #2}
\begin{center}
  ~\\
  \xiaoer\heiti\heu@appendix@etitle #1 ~ #2
\end{center}\par
\else
\addcontentsline{toc}{chapter}{\heu@appendix@ctitle #1 ~ #2}
\begin{center}
  ~\\
  \xiaoer\heiti\heu@appendix@ctitle #1 ~ #2
\end{center}\par
\fi}{%
  \let\title\@gobble}
\let\title\@gobble

\newlist{translationbib}{enumerate}{1}
\setlist[translationbib]{label=[\arabic*],align=left,nosep,itemsep=6bp,
  leftmargin=10mm,labelsep=!,before=\vspace{0.5\baselineskip}\wuhao[1.3]}

\newenvironment{conclusions}{%
\ifheu@openright\cleardoublepage\else\clearpage\fi
\ifheu@english
```

```
%\heu@appendix@chapter*{\heu@conclusion@etitle}
\addcontentsline{toc}{chapter}{\heu@conclusion@etitle}
\begin{center}
  ~\\
  \xiaoer\heiti\heu@conclusion@etitle
\end{center}\par
\else
%\heu@appendix@chapter*{\heu@conclusion@ctitle}[\heu@conclusion@etitle]
\addcontentsline{toc}{chapter}{\heu@conclusion@ctitle}
\begin{center}
  ~\\
  \xiaoer\heiti\heu@conclusion@ctitle
\end{center}\par
\fi}{}

\newenvironment{acknowledgements}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  \ifheu@english
  %\heu@appendix@chapter*{\heu@acknowledgement@etitle}
  \addcontentsline{toc}{chapter}{\heu@acknowledgement@etitle}
  \begin{center}
    ~\\
    \xiaoer\heiti\heu@acknowledgement@etitle
  \end{center}\par
  \else
  %\heu@appendix@chapter*{\heu@acknowledgement@ctitle}[\heu@acknowledgement@etitle]
  \addcontentsline{toc}{chapter}{\heu@acknowledgement@ctitle}
  \begin{center}
    ~\\
    \xiaoer\heiti\heu@acknowledgement@ctitle
  \end{center}\par
  \fi}{}

\newenvironment{correspondingaddr}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  %\heu@appendix@chapter*{\heu@correspondingaddr@ctitle}
  \addcontentsline{toc}{chapter}{\heu@correspondingaddr@ctitle}
  \begin{center}
    ~\\
    \xiaoer\heiti\heu@correspondingaddr@ctitle
  \end{center}\par
  }{}

```

博后用到的通讯页面

```
\newenvironment{resume}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  %\heu@appendix@chapter*{\heu@resume@ctitle}[\heu@resume@etitle]
  \addcontentsline{toc}{chapter}{\heu@resume@ctitle}
  \begin{center}
```

```
    ~\\
    \xiaoer\heiti\heu@resume@ctitle
  \end{center}\par
  }{}

\newenvironment{publication}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  %\heu@appendix@chapter*{\heu@publication@ctitle}[\heu@publication@etitle]
  \addcontentsline{toc}{chapter}{\heu@publication@ctitle}
  \begin{center}
    ~\\
    \xiaoer\heiti\heu@publication@ctitle
  \end{center}\par
  }{}

\newenvironment{doctorpublication}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  %\heu@appendix@chapter*{\heu@doctorpublication@ctitle}
  \addcontentsline{toc}{chapter}{\heu@doctorpublication@ctitle}
  \begin{center}
    ~\\
    \xiaoer\heiti\heu@doctorpublication@ctitle
  \end{center}\par
  }{}

```

此处添加博后用到的博士生发表页面此处中英文索引的格式设置尽量符合《哈尔滨工程大学研究生学位论文撰写规范》中给出的示例的格式。此处间距常数是人工调节的。

```
\newenvironment{ceindex}{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi
  %\heu@appendix@chapter*{\indexname}[\heu@index@etitle]
  \addcontentsline{toc}{chapter}{\indexname}
  \begin{center}
    ~\\
    \xiaoer\heiti\indexname
  \end{center}\par
  \setlength{\columnsep}{4em}
  \begin{adjustwidth}{2em}{2em}
  \begin{multicols*}{2}}{\end{multicols*}
  \end{adjustwidth}
  }

\newlist{idxwordlist}{description}{3}
\setlist[idxwordlist, 1]{%
  itemsep=\baselineskip,
  labelindent=8em,
  font=\normalsize\bfseries,
}
\setlist[idxwordlist, 2]{%
  nosep,
```

```
  labelindent=2em,
  font=\wuhao\rm,
}
\setlist[idxwordlist, 3]{%
  nosep,
  labelindent=4em,
  font=\wuhao\rm,
}

\newcommand\bibstyle@numerical{\bibpunct{[}{]}{,}{s}{,}{\textsuperscript{,}}}
\citestyle{numerical}
\DeclareRobustCommand\inlinecite{\@inlinecite}
\def\@inlinecite#1{\begingroup\citestyle{inline}\let\@cite\NAT@citenum\citep{#1}\endgro
\let\onlinecite\inlinecite
\newcommand{\lcite}{\inlinecite}
\renewenvironment{thebibliography}[1]{%
  \ifheu@openright\cleardoublepage\else\clearpage\fi\phantomsection%
  %\heu@appendix@chapter*{\bibname}[\heu@bibname@etitle]
  \addcontentsline{toc}{chapter}{\bibname}
  \begin{center}
    ~\\
    \xiaoer\heiti\bibname
  \end{center}\par
  \normalsize
  \list{\@biblabel{\@arabic\c@enumiv}}%
  {\renewcommand{\makelabel}[1]{##1\hfill}
    \settowidth{\labelwidth}{\@biblabel{#1}}
    \setlength{\labelsep}{0.5em}
    \setlength{\itemindent}{0pt}
    \setlength{\leftmargin}{\labelsep+\labelwidth}
    \addtolength{\itemsep}{-0.8em}
    \usecounter{enumiv}%
    \let\p@enumiv\@empty
  \renewcommand\theenumiv{\@arabic\c@enumiv}}%
  \sloppy\frenchspacing
  \flushbottom
  \ifheu@splitbibitem
  \clubpenalty0
  \@clubpenalty \clubpenalty
  \widowpenalty0%
  \interlinepenalty-50%
  \else
  \clubpenalty4000
  \@clubpenalty \clubpenalty
  \widowpenalty4000%
  \interlinepenalty4000%
  \fi
\sfcode`\.\@m}
{\def\@noitemerr
  {\@latex@warning{Empty `thebibliography' environment}}%
```

```
\endlist\frenchspacing}
\patchcmd\NAT@citexnum{%
  \@ifnum{\NAT@ctype=\z@}{%
    \if*#2*\else\NAT@cmt#2\fi
  }{}%
  \NAT@mbox{\NAT@@close}%
}{%
  \NAT@mbox{\NAT@@close}%
  \@ifnum{\NAT@ctype=\z@}{%
    \if*#2*\else\textsuperscript{#2}\fi
  }{}%
}{}{}
\renewcommand\NAT@citesuper[3]{\ifNAT@swa
  \if*#2*\else#2\NAT@spacechar\fi
\unskip\kern\p@\textsuperscript{\NAT@@open#1\NAT@@close\if*#3*\else#3\fi}%
   \else #1\fi\endgroup}
\patchcmd{\NAT@citex}{%
  \if*#2*\else\NAT@cmt#2\fi
  \if\relax\NAT@date\relax\else\NAT@@close\fi
}{%
  \if\relax\NAT@date\relax\else\NAT@@close\fi
  \if*#2*\else\textsuperscript{#2}\fi
}{}{}
\renewcommand\NAT@cite%
    [3]{\ifNAT@swa\NAT@@open\if*#2*\else#2\NAT@spacechar\fi
        #1\NAT@@close\if*#3*\else\textsuperscript{#3}\fi\else#1\fi\endgroup}
⟨/bookcls⟩
```

4.6 其它

在模板文档结束时即装入配置文件，这样用户就能在导言区进行相应的修改。

```
⟨*bookcls⟩
\AtEndOfClass{\input{heuthesisbook.cfg}}
\AtEndOfClass{\sloppy}
⟨/bookcls⟩
```

5 修改记录

v1.0.1
General: 矫正英文目录缩进与中文目录一致的 bug 127
v1.0.2
General: 移植自 hitThesis 论文模板 29
v1.0.3
General: 删除文献格式定义，texlive 已包含 GBT7714-2015 29
v1.0.4
General: 修改附录格式 127
v1.0.5
General: 根据 2026.02.28 哈尔滨工程大学最新要求修改原创型声明和授权使用声明 . 122

6 索引

Symbols

\- 3137
\. 301, 2393, 4665
\@ 91, 92
\@@italiccorr 2944, 2946, 2949, 2950
\@afterheading 3348
\@arabic 2377, 2386, 2938, 4642, 4651
\@biblabel 2377, 2379, 4642, 4644
\@cite 2373, 4630
\@citea 2637
\@clubpenalty 2390, 4656, 4661
\@cont@LT@nonumintoc 3206
\@cont@oldLT@c@ption 3210
\@cont@oldtablename 3204, 3209
\@currentlabel 2940
\@dotsep 4447, 4464, 4472, 4490
\@dottedtocline 4406, 4449, 4450, 4469, 4470, 4471, 4482
\@dottedtoclinebold . 4482
\@for 2227, 3682
\@gobble 4513, 4514
\@height 2795, 2797, 2843, 2845, 2896, 2898, 2925
\@highpenalty ... 4453, 4466
\@if@contemptyarg 3203, 3208
\@ifnum 4670, 4676
\@ifstar 2939
\@inlinecite 2372, 2373, 4629, 4630
\@latex@warning . 2395, 4667
\@mainmatterfalse ... 2721
\@mainmattertrue 2739, 2745
\@makecaption 3157
\@makefnmark 2929, 2931, 2932
\@makefntext 2931, 2932
\@minipagefalse . 3175, 3197
\@mkboth 2903, 4486
\@nobreakfalse 3352
\@nobreaktrue 3349
\@noitemerr 2394, 4666
\@pnumwidth 4445, 4446, 4450, 4456, 4457
\@restonecolfalse ... 4483
\@starttoc 2346, 4393, 4428, 4488
\@tempboxa .. 3159, 3160, 3176, 3181, 3182, 3198
\@tempdima . 4448, 4459, 4461
\@thefnmark 2930
\@tocrmarg 4446
\@width . 2795, 2797, 2843, 2845, 2896, 2898, 2925
\{ 2683
\} 2684
\␣ 3944

A

a3cover.tex (file) *5*
\abovecaptionskip 3158, 3226
\abovedisplayshortskip ... 2146, 2148, 2754, 2756
\abovedisplayskip 2145, 2147, 2753, 2755
acknowledgement (env.) .. *28*
\addcontentsline . 3513, 3515, 3531, 3532, 3533, 3541, 3552, 3560, 3572, 3580, 3592, 3600, 3612, 3620, 4342, 4344, 4411, 4501, 4507, 4524, 4531, 4542, 4549, 4559, 4569, 4579, 4589, 4599, 4636
\addpenalty 4453
\addtocontents 4359, 4360, 4361
\addtolength 2383, 4648
\addvspace 4406
\advance 4459
\allowdisplaybreaks . 2935
\Alph 4492
\alph 3215, 3217
amsmath (package) *81*, *91*
\appendix 3150, 3151, 3152, 4494
appendix (env.) *27*
\apptocmd 2932
\arabic 2921, 3150, 3151, 3152, 3216, 3219, 3231, 4387, 4388, 4492, 4497, 4517
\arraycolsep 3253
\arraystretch 2267, 2275, 2320, 2327, 3944, 3948, 3953, 3960, 3966, 3971, 4062, 4068, 4100, 4111, 4145, 4161, 4193, 4212, 4216, 4242, 4260
\AtEndOfClass 2397, 2398, 4696, 4697
auscan(empty).pdf (file) *21*
auscan.pdf (file) *21*, *21*
\authorization ... *21*, 4331

B

back/cover.tex (file) ... *20*
\backmatter 2743
\bahao 2760
\banxiaosi 2760
\baselineskip 4300, 4304, 4307, 4309, 4311, 4313, 4316, 4319, 4321, 4323, 4329, 4518, 4612
\belowcaptionskip 3178, 3200, 3227
\belowdisplayshortskip 2148, 2756
\belowdisplayskip 2147, 2755
\bf 2957
\bfseries 3328, 3340, 3380, 3389, 3426, 3435, 3515, 3533, 3541, 3552, 3560, 4344, 4372, 4463, 4480, 4614
\BiAppChapter 3538
\bib 572
\bibaddress 111
\bibauthor 100
\bibbooktitle 104
\bibcitedate 120
\bibcountry 107
\bibdate 122
\bibedition 109
\bibeditor 101
\bibinstitution 113
\bibitem 446
\bibjournal 105
\bibmark 106
\bibmodifydate 119

\bibname ... 4635, 4636, 4639
\bibnumber 116
\biborganization 110
\bibpages 118
\bibpatentid 108
\bibpublisher 112
\bibpunct .. 2368, 2369, 4627
\bibschool 114
\bibstyle@inline 2369
\bibstyle@numerical 2368, 4627
\bibtitle 103
\bibtranslator 102
\biburl 123, 819
\bibversion 117
\bibvolume 115
\bibyear 121
booktabs (package) *83*
\bottomfraction 3155
\box 3176, 3198, 4354
build.bat (file) *5*
build_all.bat (file) *5*

C

\c@chapter 2938
\c@enumiv 2377, 2386, 4642, 4651
\c@equation 2938
\c@tocdepth 4452
cabstract (env.) *21*
\cabstractcname 3297, 3511, 4363, 4368
\cabstractcname . 3294, 4363
\caffil *19*
\caption 3203, 3208
\captionnamefont 3223
\captiontitlefont ... 3224
\cassosupervisor *20*
\cauthor *19*
\ccosupervisor *20*
\ccwd ... 3017, 3054, 3055, 3056, 3057, 3062, 3066, 3067, 3068, 3069, 3070, 3071, 3074, 3075, 3076, 3081, 3124, 3159, 3163, 3171, 3181, 3185, 3193, 3271, 3732, 3737, 3751, 3752, 3759, 3770, 3811, 3814, 3817, 3820, 3886, 3945, 3946, 3947, 4471
\cdate *20*, 3033, 3314
ceindex (env.) *27*
\center 3370, 3415, 3460
\changes 3898
\chapter ... 3540, 3543, 4484
\chaptermark 2903
\chuhao 2760
\cite 3222
\citep 2373, 4630
\citestyle 2371, 2373, 4628, 4630
\citeup 3222
\CJK@today .. 3026, 3027, 3029, 3030, 3031, 3307, 3308, 3310, 3311, 3312
\CJK@todaybig 3025, 3031, 3306, 3312
\CJK@todaybig@short 3024, 3305
\CJK@todaysave 3021, 3029, 3302, 3310
\CJK@todaysmall .. 3023, 3026, 3030, 3033, 3304, 3307, 3311, 3314
\CJK@todaysmall@short ... 3022, 3033, 3303, 3314
\CJKtoday 3028, 3309
\ckeywords *21*
\ClassError 2013, 2026, 2038, 2066, 2532, 2558, 2571, 2643, 2915
\cleardoublepage . 2689, 2720, 2724, 2735, 2744, 3545, 4274, 4278, 4281, 4291, 4296, 4298, 4326, 4338, 4357, 4373, 4384, 4397, 4419, 4498, 4521, 4539, 4557, 4567, 4577, 4587, 4597, 4634
\clearpage 2356, 2359, 2364, 2715, 2720, 2724, 2735, 2737, 2744, 3545, 4274, 4278, 4281, 4291, 4298, 4326, 4338, 4357, 4373, 4374, 4384, 4396, 4397, 4419, 4498, 4521, 4539, 4557, 4567, 4577, 4587, 4597, 4634
\clubpenalty 2389, 2390, 3353, 3358, 4655, 4656, 4660, 4661
\cmidrulewidth 2618
\cnumber *18*
\Collect@Body ... 3675, 3677
\columnsep 4604
conclusion (env.) *26*
\contentsname 4421, 4422, 4425, 4427
\coralexdate *20*
\cprofessional *20*
\creviewer *20*
\cstudentid *21*
\cstudenttype *20*
\csubject *19*
\csubmitdate *20*
\csubtitle *19*
\csupervisor *20*
ctex (package) *80*
\CTEX@chapter@aftername 3329, 3330
\CTEX@chaptername 3329, 3330
ctexbook (package) . *79*, *80*, *80*
\ctexset 2174, 2906, 3016, 3257, 3268, 3281, 3362, 3408, 3453, 4491
\CTEXthechapter 2903
\ctitle *19*
\ctitleone *19*
\ctitletwo *19*
\CurrentOption .. 2005, 2501
\cxueke *19*

D

\dachu 2760
\dawu 2760
\day ... 3023, 3025, 3304, 3306
\DeclareBoolOption 1997, 1998, 1999, 2000, 2001, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499
\DeclareDefaultOption 2005, 2501
\DeclareRobustCommand 2372, 4629
\DeclareStringOption 2003, 2500

\def@NAT@last@yr 2634, 2637, 2639
\define@key 1952, 1962, 1979, 1992, 2207, 2224, 2411, 2421, 2437, 2449, 2459, 2466, 2473, 3625, 3679
denotation (env.) *21*
description (env.) *26*
\df@tag 2940
\dimexpr 2152, 2759
\ding 2919
\do 2227, 3682

E

eabstract (env.) *21*
\eabstractcname .. 3295, 3327, 3530, 4375, 4378
\eabstractename . 3296, 4375
\eaffil *19*
\eassosupervisor *20*
\eauthor *19*
\ecosupervisor *20*
\edate *20*, 3315
\ekeywords *21*
\emultiline 4216
\encodingdefault 2583, 2587, 2592
\endgraf 3200
\endlist 2396, 4668
\engcontentsname 4480, 4484, 4487
\enskip 3159, 3163, 3171, 3181, 3185, 3193, 3258, 4382
\enspace 2180, 2189, 2198, 2903, 3381, 3390, 3399, 3414, 3427, 3436, 3445, 3459, 3472, 3481, 3490
\ensuremath 2955, 2999
enumerate (env.) *26*
enumitem (package) . *26*, *26*, *91*
environments:
  acknowledgement ... *28*
  appendix *27*
  cabstract *21*
  ceindex *27*
  conclusion *26*
  denotation *21*
  description *26*
  eabstract *21*
  enumerate *26*
  itemize *26*
  publication *27*
  resume *28*
\eqref 2951
\equal .. 2047, 2053, 2239, 2293, 2514, 2520, 3327, 3511, 3530, 3995, 4038, 4119
\equationname 2944, 2946, 3286, 3288
\equcaption 2946, 2950, 4410
\erhao 2248, 2760, 4057, 4092, 4094, 4140, 4186, 4240
\estudenttype *20*
\esubject *19*
\esubtitle *19*
\esupervisor *20*
\etitle *19*
\everypar 3350, 3359
\ext@equation ... 4409, 4411
\ext@figure 4361
\ext@table 4360
\exueke *19*

F

\fancyfoot . 2792, 2840, 2891
fancyhdr (package) *84*
\fancyhead 2791, 2801, 2802, 2805, 2807, 2810, 2813, 2817, 2819, 2822, 2825, 2829, 2831, 2834, 2837, 2853, 2854, 2856, 2860, 2863, 2866, 2871, 2874, 2877, 2882, 2885, 2888
\fancyhf ... 2746, 2783, 2789
\fancypagestyle 2782, 2788, 2851
file:
  a3cover.tex *5*
  auscan(empty).pdf . *21*
  auscan.pdf *21*, *21*
  back/cover.tex *20*
  build.bat *5*
  build_all.bat *5*
  front/cover.tex ... *18*
  front/denotation.tex *21*
  heuthesis.bst .. *4*, *6*, *25*
  heuthesis.dtx *4*, *4*
  heuthesis.ins *4*, *4*
  heuthesis.ist *5*
  heuthesis.pdf *4*
  heuthesisbook.cfg *4*, *5*, *6*, *28*
  heuthesisbook.cls *4*, *5*, *6*
  latexmkrc *5*
  main.pdf *4*, *5*, *5*
  main.tex *6*, *18*, *22*, *22*, *26*, *26*
  Makefile *4*, *5*, *5*, *5*
  README.md *4*, *5*
  spine.tex *5*, *5*
  thesis.tex *3*
\FloatBarrier命令 2136, 2669
\floatpagefraction .. 3156
\floatsep 3144, 3148
\flushbottom 2388, 4653
\fontsize 2152, 2759, 3380, 3389, 3426, 3435, 3471, 3480
fontspec (package) *80*, *80*
\footnotemargin 2928
\footnoterule 2925
\footrulewidth 2786, 2848, 2893
\frenchspacing 2387, 2396, 4652, 4668
front/cover.tex (file) .. *18*
front/denotation.tex (file) *21*
\frontmatter 2719

G

\g@addto@macro 2229, 2232, 3150, 3151, 3152, 3684, 3687
\gdef 2208, 2209, 2225, 2226, 2940, 3035, 3036, 3037, 3041, 3042, 3043, 3047, 3205, 3626, 3627, 3674, 3676, 3680, 3681, 3695, 3696, 3697, 3698, 3699, 3700, 3706, 3707, 3708, 3709, 3710, 3711, 3716, 3717, 3718, 3719, 3720, 3721, 3726
\genacrfullformat ... 2703

\geometry 2111, 2646
\global 3175, 3197, 3204, 3206, 3209, 3210
\glsdescwidth 2695
\glsentrylong 2704

H

\hangafter 3330, 4353
\hangindent 3330, 4353
\hb@xt@ 3176, 4450
\hbox 2930, 3167, 3180, 3189, 3198, 4352, 4450, 4464
\headrule 2781, 2784, 2793, 2841, 2894
\headrulewidth .. 2785, 2892
\headwidth .. 2795, 2797, 2843, 2845, 2896, 2898
\heavyrulewidth 2616
\heiti 2179, 2188, 2197, 2960, 2998, 3248, 3321, 3513, 3531, 3532, 3950, 4057, 4058, 4091, 4092, 4093, 4098, 4140, 4141, 4187, 4188, 4303, 4315, 4368, 4372, 4401, 4425, 4463, 4504, 4510, 4527, 4534, 4545, 4552, 4562, 4572, 4582, 4592, 4602, 4639
\heu 3138
\heu@@cabstract . 3674, 3675
\heu@@eabstract . 3676, 3677
\heu@@makefnmark 2930, 2931
\heu@acknowledgement@ctitle ... 3831, 4548, 4549, 4552
\heu@acknowledgement@etitle 3832, 4541, 4542, 4545, 4548
\heu@appendix ... 4494, 4496
\heu@appendix@chapter 3500, 3520, 4299, 4327, 4363, 4375, 4385, 4398, 4523, 4530, 4541, 4548, 4558, 4568, 4578, 4588, 4598, 4635
\heu@appendix@ctitle 3828, 4507, 4510
\heu@appendix@etitle 3829, 4501, 4504
\heu@applicationresearchfalse 1966, 2425
\heu@applicationresearchtrue 1971, 2430
\heu@authorization@bachelor 4325, 4335
\heu@authorization@ctitle 3843, 4299, 4342
\heu@authorization@etitle 3844, 4299, 4344
\heu@authorization@other 4297, 4335, 4336
\heu@authorizationtext 3900, 4320
\heu@authorizationtext@bachelor 3906, 4328
\heu@authorizationtitle 3100, 3851, 4317
\heu@authorsig ... 3097, 3846, 4310, 4322, 4330
\heu@bachelor@caffiltitle 3058, 3760, 4064
\heu@bachelor@caffiltitleqd 3060, 3762
\heu@bachelor@caffiltitleya 3763
\heu@bachelor@caffiltitleyt 3059
\heu@bachelor@cclass 3057
\heu@bachelor@ccolegetitle 3753, 4103
\heu@bachelor@cdatetitle 2326, 3062, 3759
\heu@bachelor@checkdate 2341, 3065, 3768
\heu@bachelor@cmajortitle 2322, 3054
\heu@bachelor@cmajortitlecov 3754, 4065
\heu@bachelor@cmajortitlesec 3755, 4104
\heu@bachelor@cprofessionaltitle 3757, 4106
\heu@bachelor@cstudentidtitle 2324, 3056, 3751, 4049, 4085
\heu@bachelor@cstudenttitle 2323, 3055, 3750, 4066, 4102
\heu@bachelor@csupervisortitle 2325, 3061, 3756, 4067, 4105
\heu@bachelor@cthesisname 2301, 2831, 2834, 2837, 2882, 2885, 2888, 3053, 3749, 4054
\heu@bachelor@cthesistitle 3758
\heu@bachelor@cupervisoraffiltitle 3761, 4107
\heu@bachelor@cxuewei 2831, 2834, 2837, 2882, 2885, 2888, 3052, 3748, 4054
\heu@bachelor@secretlevel 3752, 4050, 4086
\heu@bachelor@teachercomment 2340, 3063, 3766
\heu@bachelor@teachersign 2341, 3064, 3767
\heu@bachelorfalse 1953, 2412
\heu@bibname@etitle 3826, 4635
\heu@cabstract .. 3674, 4370
\heu@caffil 2269, 4064, 4103, 4203, 4207
\heu@caffiltitle 3085, 3782, 4207
\heu@cassosupervisor 4149, 4150, 4197, 4198
\heu@cassosupervisortitle ... 3089, 3789, 4150, 4198
\heu@cauthor 2272, 2323, 3963, 4008, 4066, 4102, 4147, 4195
\heu@cauthortitle 3038, 3044, 3701, 3712, 3722, 4147, 4195
\heu@cbraceleft 3927
\heu@cbraceright 3928
\heu@ccolegetitle 3783, 4203
\heu@ccosupervisor 4153, 4201
\heu@ccosupervisortitle ... 3090, 3790, 4153, 4201
\heu@cdate .. 2274, 2326, 3989, 4033, 4072, 4165

\heu@cdegree 2807, 2810, 2813, 2819, 2825, 2860, 2863, 2866, 2871, 2877, 3037, 3043, 3699, 3710, 3720, 4136, 4183, 4205
\heu@cdegreetitle 3087, 3787, 4205
\heu@cenddate 4013
\heu@cfinishdate 3969
\heu@chapter 3516, 3534, 3543, 3549, 3550, 3557, 3558, 4422
\heu@chapter@engformat 3333, 3369
\heu@chapter@engnameformat 3343, 3370
\heu@chapter@postdocformat 3347, 3415
\heu@chapter@titleformat ... 3322, 3372, 3418, 3463
\heu@chinesefalse ... 2451
\heu@chinesetrue 2454
\heu@citetwocommafalse 2474
\heu@citetwoendashfalse 2475
\heu@citetwoendashtrue 2472
\heu@ckeywords 4372
\heu@ckeywords@separator 3300
\heu@ckeywords@title 3299, 4372
\heu@cleardoublepage 2713, 2715
\heu@clearemptydoublepage 2714, 2717
\heu@cnumber 3947, 4131, 4179
\heu@cnumbertitle 3096, 3809, 4131, 4179
\heu@conclusion@ctitle ... 3823, 4530, 4531, 4534
\heu@conclusion@etitle 3825, 4523, 4524, 4527, 4530
\heu@coralexdate 4109, 4210
\heu@coralexdatetitle ... 3084, 3781, 4109, 4210
\heu@correspondingaddr@ctitle ... 3840, 4558, 4559, 4562
\heu@cpositionname .. 4010
\heu@cprodrgreetitle 3785, 4154
\heu@cprofessional 4067, 4106
\heu@cprosubjecttile 3786, 4202
\heu@creviewer .. 4158, 4159
\heu@creviewertitle 3091, 3791, 4159
\heu@cschoolname 2244, 2802, 2807, 2810, 2813, 2819, 2822, 2825, 2831, 2834, 2837, 2856, 2860, 2863, 2866, 2871, 2874, 2877, 2882, 2885, 2888, 3077, 3772, 3901, 3903, 3905, 3907, 3950, 3977, 4021, 4054, 4071, 4110, 4164, 4211, 4303, 4315
\heu@cschoolnametitle ... 3082, 3779, 4110, 4211
\heu@cstartdate 4012
\heu@cstudentid 2273, 2324, 4049, 4085
\heu@cstudenttype ... 4154
\heu@csubject 2270, 2322, 4011, 4065, 4104, 4156, 4202, 4206
\heu@csubjecttitle 3086, 3784, 4156, 4206
\heu@csubmitdatatitle ... 3083, 3780, 4108, 4209
\heu@csubmitdate 3970, 4108, 4209
\heu@csubtitle 4058, 4093, 4141, 4188
\heu@csupervisor 2271, 2325, 4009, 4067, 4105, 4148, 4196
\heu@csupervisorcaffil 4107
\heu@csupervisortitle ... 3088, 3788, 4148, 4196
\heu@cthesismidtermcheck 3081
\heu@cthesisname . 2248, 2807, 2810, 2813, 2819, 2822, 2825, 2860, 2863, 2866, 2871, 2874, 2877, 3079, 3771, 4136, 4183
\heu@cthesistitleprefix ... 2262, 2315, 3066, 3770
\heu@ctitle 2854, 3907, 4266
\heu@ctitlecover 2262, 2315, 4001, 4057, 4092, 4140, 4187
\heu@ctitleone 3956
\heu@ctitletwo 3958
\heu@cxueke 3037, 3043, 3699, 3710, 3720
\heu@cxuewei 2248, 3036, 3037, 3042, 3043, 3047, 3698, 3699, 3709, 3710, 3719, 3720, 3726, 3918
\heu@cxueweishort 3035, 3036, 3038, 3041, 3042, 3044, 3081, 3695, 3698, 3701, 3706, 3709, 3712, 3716, 3719, 3722
\heu@datefill 3135, 3923, 4312, 4324, 4330
\heu@declarename 3908, 4305
\heu@declarename@bachelor 3905, 4327, 4342
\heu@declaretext 3909, 4308
\heu@def@fontsize 2150, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2757, 2760, 2761, 2762, 2763, 2764, 2765, 2766, 2767, 2768, 2769, 2770, 2771, 2772, 2773, 2774, 2775, 2776, 2777, 2778
\heu@def@term 2206, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 3624, 3630, 3631, 3632, 3633, 3634, 3635, 3636, 3637, 3638, 3639, 3640, 3641, 3642, 3643, 3644, 3645, 3646, 3647, 3648, 3649, 3650, 3651, 3652, 3653, 3655, 3656, 3657, 3658, 3660, 3661, 3662, 3663, 3664,

3665, 3666, 3667, 3668, 3669, 3670, 3671, 3672
\heu@denotation@ctitle 3849, 4385
\heu@denotation@etitle 3850, 4385
\heu@doctor@midterm@note 3121, 3883
\heu@doctorfalse 1955, 2414
\heu@doctorpublication@ctitle ... 3921, 4588, 4589, 4592
\heu@eabstract .. 3676, 4380
\heu@eaffil 4252
\heu@eaffiltitle 3799, 4252
\heu@eassosupervisor 4246, 4247, 4250
\heu@eassosupervisortitle 3795, 4247, 4250
\heu@eauthor 4244
\heu@eauthortitle 3793, 4244
\heu@ebraceleft 3929
\heu@ebraceright 3930
\heu@ecategorytitle 3800, 4251
\heu@ecosupervisortitle 3796
\heu@edegree 3700, 3711, 3721, 4254
\heu@edegreetitle 3797, 4254
\heu@ekeywords 4382
\heu@ekeywords@separator 3301
\heu@empty 2780
\heu@engcover ... 4217, 4290
\heu@englishfalse ... 2450
\heu@eoralexdate 4258
\heu@eoralexdatetitle 3802, 4258
\heu@eschoolname 3804, 4259
\heu@eschoolnametitle 3803, 4259
\heu@estudenttype ... 4234
\heu@esubject ... 4251, 4255
\heu@esubjecttitle 3798, 4255
\heu@esubmitdate 4257
\heu@esubmitdatetitle 3801, 4257
\heu@esubtitle .. 4095, 4241
\heu@esupervisor 4245
\heu@esupervisortitle 3794, 4245
\heu@etitle 4003, 4094, 4240, 4289
\heu@exueke 3700, 3711, 3721
\heu@exuewei 3696, 3700, 3707, 3711, 3717, 3721
\heu@exueweier 3697, 3708, 3718, 4235
\heu@first@titlepage 3937, 4273, 4277
\heu@first@titlepage@bachelor 3938, 4036
\heu@first@titlepage@other 3938, 4117
\heu@first@titlepage@postdoc 3938, 3943
\heu@fontset 2047, 2053, 2058, 2239, 2293, 2514, 2520, 2525, 3995, 4038, 4119
\heu@footnotesize 2926, 2927
\heu@frontdate ... 3099, 3848, 4312, 4324, 4330
\heu@geometrynewnofalse 2462
\heu@geometrynewonefalse 2460
\heu@geometrynewtwofalse 2461
\heu@geometrynewtwotrue 2547
\heu@graduate@caffiltitle 2269, 3067
\heu@graduate@cafflimajor 3076
\heu@graduate@cmajortitle 2270, 3068
\heu@graduate@datetitle 2274, 3072
\heu@graduate@enrolldate 3074
\heu@graduate@studentid 2273, 3071
\heu@graduate@studenttitle 2272, 3070
\heu@graduate@supervisor 2271, 3069
\heu@graduate@thesistitle 3075
\heu@harbin@bachelor@schoolbottommar 2334, 3116, 3877
\heu@harbin@schoolbottommark 2280, 3111, 3869
\heu@harbincampus 3777, 3979, 4023
\heu@harbinfalse 1982, 2440
\heu@harbintrue . 1986, 2444
\heu@headings 2780
\heu@headrule ... 2781, 2784
\heu@index@etitle 3925, 4598
\heu@inline@sep 2369
\heu@intclassifiedindex ... 3947, 4126, 4174, 4221
\heu@internatclassifiedindextitle ... 3094, 3807, 4126, 4174
\heu@kw 4349, 4352, 4353, 4354
\heu@listof 4395, 4405, 4407, 4413
\heu@LT@array 3212, 3213, 3214
\heu@make@df@tag@@ 2939, 2940
\heu@makefnmark . 2929, 2932
\heu@maketag 2940, 2943, 2949
\heu@masterfalse 1954, 2413
\heu@midtermfalse ... 1994
\heu@motto 2241, 2295, 3136, 3926, 3997, 4039, 4120
\heu@natclassifiedindex ... 3946, 4125, 4173, 4220
\heu@natclassifiedindextitle ... 3093, 3806, 4125, 4173
\heu@openingfalse ... 1993
\heu@openrightfalse . 2711
\heu@parse@keywords 2223, 3678, 3689, 3690
\heu@plain 2780
\heu@postdoc@authorname 3741, 4008
\heu@postdoc@classifiedindex 3729, 3946
\heu@postdoc@documenttitle 3737, 3951
\heu@postdoc@documenttitlenospace 2802, 2856, 3738
\heu@postdoc@enddate 3746, 4013
\heu@postdoc@finishdate 3739, 3969

\heu@postdoc@major 3744, 4011
\heu@postdoc@number 3731, 3947
\heu@postdoc@positionname 3743, 4010
\heu@postdoc@secretlevel 3730, 3946
\heu@postdoc@separator 3747, 4007
\heu@postdoc@startdate 3745, 4012
\heu@postdoc@submitdate 3740, 3970
\heu@postdoc@supervisor 3742, 4009
\heu@postdoc@UDC 3732, 3947
\heu@postdocfalse ... 2415
\heu@productdevelopmentfalse 1963, 2422
\heu@projectdesignfalse 1965, 2424
\heu@projectdplanningfalse 1964, 2423
\heu@publication@ctitle 3913, 3916, 3918, 4578, 4579, 4582
\heu@publication@etitle 3924, 4578
\heu@put@keywords 4350, 4372, 4382
\heu@put@title 3934, 4064, 4065, 4066, 4067, 4102, 4103, 4104, 4105, 4106, 4107, 4108, 4109, 4110, 4147, 4148, 4150, 4153, 4154, 4156, 4159, 4195, 4196, 4198, 4201, 4202, 4203, 4205, 4206, 4207, 4209, 4210, 4211, 4247
\heu@qingdao@schoolbottommark 2283, 3101, 3853
\heu@qingdaocampus 2244, 2810, 2822, 2834, 2863, 2874, 2885, 3078, 3775, 3985, 4029
\heu@qingdaofalse 1980, 2438
\heu@report@backpage@bachelor 2338, 2365
\heu@report@titlepage@bachelor 2291, 2352
\heu@report@titlepage@graduate 2237, 2354
\heu@resume@ctitle 3835, 3837, 4568, 4569, 4572
\heu@resume@etitle 3842, 4568
\heu@searcetype .. 3811, 3814, 3817, 3820, 4189
\heu@second@titlepage 3940, 4280
\heu@second@titlepage@bachelor 3941, 4076
\heu@second@titlepage@other 3941, 4169
\heu@second@titlepage@postdoc 3941, 3993
\heu@secretlevel 3095, 3808, 4130, 4178
\heu@section 3564, 3569, 3570, 3577, 3578
\heu@stage@doctype 2256, 2309, 3051, 3073, 3735
\heu@stage@midterm 2253, 2306, 3050, 3073, 3734
\heu@stage@opening 2250, 2303, 3049, 3072, 3733
\heu@starttoc ... 4390, 4403
\heu@statesecrets 3946, 4050, 4086, 4130, 4178
\heu@subsection .. 3584, 3589, 3590, 3597, 3598
\heu@subsubsection 3604, 3609, 3610, 3617, 3618
\heu@teachersig 3098, 3847, 4322
\heu@textcircled 2913, 2923, 2924
\heu@title@csep .. 2262, 2315, 3092, 3792, 4063, 4101, 4146, 4194
\heu@title@esep .. 3805, 4095, 4241, 4244, 4245, 4250, 4251, 4252, 4254, 4255, 4257, 4258, 4259
\heu@title@font .. 3320, 3335, 3347, 3380, 3389, 3398, 3426, 3435, 3444, 3460, 3471, 3480, 3489
\heu@title@width 3933, 3934, 4061, 4099, 4144, 4192
\heu@titleheadings .. 2780
\heu@toc@font 4443
\heu@yantai@schoolbottommark 2286, 3106, 3861
\heu@yantaicampus 2244, 2813, 2825, 2837, 2866, 2877, 2888, 3080, 3776, 3982, 4026
\heu@yantaifalse 1981, 2439
\heu@zijvregufalse .. 2467
\heu@zijvregutrue ... 2541
\heu@zijvwordfalse .. 2468
\heusetup *18*, 2235, 3691
\heuthesis 3137
heuthesis.bst (file) . *4*, *6*, *25*
heuthesis.dtx (file) *4*, *4*
heuthesis.ins (file) *4*, *4*
heuthesis.ist (file) *5*
heuthesis.pdf (file) *4*
heuthesisbook.cfg (file) *4*, *5*, *6*, *28*
heuthesisbook.cls (file) *4*, *5*, *6*
\hfil 3176, 3198
\hfill 2341, 2378, 3167, 3189, 3970, 4046, 4082, 4127, 4175, 4222, 4330, 4359, 4360, 4361, 4464, 4643
\hline 3957, 3959
\hrule .. 2795, 2797, 2843, 2845, 2896, 2898, 2925
\hsize . 3160, 3176, 3182, 3198
\hskip 2695, 2944, 2946, 3159, 3163, 3171, 3181, 3185, 3193, 3245, 3246, 4392, 4460
\hspace . 3017, 3054, 3055, 3056, 3057, 3062, 3066, 3067, 3068, 3069, 3070, 3071, 3074, 3075, 3076,

3081, 3124, 3135, 3271, 3560, 3732, 3737, 3751, 3752, 3759, 3770, 3811, 3814, 3817, 3820, 3886, 3903, 3923, 3945, 4058, 4093, 4141, 4188, 4310, 4312, 4322, 4324, 4330
\hss 3180, 3201
\hypersetup 2098, 2621

I

\if@afterindent 3354
\if@mainmatter 3336
\if@nobreak 3351
\if@restonecol 4489
\IfBooleanT 3501, 3521
\IfBooleanTF 3546, 3566, 3586, 3606
\iffalse 2941
\ifheu@absupper 3531
\ifheu@applicationresearch 1961, 2025, 2420, 2557, 3819
\ifheu@arialtitle 3321, 3513, 3531, 3532
\ifheu@arialtoc 4442
\ifheu@bachelor .. 1949, 2010, 2351, 2407, 2504, 2528, 2828, 2880, 2942, 3033, 3046, 3159, 3163, 3171, 3181, 3185, 3193, 3314, 3725, 3938, 3941, 4270, 4285, 4335, 4342, 4417, 4462, 4492, 4497
\ifheu@capcenterlast 3161, 3183
\ifheu@chapterhang .. 3330
\ifheu@chinese 2448
\ifheu@citetwocomma 2470, 2633
\ifheu@citetwoendash 2471
\ifheu@debug 2125, 2645, 2659
\ifheu@doctor 1951, 2012, 2409, 2510, 2530, 2734, 2804, 2858, 3034, 3705, 4343, 4429
\ifheu@english 2447, 2453, 2690, 2706, 2790, 2956, 2966, 3256, 3285, 3323, 3361, 3499, 3912, 4358, 4436, 4463, 4500, 4522, 4540
\ifheu@engtoc 4432
\ifheu@geometrynewno 2458, 2546
\ifheu@geometrynewone 2456, 2544
\ifheu@geometrynewtwo 2457, 2545
\ifheu@glue 2752, 3146, 3147, 3148, 3378, 3379, 3380, 3387, 3388, 3389, 3396, 3397, 3424, 3425, 3426, 3433, 3434, 3435, 3442, 3443, 3469, 3470, 3471, 3478, 3479, 3480, 3487, 3488
\ifheu@harbin 1978, 2279, 2436, 2806, 2818, 2830, 2859, 2870, 2881, 3978, 4022
\ifheu@library 2711, 2735, 4274, 4278, 4281, 4291
\ifheu@master 1950, 2011, 2408, 2507, 2529, 2816, 2869, 3040, 3715
\ifheu@midterm ... 1991, 2037, 2252, 2305, 3073
\ifheu@newtxmath 2074, 2582
\ifheu@opening ... 1990, 2036, 2249, 2302, 3072
\ifheu@openright . 2720, 2724, 2744, 3545, 4298, 4326, 4338, 4357, 4373, 4384, 4397, 4419, 4498, 4521, 4539, 4557, 4567, 4577, 4587, 4597, 4634
\ifheu@pageempty 2712
\ifheu@pifootnote 2089, 2609, 2918
\ifheu@postdoc 2410, 2531, 2800, 2855, 3141, 3280, 3407, 3694, 3834, 3915, 3938, 3941, 4282, 4372, 4463
\ifheu@productdevelopment 1958, 1968, 2022, 2417, 2427, 2554, 3810
\ifheu@profdegree 2553, 4152, 4189, 4200, 4234, 4249
\ifheu@projectdesign 1960, 1970, 2024, 2419, 2429, 2556, 3816
\ifheu@projectdplanning 1959, 1969, 2023, 2418, 2428, 2555, 3813
\ifheu@qingdao 1976, 1985, 2244, 2282, 2434, 2443, 2809, 2821, 2833, 2862, 2873, 2884, 3984, 4028
\ifheu@raggedbottom 2082, 2602
\ifheu@splitbibitem . 4654
\ifheu@subcapcenterlast 2672
\ifheu@subtitle 4058, 4093, 4095, 4141, 4188, 4241
\ifheu@toc 2357
\ifheu@tocfour 4417
\ifheu@yantai 1977, 1984, 2244, 2285, 2331, 2435, 2442, 2812, 2824, 2836, 2865, 2876, 2887, 3981, 4025, 4269
\ifheu@zijvregu 2464, 2539, 2728
\ifheu@zijvword 2465, 2540, 2725
\ifNAT@swa 4680, 4692
\IfNoValueTF 3548, 3556, 3568, 3576, 3588, 3596, 3608, 3616, 4332
\ifthenelse 2046, 2052, 2238, 2292, 2513, 2519, 3326, 3510, 3529, 3994, 4037, 4118
\IfValueT ... 3515, 3533, 3551, 3559, 3571, 3579, 3591, 3599, 3611, 3619
\IfValueTF 3504, 3524
\ifxetex 2064, 2569
\ignorespaces 2230, 2944, 2946, 2949, 2950, 3560, 3580, 3600, 3620, 3685
\includegraphics 2298

\includepdf 4339
\includepdfset .. 2079, 2599
\indexname . 4598, 4599, 4602
\inlinecite *25*, 2372, 2374, 2375, 4629, 4631, 4632
\input 2397, 4696
\intclassifiedindex .. *18*
\interlinepenalty 2392, 4658, 4663
\intextsep . 2695, 3142, 3146
\item 3129, 3130, 3131, 3132, 3245, 3246, 3891, 3892, 3893, 3894
\itemindent 2381, 3234, 3235, 4646
itemize (env.) *26*
\itemsep ... 2383, 3240, 4648

K

\kaishu 2244, 4071, 4164
\kern 4682
kvoptions (package) *18*
\kvsetkeys 1947, 2235, 2406, 3691

L

\l@chapter 4451, 4482
\l@equation 4414
\l@figure .. 4406, 4408, 4414
\l@section 4469
\l@subsection 4470
\l@subsubsection 4471
\l@table 4408
\labelsep 2380, 2382, 3234, 3236, 3245, 3246, 4645, 4647
\labelwidth 2379, 2382, 3234, 4644, 4647
\lastbox ... 3166, 3188, 3355
latexmkrc (file) *5*
\lcite 2375, 4632
\leaders 4464
\leavevmode 4458
\leftmargin 2382, 3234, 4647
\leftmargini 4473
\leftmarginii 4474
\leftmarginiii 4475
\leftmarginiv 4476
\leftmarginv 4477
\leftmarginvi 4478
\leftmark 2791, 2801, 2805, 2817, 2829
\leftskip 4459, 4460
\lightrulewidth 2617
\list 2377, 4642
\listequationename 3292, 4413
\listequationname ... 3293
\listfigureename 3290, 4405
\listof 4390
\listofequations 4413
\listoffigures .. 4405, 4439
\listoftables ... 4407, 4438
\listparindent 3241
\listtableename . 3291, 4407
\liuhao 2760
\long .. 3157, 3179, 3674, 3676
\longbionenumcaption 3202
longtable (package) *98*
\LT@array .. 3212, 3213, 3214
\LT@c@ption 3206, 3210
\LT@cols 3180
\LT@makecaption 3179
\LT@mcol 3180
\LTcapwidth 3180
\ltfontsize 3212

M

\m@th 4464
main.pdf (file) *4*, *5*, *5*
main.tex (file) *6*, *18*, *22*, *22*, *26*, *26*
\mainmatter 2733
\make@df@tag 2939
\make@df@tag@@@ 2939
\makeabstract 4356
\makebackcover 2363
\makebox 2269, 2270, 2271, 2272, 2273, 2274, 2322, 2323, 2324, 2325, 2326, 2341, 3946, 3947, 3956, 3958, 3969, 3970, 4008, 4009, 4010, 4011, 4012, 4013, 4049, 4050, 4085, 4086, 4125, 4126, 4130, 4131, 4173, 4174, 4178, 4179
\makecover 2349, 4264
Makefile (file) *4*, *5*, *5*, *5*
\makeglossaries 2709
\makelabel 2378, 4643
\makenoidxglossaries 2707
\maketag@@@ 2943, 2945, 2949, 2950
\MakeUppercase 3324, 3345, 3531, 4003
\markboth 3505, 3507, 3525, 3527, 4421
\mbox 106, 717, 718
\MessageBreak 2015, 2016, 2028, 2029, 2040, 2041, 2067, 2533, 2559, 2572
\mkern 4464
\month 3022, 3023, 3024, 3025, 3303, 3304, 3305, 3306, 3315

N

\n 2684
\NAT@@close . 4673, 4675, 4682, 4686, 4688, 4693
\NAT@@open 4682, 4692
\NAT@cite 4691
\NAT@citenum 2373, 4630
\NAT@citesuper 4680
\NAT@citex 4684
\NAT@citexnum 2634, 2643, 4669
\NAT@cmt 4671, 4685
\NAT@ctype 4670, 4676
\NAT@date 4686, 4688
\NAT@last@yr 2636
\NAT@mbox 4673, 4675
\NAT@penalty 2634, 2639
\NAT@spacechar .. 4681, 4692
\natclassifiedindex .. *18*
\NeedsTeXFormat 1934
\newbox 4349
\NewDocumentCommand 3500, 3520, 4331
\newglossarystyle ... 2692
\newif 1949, 1950, 1951, 1958, 1959, 1960, 1961, 1976, 1977, 1978, 1990, 1991, 2407, 2408, 2409, 2410, 2417, 2418, 2419, 2420, 2434, 2435, 2436, 2447, 2448, 2456, 2457, 2458, 2464, 2465, 2470, 2471
\newindex 2678, 2679

\newlength 3933
\newline 123
\newlist 4516, 4610
\newtheorem 2958, 2961, 2967, 2968, 2969, 2970, 2971, 2972, 2973, 2974, 2975, 2976, 2977, 2978, 2979, 2981, 2982, 2983, 2984, 2985, 2986, 2987, 2988, 2989, 2990, 2991, 2992, 2993, 3000, 3004, 3005, 3006, 3007, 3008, 3009, 3010, 3011, 3012, 3013, 3014, 3015
newtxmath (package) *82*
\nobreak 4465
\normalcolor 4465
\normalsize 2750
\numberline 3580, 3600, 3620, 4391, 4392, 4394, 4412
\numexpr 2919

O

\oldencodingdefault 2583, 2592
\oldnumberline 4391, 4392, 4394
\oldrmdefault ... 2584, 2593
\oldsfdefault ... 2585, 2594
\oldttdefault ... 2586, 2595
\onlinecite 2374, 4631

P

\p@enumiv 2385, 4650
\p@subfigure 3218
package:
- amsmath *81*, *91*
- booktabs *83*
- ctex *80*
- ctexbook *79*, *80*, *80*
- enumitem *26*, *26*, *91*
- fancyhdr *84*
- fontspec *80*, *80*
- kvoptions *18*
- longtable *98*
- newtxmath *82*
- pdfpages *82*
- pifont *79*
- xeCJK *3*, *18*, *80*
- xeCJKfntef *82*

\pagenumbering 2722, 2740, 4365
\pagestyle .. 2715, 2723, 2741, 2747, 2902, 4364
\parbox 2260, 2265, 2313, 2318, 2340, 2341, 3180, 3950, 3952, 3965, 4000, 4005, 4052, 4053, 4055, 4056, 4059, 4070, 4090, 4097, 4135, 4138, 4139, 4142, 4163, 4181, 4182, 4185, 4186, 4190, 4231, 4232, 4238, 4239
\parfillskip 4457
\parindent 3252, 4456
\parsep 3239
\PassOptionsToClass 2005, 2058, 2501, 2505, 2508, 2511, 2525
\PassOptionsToPackage 2008, 2049, 2056, 2503, 2516, 2523, 2673
\patchcmd 2634, 4449, 4450, 4669, 4684
\pdfbookmark 4266, 4289, 4427, 4485
pdfpages (package) *82*
\penalty 4466
pifont (package) *79*
\postdisplaypenalty . 2937
\pozhehao 4058, 4093, 4141, 4188
\predisplaypenalty .. 2936
\pretocmd 2931
\printnoidxglossary . 4437
\ProcessKeyvalOptions 2007, 2502
\profdegree *20*
\ProvidesClass .. 1935, 1936
\ProvidesFile ... 1937, 1938
publication (env.) *27*

Q

\qihao 2760
\qquad .. 4043, 4048, 4079, 4084, 4124, 4129, 4172, 4177, 4219, 4224, 4238
\quad 4067, 4138, 4185, 4189, 4302

R

\raggedbottom ... 2084, 2604
README.md (file) *4*, *5*
\RenewDocumentCommand ... 3544, 3565, 3585, 3605
\renewenvironment 2376, 2694, 4495, 4633
\renewtheoremstyle .. 3244
\researchtype *20*
\reserved@a 2227, 2233, 3682, 3688
resume (env.) *28*
\rightmargin 3237
\rightskip 4456
\rm 4619, 4624
\rmdefault . 2584, 2588, 2593
\rmfamily .. 3247, 3248, 4462

S

\sanhao 2266, 2319, 2760, 3103, 3108, 3113, 3124, 3339, 3855, 3858, 3863, 3866, 3871, 3874, 3886, 4001
\sbox 3159, 3181
\section 3564
\selectfont 2152, 2759, 3380, 3389, 3426, 3435, 3471, 3480
\setabbreviationstyle 2697
\setacronymstyle 2700
\setbox 3163, 3164, 3166, 3185, 3186, 3188, 3355, 4352
\setenumerate 3221
\setglossarystyle ... 2693
\setitemize 3220
\setlist 2910, 4517, 4611, 4616, 4621
\setmainfont 2579
\setmonofont 2581
\setsansfont 2580
\settowidth 2379, 3330, 4644
\SetupKeyvalOptions 1944, 2403
\sfdefault . 2585, 2589, 2594
\sffamily 3321, 3513, 3531, 3532, 4443
\sihao .. 2188, 2760, 3127, 3889, 3944, 3963, 3966,

3975, 4003, 4006, 4019, 4189, 4191, 4360, 4361
\sloppy 2387, 2398, 4652, 4697
\songti . 2241, 2248, 2262, 2266, 2295, 2301, 2315, 2319, 2801, 2802, 2805, 2807, 2810, 2813, 2817, 2819, 2822, 2825, 2829, 2831, 2834, 2837, 2854, 2856, 2860, 2863, 2866, 2871, 2874, 2877, 2882, 2885, 2888, 3103, 3108, 3113, 3118, 3124, 3127, 3247, 3855, 3858, 3863, 3866, 3871, 3874, 3879, 3886, 3889, 3944, 3953, 3963, 3966, 3975, 3997, 4001, 4006, 4019, 4039, 4042, 4047, 4054, 4062, 4064, 4065, 4066, 4067, 4072, 4078, 4083, 4100, 4120, 4123, 4128, 4136, 4145, 4147, 4148, 4150, 4153, 4154, 4156, 4159, 4165, 4171, 4176, 4183, 4189, 4193, 4218, 4223
spine.tex (file) 5, 5
\Square 3811, 3814, 3817, 3820, 3903
\square 2955, 2999, 3249
\SquareSolid 3811, 3814, 3817, 3820
\statesecrets 18
\subcapsize 3225
\subsection 3584
\subsubsection 3604

T

\tablename . 3204, 3205, 3209
\tableofcontents 2344, 2358, 4418
\tableofengcontents 4430, 4433, 4481
\tagform@ 2945, 2950
\texorpdfstring .. 3513, 3515, 3531, 3532, 3533, 3541, 3552, 3560, 4344
\text 2944, 2946
\textbf 344, 2244, 2248, 2262, 2269, 2270, 2271, 2272, 2273, 2274, 2301, 2315, 2322, 2323, 2324, 2325, 2326, 3103, 3108, 3113, 3118, 3124, 3347, 3855, 3858, 3863, 3866, 3871, 3874, 3879, 3886, 4095, 4102, 4103, 4104, 4105, 4106, 4107, 4108, 4109, 4110, 4195, 4196, 4198, 4201, 4202, 4203, 4205, 4206, 4207, 4209, 4210, 4211, 4244, 4245, 4247, 4250, 4251, 4252, 4254, 4255, 4257, 4258, 4259, 4359, 4360, 4361, 4378, 4382
\textcircled 2921
\textcolor 2241, 2295, 3997, 4039, 4120
\textfloatsep ... 3143, 3147
\textfraction 3153
\textit 338
\textnormal 93, 95
\textsc 3137
\textsuperscript 2368, 3222, 4627, 4677, 4682, 4689, 4693
\textup 2951
\textwidth 2260, 2265, 2313, 2318, 2340, 2341, 2925, 3167, 3189, 3950, 3952, 3965, 4000, 4005, 4052, 4053, 4055, 4056, 4059, 4070, 4090, 4097, 4135, 4138, 4139, 4142, 4163, 4181, 4182, 4185, 4186, 4190, 4231, 4232, 4238, 4239
\the 2919, 3022, 3023, 3024, 3025, 3303, 3304, 3305, 3306, 3317
\thechapter 2938, 3150, 3151, 3152, 3541, 3560, 4497, 4499
\theenumiv 2386, 4651
\theequation 2938, 3152
\thefigure . 3150, 3216, 3218
\thefootnote 2923
\thempfootnote 2924
\theorem@headerfont 3245, 3246
\theorem@separator 3245, 3246
\theorembodyfont 2954, 2997, 3247
\theoremheaderfont 2957, 2960, 2998, 3248
\theorempostskipamount 3251
\theorempreskipamount 3250
\theoremseparator 2965, 3003
\theoremstyle ... 2963, 3001
\theoremsymbol ... 2955, 2964, 2999, 3002, 3249
\thepage ... 2792, 2840, 2891
thesis.tex (file) 3
\thesubfigure 3217
\thesubtable 3215
\thetable 3151, 3219, 4387, 4388
\thinspace 94
\thispagestyle 2339, 2345, 3122, 3884, 4333, 4340
\title 4513, 4514
\today . 3021, 3027, 3302, 3308
\topfraction 3154
\topsep 3238
\ttdefault . 2586, 2590, 2595
\twocolumn 4489
\typeout 2642

U

\underline 2269, 2270, 2271, 2272, 2273, 2274, 2322, 2323, 2324, 2325, 2326, 2341, 3903, 3946, 3947, 3969, 3970, 4049, 4050, 4085, 4086, 4125, 4126, 4130, 4131, 4173, 4174, 4178, 4179
\unhcopy 3167, 3189
\unskip 2944, 2946, 2949, 2950, 3167, 3189, 3317, 4682
\unvbox 3165, 3169, 3187, 3191
\url 123, 821
\urlstyle 2108, 2631
\usecounter 2384, 3233, 4649

V

\value 2914, 2919
\vbox .. 3163, 3164, 3185, 3186
\vfill . 2278, 2330, 3974, 4018
\vskip .. 2794, 2796, 2842, 2844, 2895, 2897, 2925, 3158, 3162, 3178, 3184, 3200, 4371, 4381, 4454
\vspace 2240, 2242, 2246, 2259, 2294, 2296, 2312, 2695, 3127, 3211, 3889, 3949, 3996, 3998, 4002, 4040, 4077, 4088, 4121, 4134, 4228, 4300, 4304, 4307, 4309, 4311, 4313, 4316, 4319, 4321, 4323, 4329, 4518

W

\wd 3160, 3182, 4353
\widowpenalty 2391, 4657, 4662
\wuhao .. 2760, 2791, 2792, 2801, 2802, 2805, 2807, 2810, 2813, 2817, 2819, 2822, 2825, 2829, 2831, 2834, 2837, 2840, 2854, 2856, 2860, 2863, 2866, 2871, 2874, 2877, 2882, 2885, 2888, 2891, 3159, 3181, 3213, 3223, 3224, 3225, 3811, 3814, 3817, 3820, 3901, 3903, 3907, 3910, 4310, 4312, 4322, 4324, 4518, 4619, 4624

X

xeCJK (package) *3*, *18*, *80*
xeCJKfntef (package) *82*
\xiaochu 2760
\xiaoer . 2244, 2262, 2315, 2760, 3118, 3337, 3347, 3460, 3879, 3950, 3953, 4054, 4058, 4071, 4093, 4095, 4135, 4136, 4141, 4164, 4182, 4188, 4232, 4238, 4241, 4303, 4315, 4368, 4378, 4401, 4425, 4504, 4510, 4527, 4534, 4545, 4552, 4562, 4572, 4582, 4592, 4602, 4639
\xiaoliu ... 2760, 2921, 3903
\xiaosan 2179, 2760, 4060, 4072, 4091, 4098, 4143, 4165
\xiaosi . 2240, 2241, 2294, 2295, 2760, 3541, 3996, 3997, 4039, 4042, 4047, 4078, 4083, 4120, 4123, 4128, 4171, 4176, 4218, 4223, 4242, 4267, 4300
\xiaowu 2760, 2927
\xiaoyi 2301, 2760

Y

\year 3022, 3023, 3024, 3025, 3303, 3304, 3305, 3306, 3317
\yihao 2760

Z

\zhdigits 3024, 3025, 3305, 3306
\zhnumber 3024, 3025, 3305, 3306
\ziju 2726, 2729